# Configuration Summary

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FORECAST ANALYSIS — CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-09-22 12:20:43
======================================================================

── Data Preparation ──

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

── Model Training ──

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0005
min_learning_rate...................... 1e-09
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 10
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

── Model Inference ──

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

── Results Analysis ──
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val_predictions_path..................... results_64x128/predictions_val.nc
test_predictions_path.................... results_64x128/predictions_test.nc
processed_data_path...................... data/processed_64x128/sst_final.nc
analyse_validation....................... True
analyse_test............................. True
std_params_path.......................... data/processed_64x128/std_params.nc
trend_params_path........................ data/processed_64x128/trend_params.nc
compute_persistence...................... True
compute_climatology...................... True
acc_clim_years........................... (1991, 2020)
acc_clim_smoothing_window................ 5
n_sample_chains.......................... 9
spatial_lead_steps....................... [1, 3, 7, 14, 21, 30, 45, 60, 90]
compute_enso34........................... True
enso34_lat_bounds........................ (-5.0, 5.0)
enso34_lon_bounds........................ (190.0, 240.0)
enso34_clim_years........................ (1991, 2020)
enso34_n_sample_chains................... 9
compute_dhw.............................. True
dhw_baseline_years....................... (1985, 2012)
dhw_output_val........................... dhw_val.nc
dhw_output_test.......................... dhw_test.nc
output_dir............................... results_64x128/
report_file.............................. results_64x128/analysis_report.txt
metrics_csv.............................. results_64x128/metrics_by_lead_step.csv
figures_dir.............................. results_64x128/figures/
analysis_pdf............................. results_64x128/forecast_analysis.pdf
generate_plots........................... True
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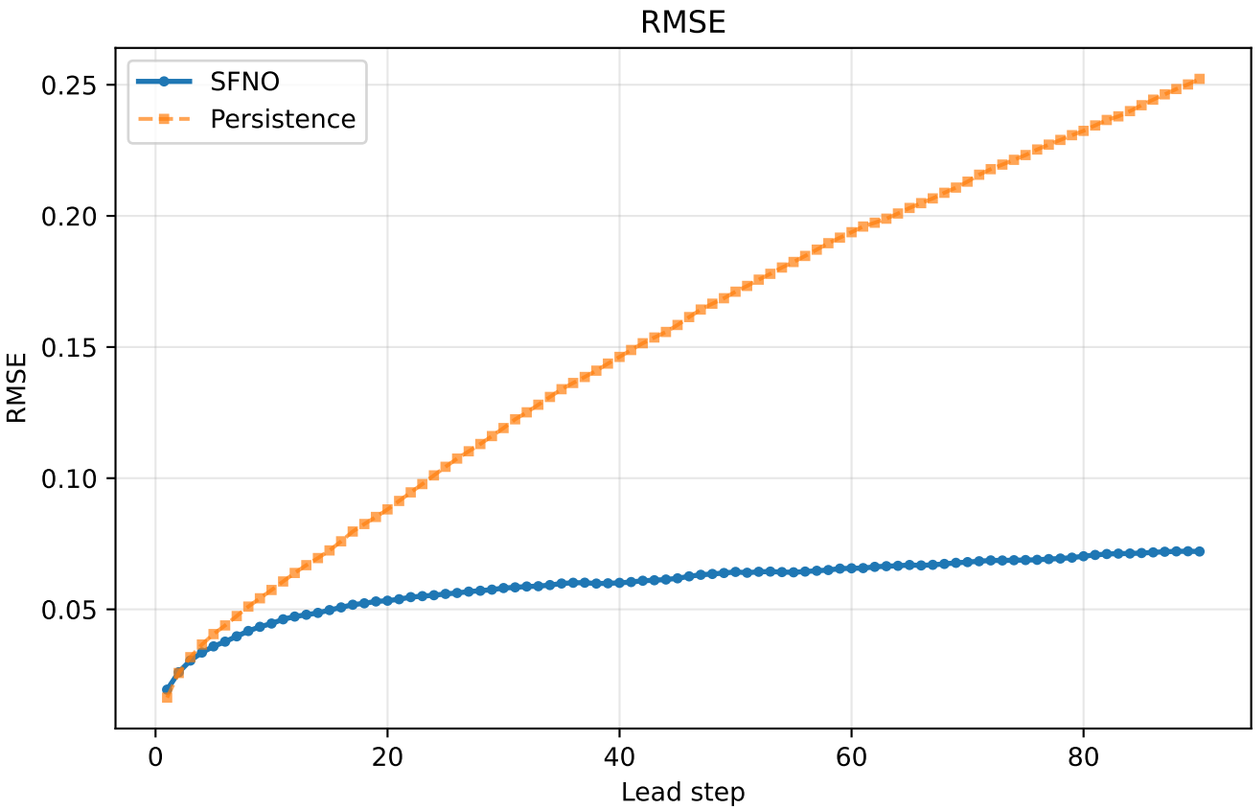
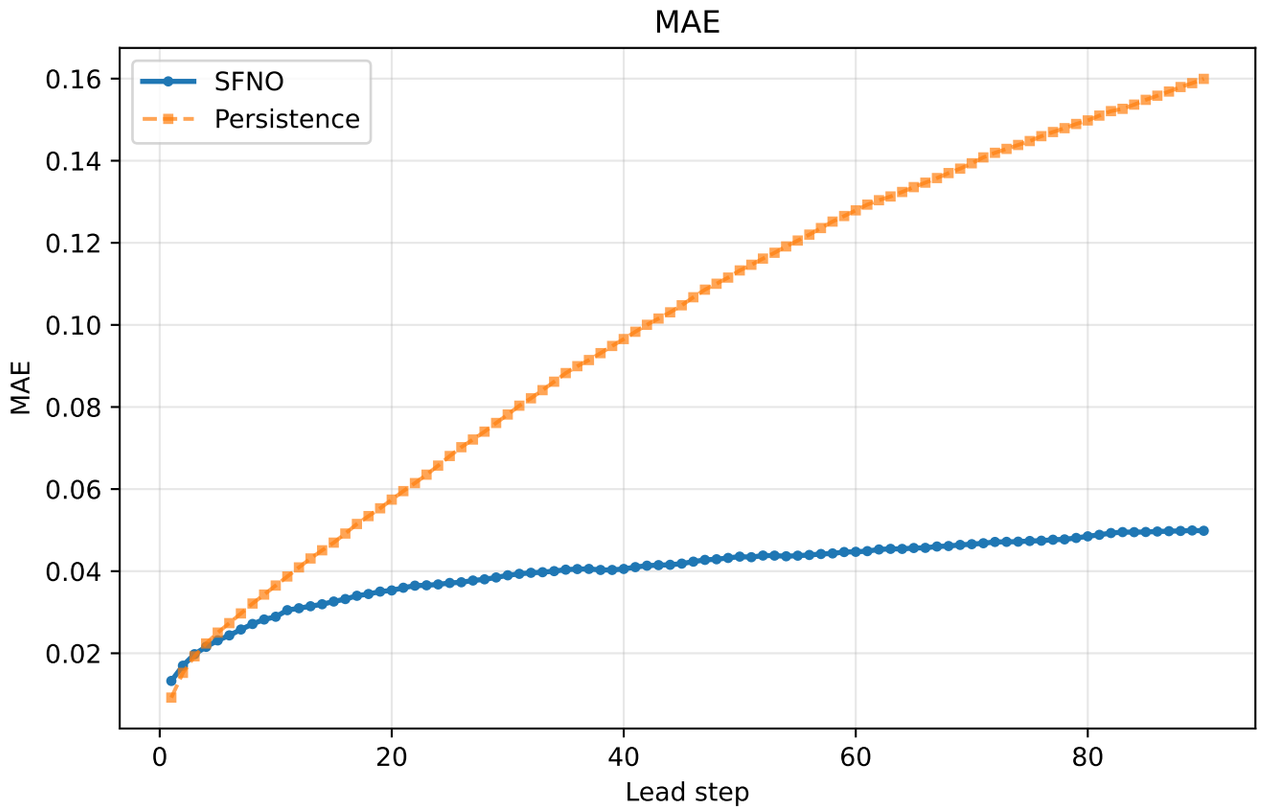
## Metrics by Lead Step — Validation

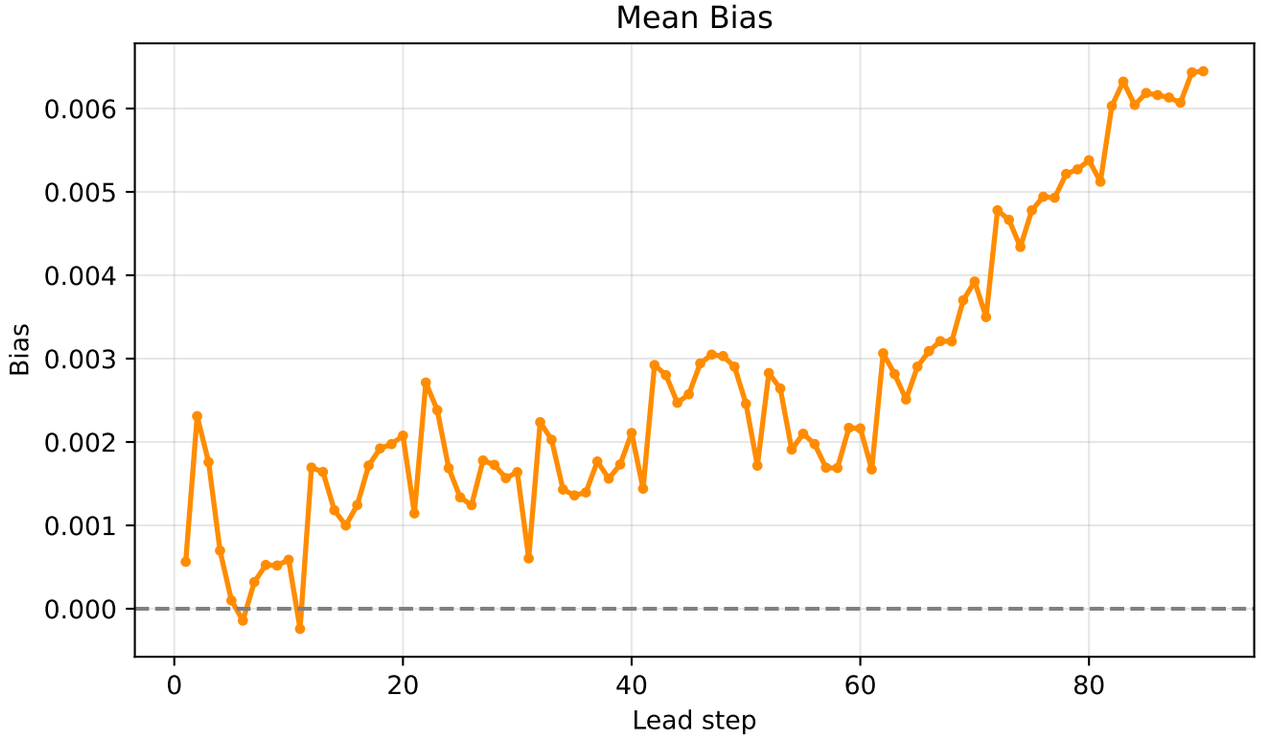
| Lead Step | SFNO RMSE | Persist RMSE | Skill % | SFNO MAE | SFNO Bias | SFNO ACC | Persist ACC | SFNO R² |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.0194 | 0.0164 | -18.5% | 0.0133 | 0.0006 | 0.9472 | 0.9592 | 0.9996 |
| 2 | 0.0260 | 0.0257 | -1.1% | 0.0170 | 0.0023 | 0.8995 | 0.8991 | 0.9993 |
| 3 | 0.0305 | 0.0318 | 4.0% | 0.0198 | 0.0018 | 0.8560 | 0.8481 | 0.9991 |
| 4 | 0.0335 | 0.0366 | 8.4% | 0.0216 | 0.0007 | 0.8218 | 0.8008 | 0.9989 |
| 5 | 0.0359 | 0.0406 | 11.6% | 0.0231 | 0.0001 | 0.7904 | 0.7574 | 0.9987 |
| 6 | 0.0377 | 0.0439 | 14.1% | 0.0244 | -0.0001 | 0.7643 | 0.7202 | 0.9986 |
| 7 | 0.0397 | 0.0474 | 16.3% | 0.0258 | 0.0003 | 0.7374 | 0.6815 | 0.9985 |
| 8 | 0.0417 | 0.0510 | 18.2% | 0.0271 | 0.0005 | 0.7101 | 0.6425 | 0.9983 |
| 9 | 0.0433 | 0.0542 | 20.1% | 0.0282 | 0.0005 | 0.6850 | 0.6072 | 0.9982 |
| 10 | 0.0446 | 0.0574 | 22.3% | 0.0289 | 0.0006 | 0.6620 | 0.5705 | 0.9981 |
| 11 | 0.0462 | 0.0606 | 23.8% | 0.0305 | -0.0002 | 0.6344 | 0.5345 | 0.9979 |
| 12 | 0.0472 | 0.0639 | 26.0% | 0.0310 | 0.0017 | 0.6153 | 0.4984 | 0.9978 |
| 13 | 0.0479 | 0.0668 | 28.3% | 0.0315 | 0.0016 | 0.5942 | 0.4642 | 0.9978 |
| 14 | 0.0486 | 0.0695 | 30.1% | 0.0319 | 0.0012 | 0.5740 | 0.4363 | 0.9977 |
| 15 | 0.0497 | 0.0725 | 31.4% | 0.0326 | 0.0010 | 0.5549 | 0.4112 | 0.9976 |
| 16 | 0.0507 | 0.0759 | 33.2% | 0.0332 | 0.0012 | 0.5374 | 0.3812 | 0.9975 |
| 17 | 0.0518 | 0.0797 | 35.0% | 0.0340 | 0.0017 | 0.5190 | 0.3482 | 0.9974 |
| 18 | 0.0523 | 0.0825 | 36.6% | 0.0344 | 0.0019 | 0.4978 | 0.3242 | 0.9973 |
| 19 | 0.0530 | 0.0852 | 37.8% | 0.0350 | 0.0020 | 0.4778 | 0.3058 | 0.9973 |
| 20 | 0.0533 | 0.0881 | 39.5% | 0.0353 | 0.0021 | 0.4603 | 0.2821 | 0.9972 |
| 21 | 0.0538 | 0.0913 | 41.0% | 0.0360 | 0.0011 | 0.4476 | 0.2589 | 0.9972 |
| 22 | 0.0547 | 0.0946 | 42.2% | 0.0365 | 0.0027 | 0.4329 | 0.2377 | 0.9971 |
| 23 | 0.0550 | 0.0978 | 43.7% | 0.0366 | 0.0024 | 0.4206 | 0.2157 | 0.9971 |
| 24 | 0.0554 | 0.1011 | 45.2% | 0.0368 | 0.0017 | 0.4075 | 0.1932 | 0.9970 |
| 25 | 0.0559 | 0.1043 | 46.4% | 0.0371 | 0.0013 | 0.3945 | 0.1725 | 0.9970 |
| 26 | 0.0562 | 0.1075 | 47.7% | 0.0373 | 0.0012 | 0.3811 | 0.1551 | 0.9969 |
| 27 | 0.0568 | 0.1103 | 48.5% | 0.0377 | 0.0018 | 0.3637 | 0.1418 | 0.9969 |
| 28 | 0.0571 | 0.1130 | 49.5% | 0.0380 | 0.0017 | 0.3517 | 0.1313 | 0.9968 |
| 29 | 0.0575 | 0.1161 | 50.4% | 0.0385 | 0.0016 | 0.3390 | 0.1164 | 0.9968 |
| 30 | 0.0581 | 0.1191 | 51.2% | 0.0390 | 0.0016 | 0.3237 | 0.1038 | 0.9967 |
| 31 | 0.0584 | 0.1225 | 52.3% | 0.0394 | 0.0006 | 0.3146 | 0.0889 | 0.9967 |
| 32 | 0.0587 | 0.1251 | 53.1% | 0.0396 | 0.0022 | 0.3066 | 0.0821 | 0.9966 |
| 33 | 0.0588 | 0.1280 | 54.1% | 0.0397 | 0.0020 | 0.3010 | 0.0694 | 0.9966 |
| 34 | 0.0593 | 0.1310 | 54.8% | 0.0400 | 0.0014 | 0.2940 | 0.0582 | 0.9966 |
| 35 | 0.0598 | 0.1339 | 55.3% | 0.0404 | 0.0014 | 0.2886 | 0.0464 | 0.9965 |
| 36 | 0.0601 | 0.1363 | 55.9% | 0.0405 | 0.0014 | 0.2819 | 0.0400 | 0.9965 |
| 37 | 0.0602 | 0.1386 | 56.6% | 0.0406 | 0.0018 | 0.2779 | 0.0378 | 0.9965 |
| 38 | 0.0598 | 0.1410 | 57.6% | 0.0403 | 0.0016 | 0.2739 | 0.0334 | 0.9965 |
| 39 | 0.0599 | 0.1437 | 58.3% | 0.0403 | 0.0017 | 0.2675 | 0.0265 | 0.9965 |
| 40 | 0.0601 | 0.1462 | 58.9% | 0.0405 | 0.0021 | 0.2586 | 0.0227 | 0.9965 |
| 41 | 0.0604 | 0.1489 | 59.4% | 0.0410 | 0.0014 | 0.2456 | 0.0178 | 0.9964 |
| 42 | 0.0609 | 0.1514 | 59.8% | 0.0413 | 0.0029 | 0.2402 | 0.0118 | 0.9964 |
| 43 | 0.0611 | 0.1536 | 60.2% | 0.0415 | 0.0028 | 0.2302 | 0.0084 | 0.9964 |
| 44 | 0.0614 | 0.1558 | 60.6% | 0.0416 | 0.0025 | 0.2213 | 0.0082 | 0.9963 |
| 45 | 0.0618 | 0.1584 | 61.0% | 0.0418 | 0.0026 | 0.2128 | -0.0016 | 0.9963 |
| 46 | 0.0626 | 0.1614 | 61.2% | 0.0423 | 0.0029 | 0.1997 | -0.0150 | 0.9962 |
| 47 | 0.0632 | 0.1643 | 61.6% | 0.0428 | 0.0030 | 0.1895 | -0.0262 | 0.9961 |
| 48 | 0.0635 | 0.1665 | 61.9% | 0.0429 | 0.0030 | 0.1797 | -0.0296 | 0.9961 |
| 49 | 0.0638 | 0.1686 | 62.1% | 0.0432 | 0.0029 | 0.1682 | -0.0316 | 0.9960 |
| 50 | 0.0643 | 0.1711 | 62.4% | 0.0435 | 0.0025 | 0.1604 | -0.0359 | 0.9960 |
| 51 | 0.0639 | 0.1733 | 63.1% | 0.0434 | 0.0017 | 0.1545 | -0.0392 | 0.9960 |
| 52 | 0.0643 | 0.1757 | 63.4% | 0.0438 | 0.0028 | 0.1474 | -0.0454 | 0.9960 |
| 53 | 0.0644 | 0.1779 | 63.8% | 0.0438 | 0.0026 | 0.1440 | -0.0488 | 0.9960 |
| 54 | 0.0642 | 0.1803 | 64.4% | 0.0436 | 0.0019 | 0.1528 | -0.0525 | 0.9960 |
| 55 | 0.0641 | 0.1824 | 64.9% | 0.0438 | 0.0021 | 0.1521 | -0.0540 | 0.9960 |
| 56 | 0.0644 | 0.1847 | 65.1% | 0.0439 | 0.0020 | 0.1465 | -0.0585 | 0.9960 |
| 57 | 0.0647 | 0.1871 | 65.4% | 0.0442 | 0.0017 | 0.1413 | -0.0631 | 0.9959 |
| 58 | 0.0650 | 0.1896 | 65.7% | 0.0443 | 0.0017 | 0.1379 | -0.0696 | 0.9959 |
| 59 | 0.0655 | 0.1917 | 65.8% | 0.0447 | 0.0022 | 0.1285 | -0.0708 | 0.9958 |
| 60 | 0.0656 | 0.1938 | 66.1% | 0.0447 | 0.0022 | 0.1313 | -0.0716 | 0.9958 |
| 61 | 0.0657 | 0.1959 | 66.5% | 0.0449 | 0.0017 | 0.1317 | -0.0755 | 0.9958 |
| 62 | 0.0662 | 0.1974 | 66.5% | 0.0453 | 0.0031 | 0.1189 | -0.0678 | 0.9957 |
| 63 | 0.0665 | 0.1989 | 66.6% | 0.0454 | 0.0028 | 0.1145 | -0.0586 | 0.9957 |
| 64 | 0.0666 | 0.2009 | 66.9% | 0.0454 | 0.0025 | 0.1144 | -0.0567 | 0.9957 |
| 65 | 0.0669 | 0.2030 | 67.0% | 0.0456 | 0.0029 | 0.1094 | -0.0588 | 0.9956 |
| 66 | 0.0667 | 0.2049 | 67.4% | 0.0457 | 0.0031 | 0.1066 | -0.0583 | 0.9957 |
| 67 | 0.0670 | 0.2067 | 67.6% | 0.0460 | 0.0032 | 0.1041 | -0.0556 | 0.9956 |
| 68 | 0.0673 | 0.2088 | 67.8% | 0.0461 | 0.0032 | 0.1031 | -0.0542 | 0.9956 |
| 69 | 0.0677 | 0.2108 | 67.9% | 0.0464 | 0.0037 | 0.1017 | -0.0508 | 0.9955 |
| 70 | 0.0680 | 0.2131 | 68.1% | 0.0466 | 0.0039 | 0.1031 | -0.0529 | 0.9955 |
| 71 | 0.0683 | 0.2157 | 68.3% | 0.0468 | 0.0035 | 0.1000 | -0.0598 | 0.9955 |
| 72 | 0.0687 | 0.2178 | 68.5% | 0.0471 | 0.0048 | 0.0963 | -0.0609 | 0.9954 |
| 73 | 0.0686 | 0.2196 | 68.7% | 0.0471 | 0.0047 | 0.0980 | -0.0564 | 0.9954 |
| 74 | 0.0687 | 0.2214 | 69.0% | 0.0472 | 0.0043 | 0.1011 | -0.0508 | 0.9954 |
| 75 | 0.0688 | 0.2232 | 69.2% | 0.0474 | 0.0048 | 0.0989 | -0.0483 | 0.9954 |
| 76 | 0.0689 | 0.2253 | 69.4% | 0.0474 | 0.0049 | 0.0945 | -0.0507 | 0.9954 |
| 77 | 0.0692 | 0.2272 | 69.6% | 0.0477 | 0.0049 | 0.0899 | -0.0482 | 0.9953 |
| 78 | 0.0694 | 0.2290 | 69.7% | 0.0477 | 0.0052 | 0.0834 | -0.0477 | 0.9953 |
| 79 | 0.0697 | 0.2308 | 69.8% | 0.0481 | 0.0053 | 0.0788 | -0.0476 | 0.9953 |
| 80 | 0.0702 | 0.2324 | 69.8% | 0.0485 | 0.0054 | 0.0709 | -0.0445 | 0.9952 |
| 81 | 0.0707 | 0.2345 | 69.8% | 0.0489 | 0.0051 | 0.0657 | -0.0448 | 0.9951 |
| 82 | 0.0711 | 0.2366 | 69.9% | 0.0493 | 0.0060 | 0.0684 | -0.0443 | 0.9951 |
| 83 | 0.0712 | 0.2380 | 70.1% | 0.0495 | 0.0063 | 0.0669 | -0.0426 | 0.9951 |
| 84 | 0.0713 | 0.2399 | 70.3% | 0.0495 | 0.0060 | 0.0704 | -0.0439 | 0.9951 |
| 85 | 0.0715 | 0.2422 | 70.5% | 0.0496 | 0.0062 | 0.0754 | -0.0456 | 0.9950 |
| 86 | 0.0717 | 0.2443 | 70.7% | 0.0497 | 0.0062 | 0.0775 | -0.0445 | 0.9950 |
| 87 | 0.0720 | 0.2464 | 70.8% | 0.0498 | 0.0061 | 0.0753 | -0.0446 | 0.9950 |
| 88 | 0.0721 | 0.2484 | 71.0% | 0.0498 | 0.0061 | 0.0732 | -0.0447 | 0.9949 |
| 89 | 0.0721 | 0.2501 | 71.2% | 0.0499 | 0.0064 | 0.0708 | -0.0431 | 0.9949 |
| 90 | 0.0720 | 0.2522 | 71.4% | 0.0498 | 0.0064 | 0.0727 | -0.0441 | 0.9949 |

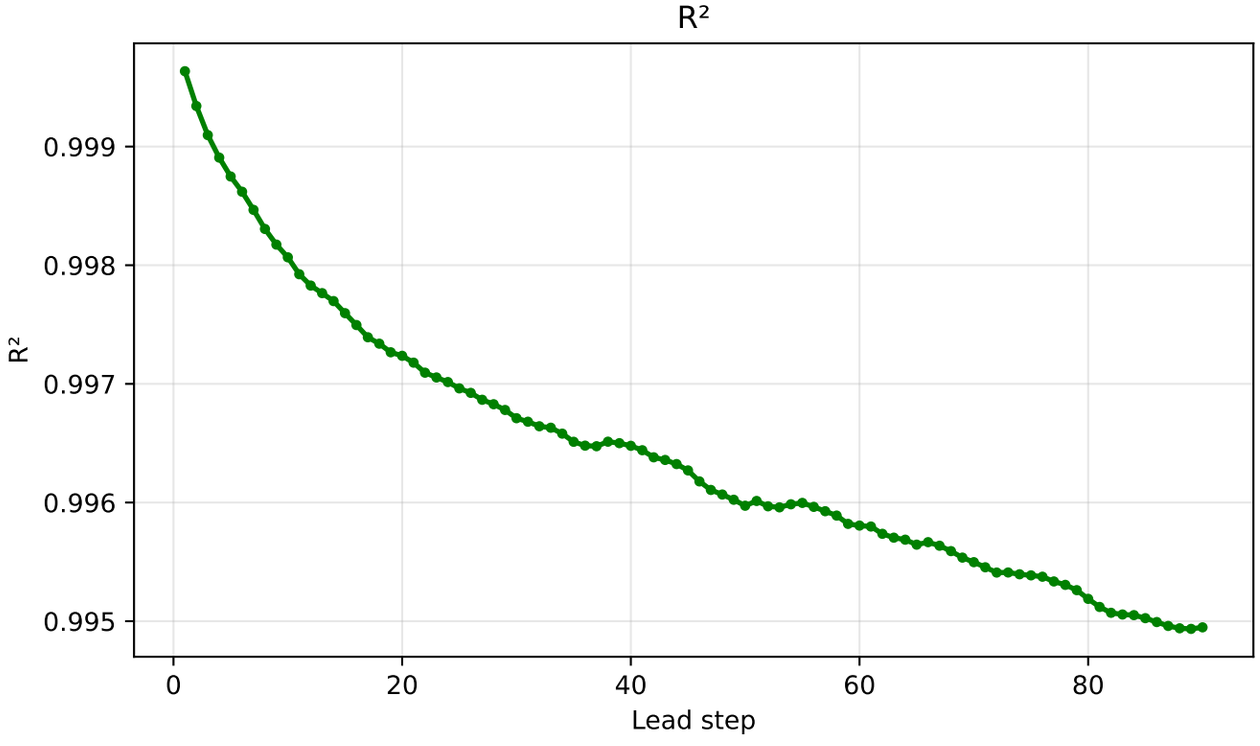
# Forecast Skill vs Lead Step — Validation



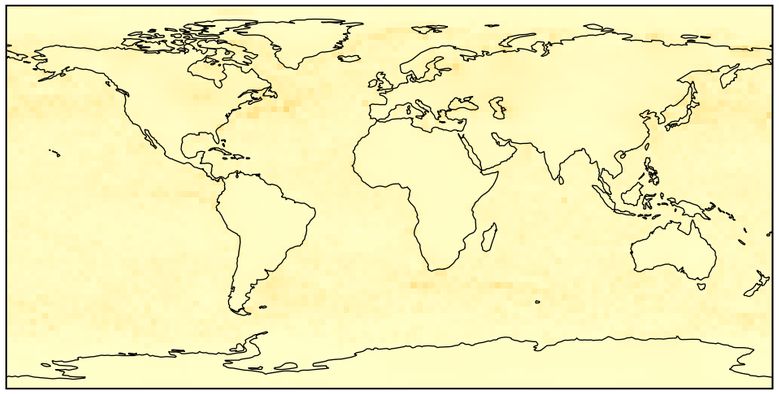
# Bias and R² — Validation



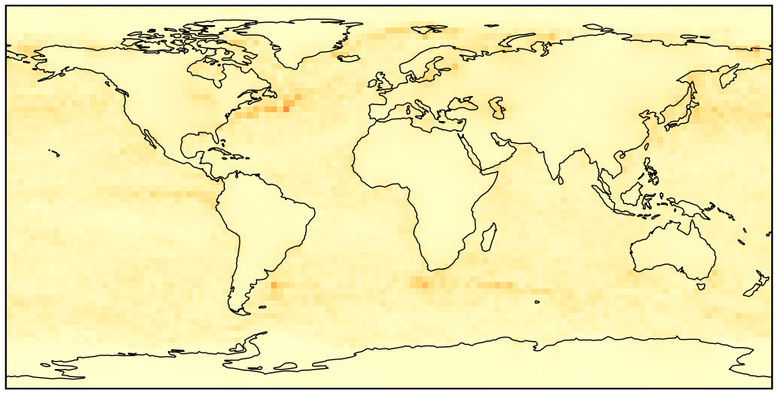



# Spatial RMSE (all grid) — Validation

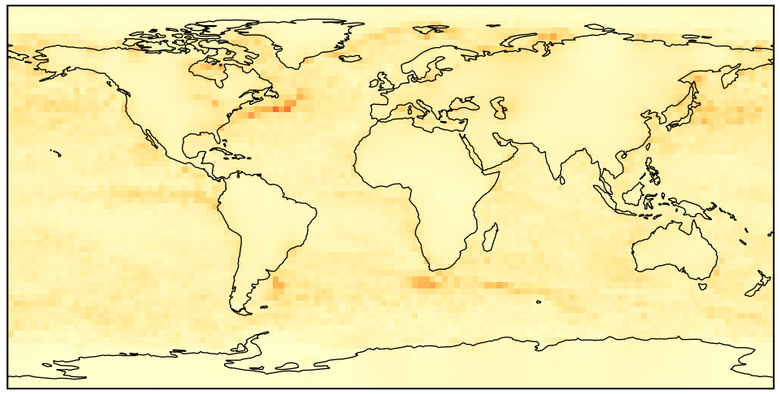
Step 1



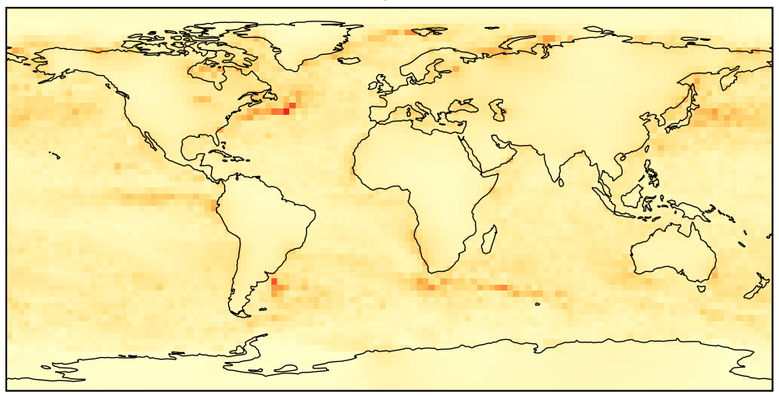
Step 3



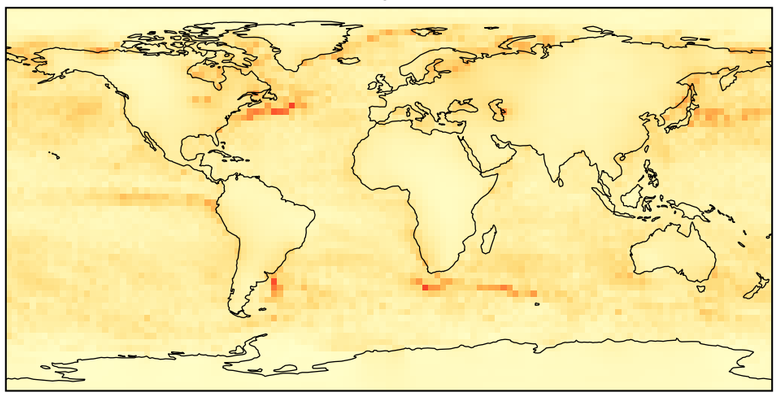
Step 7



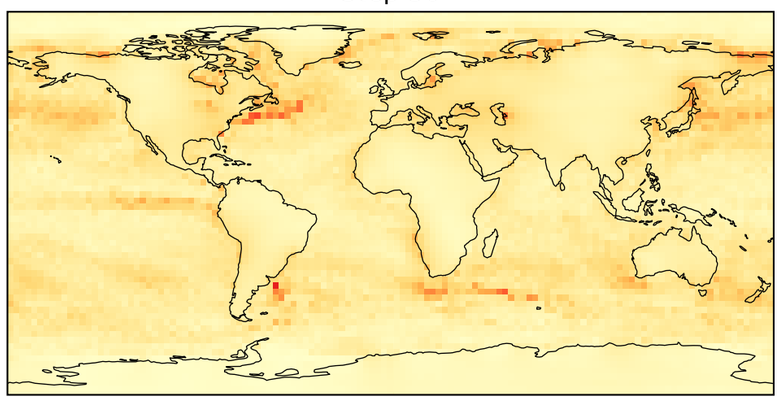
Step 14



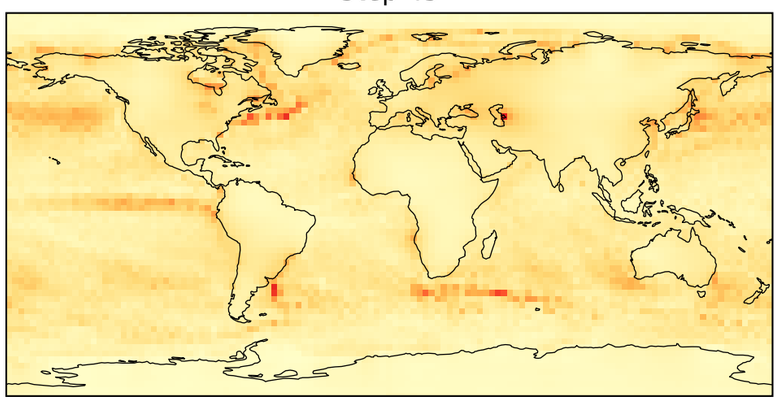
Step 21



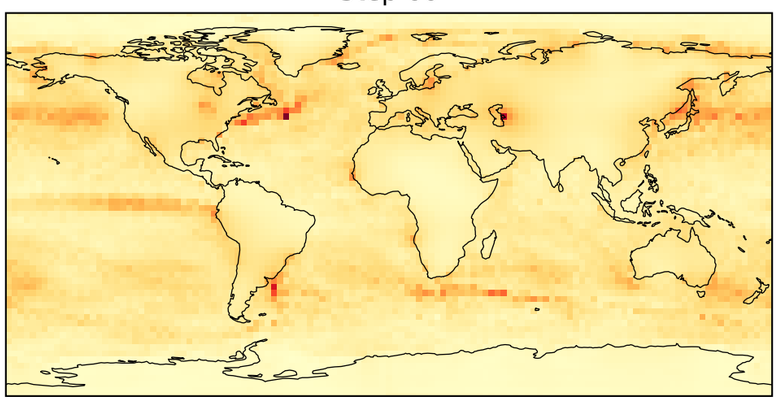
Step 30



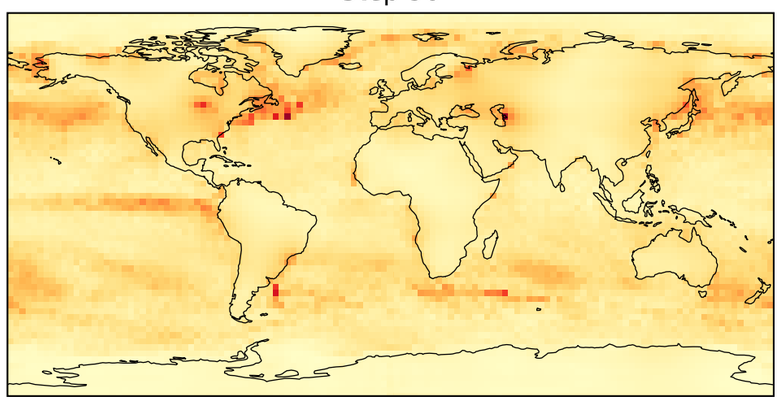
Step 45

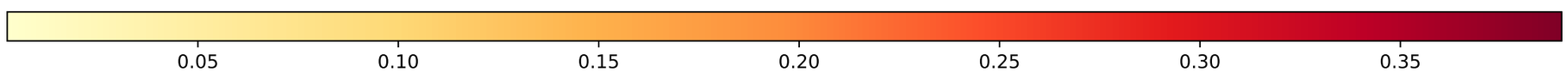


Step 60



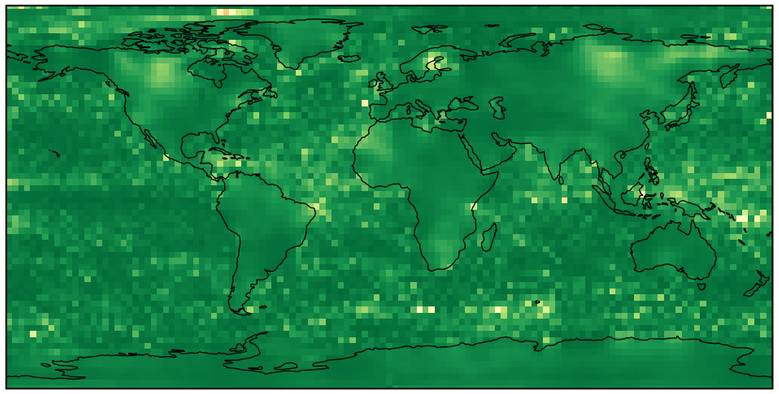
Step 90



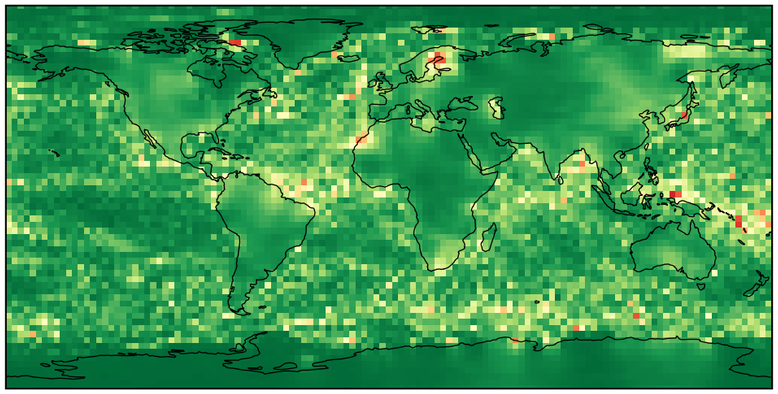


# Spatial ACC (all grid) — Validation

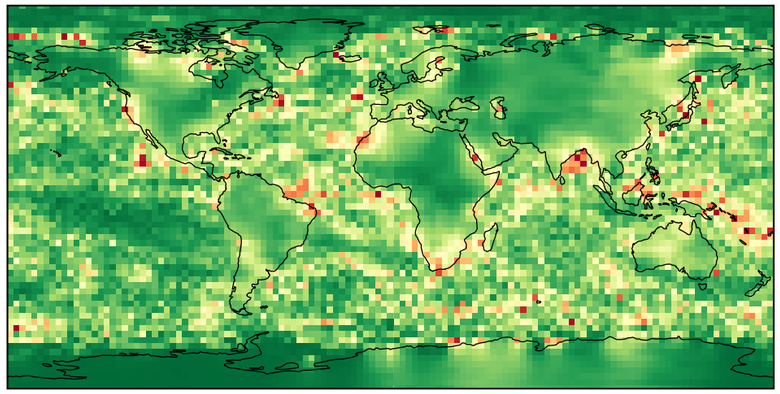
Step 1



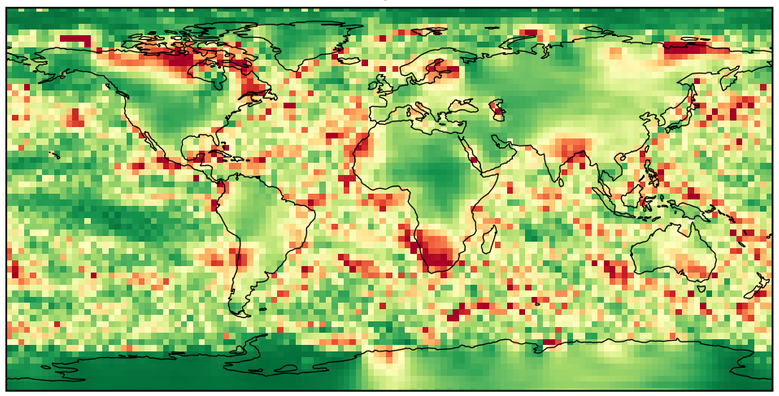
Step 3



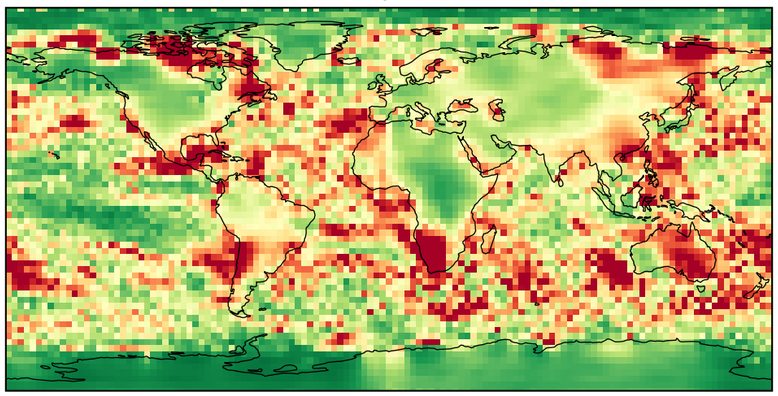
Step 7



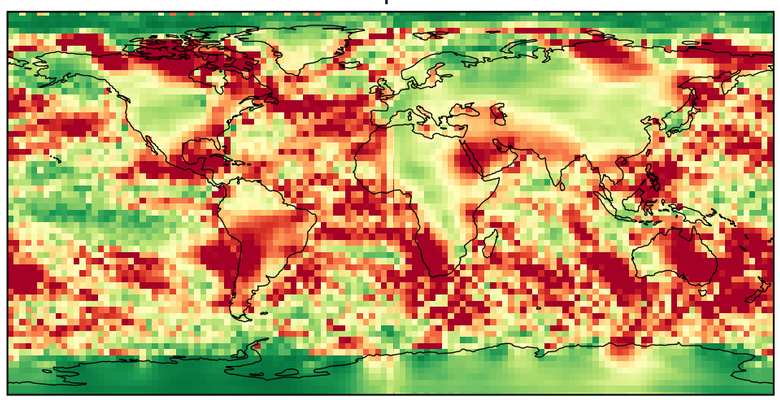
Step 14



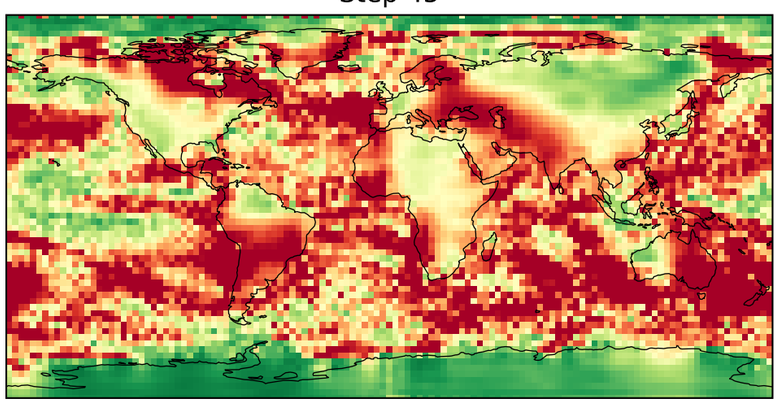
Step 21



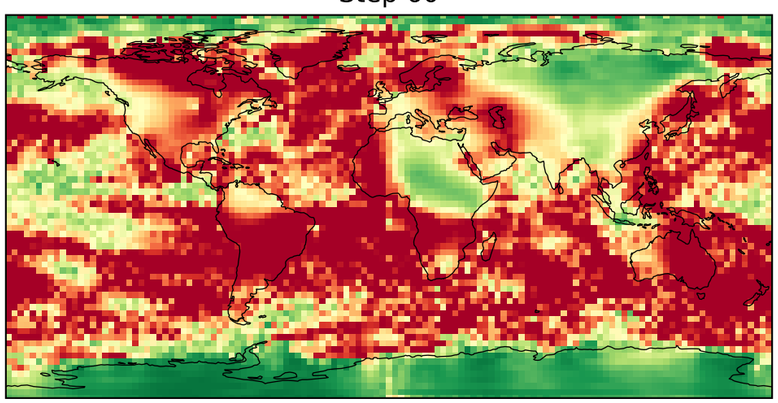
Step 30



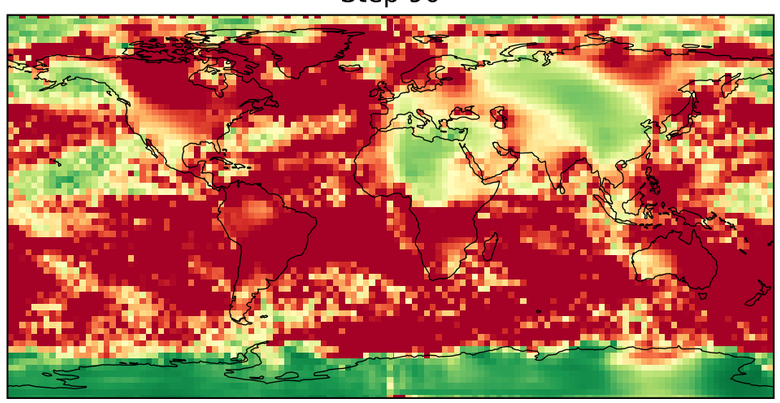
Step 45



Step 60

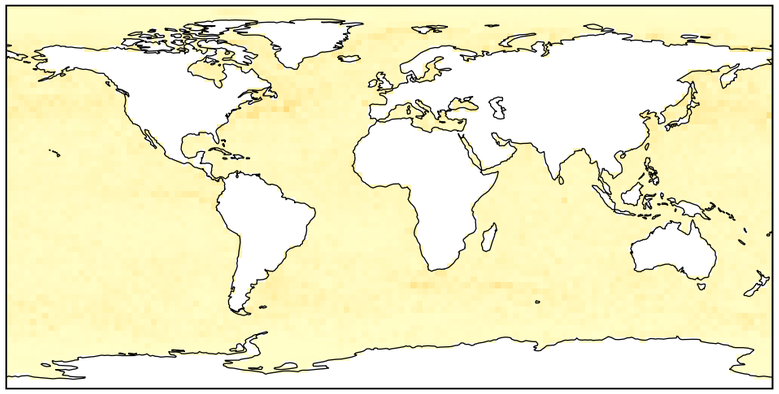


Step 90

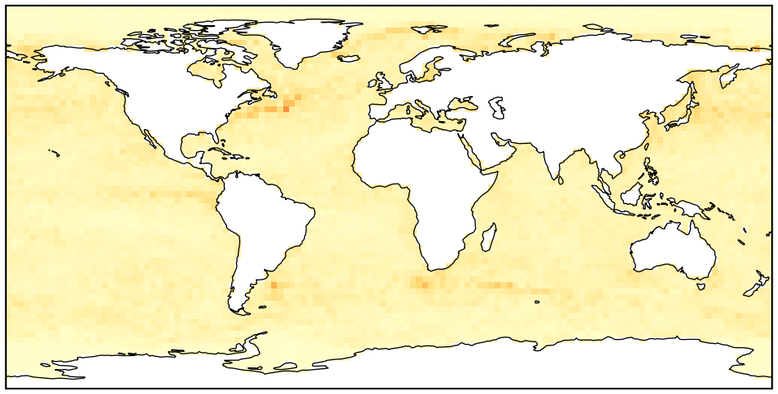


**Spatial RMSE (ocean only) — Validation**

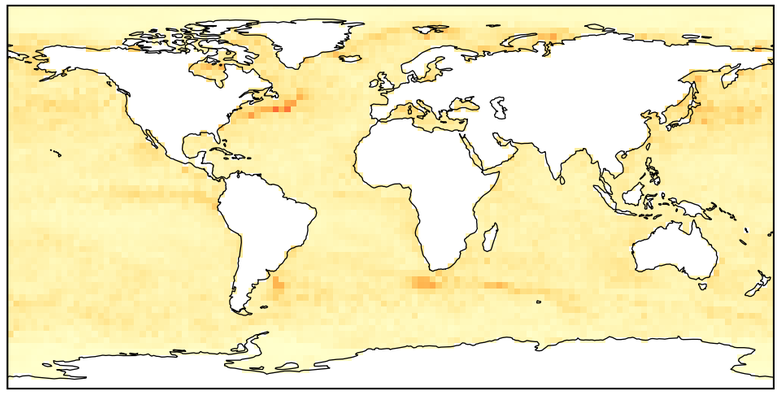
Step 1



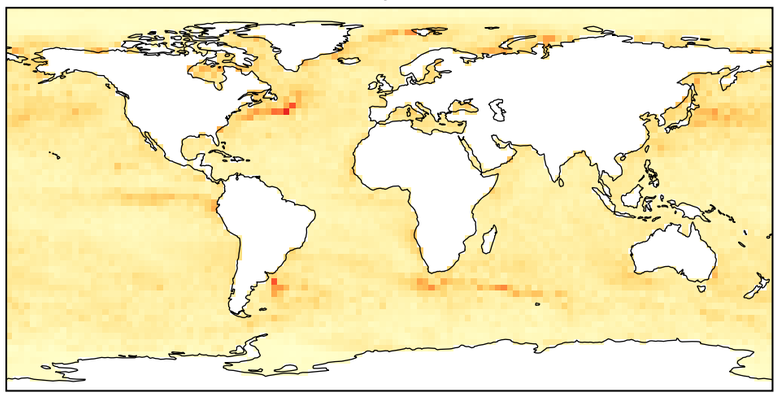
Step 3



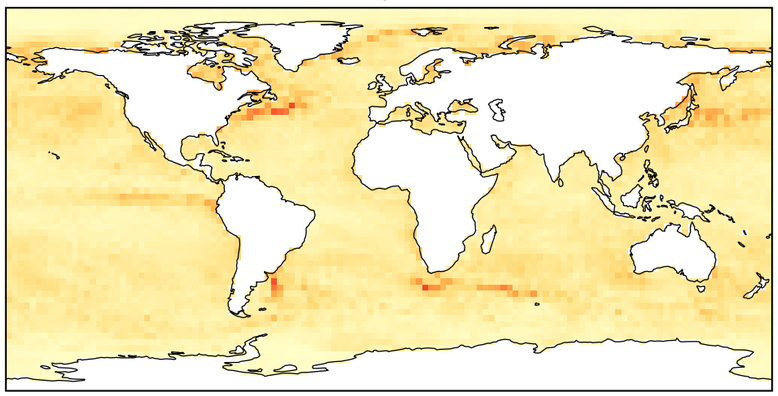
Step 7



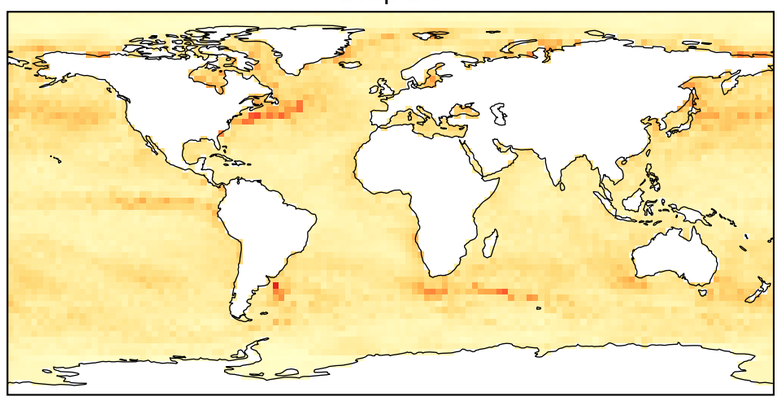
Step 14



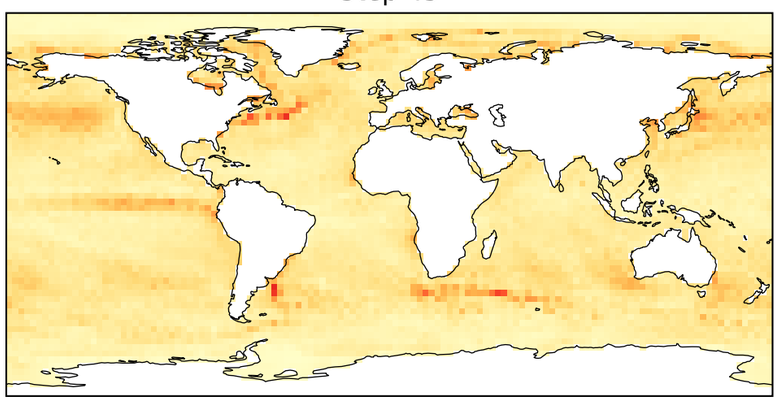
Step 21



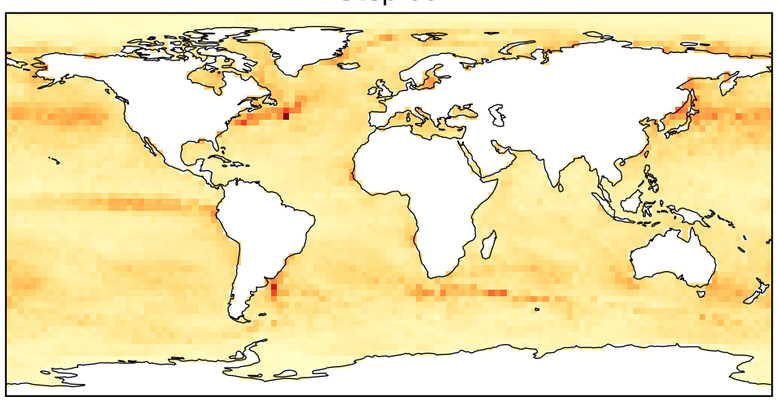
Step 30



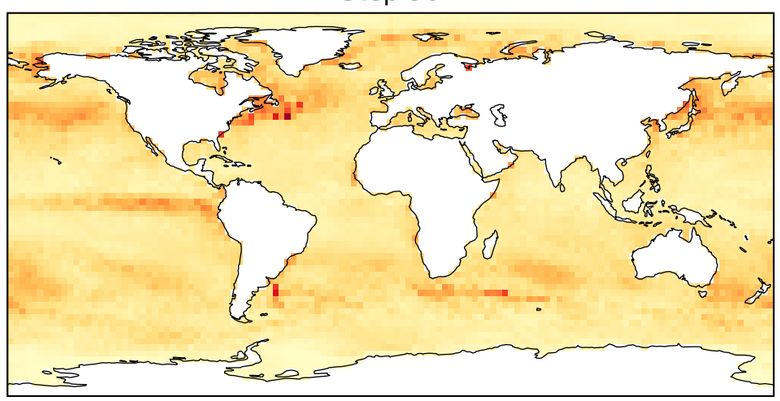
Step 45

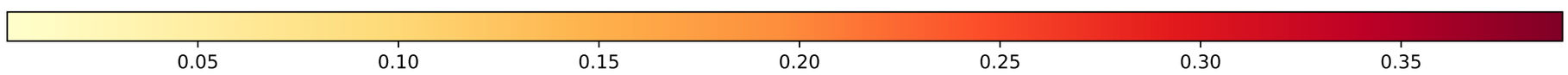


Step 60



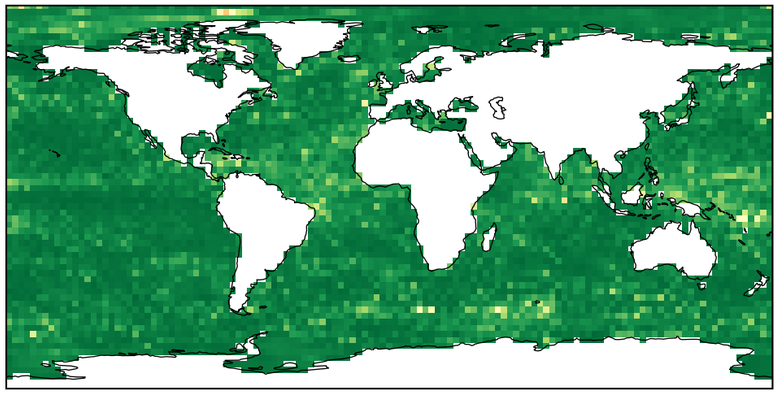
Step 90



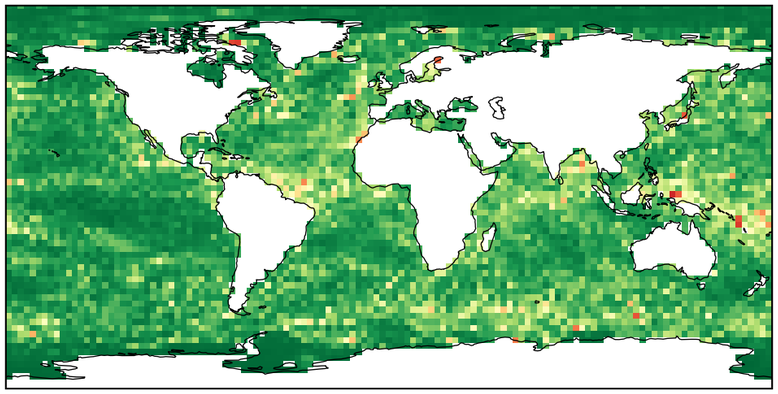


# Spatial ACC (ocean only) — Validation

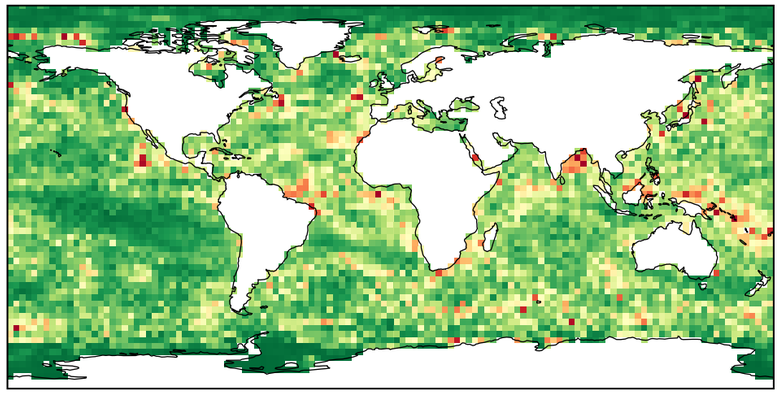
Step 1



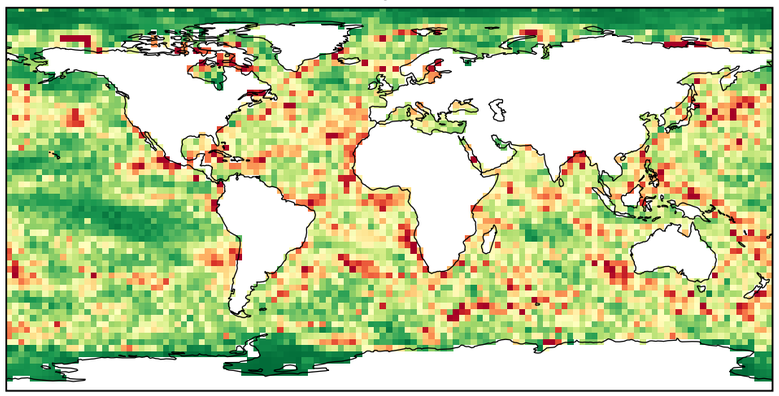
Step 3



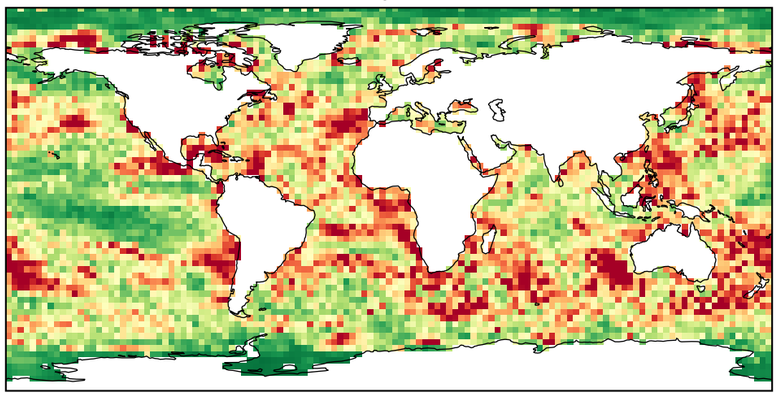
Step 7



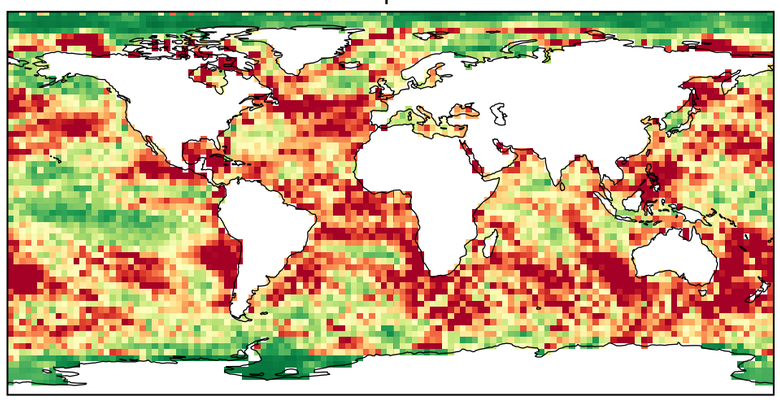
Step 14



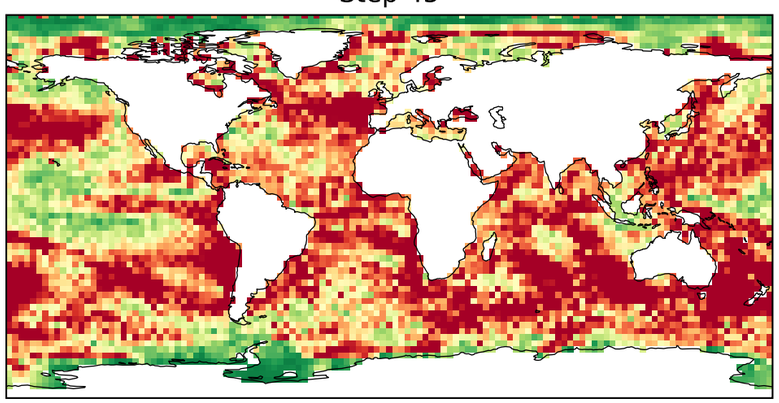
Step 21



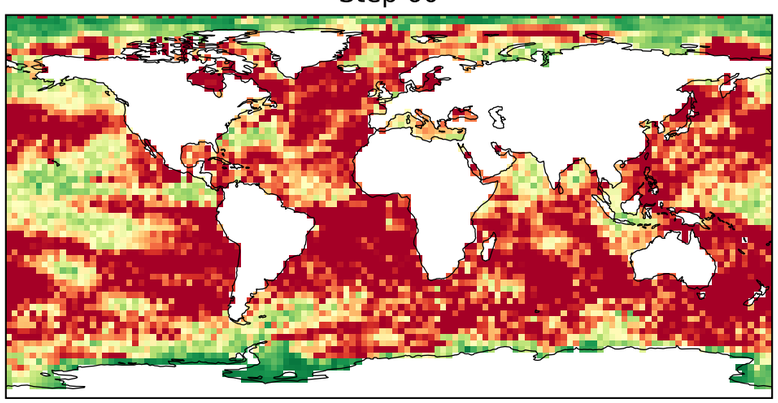
Step 30



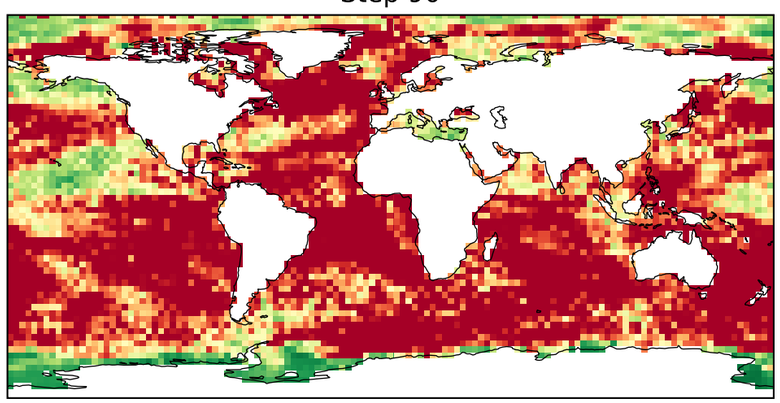
Step 45

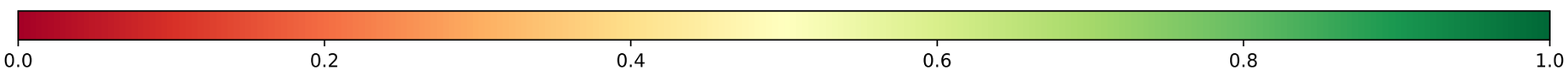


Step 60



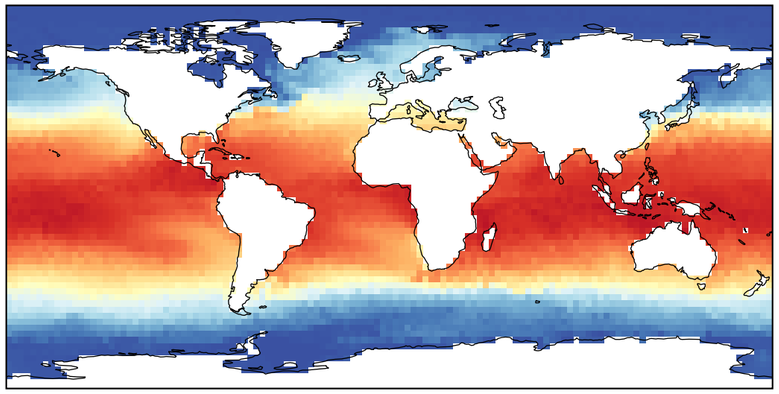
Step 90



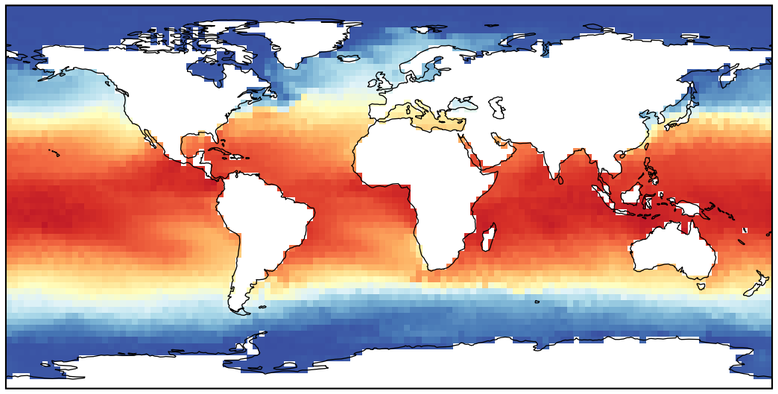


# Chain init: 2016-01-11 — Validation (°C)

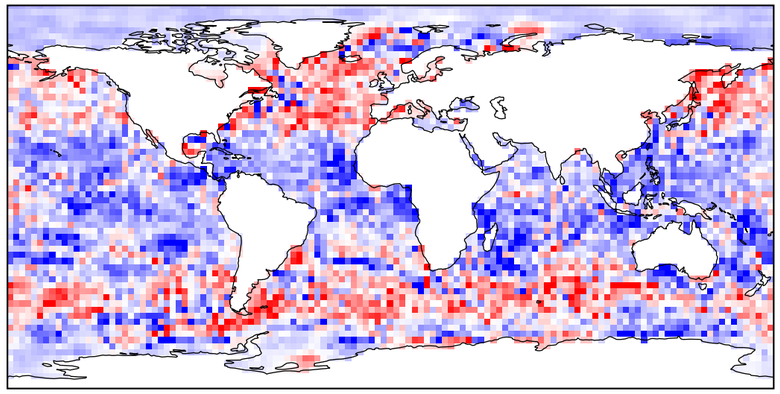
Actual — Step 1 (°C)



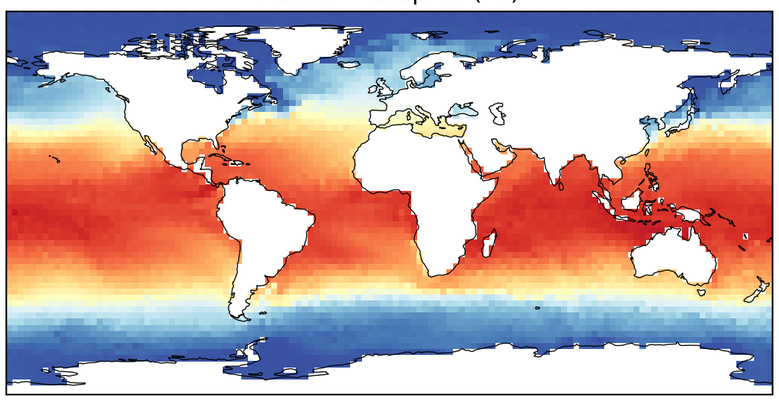
SFNO — Step 1 (°C)



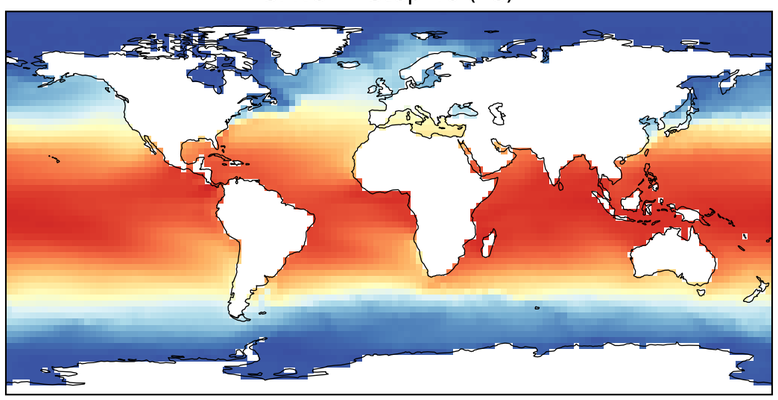
Error — Step 1 (°C)



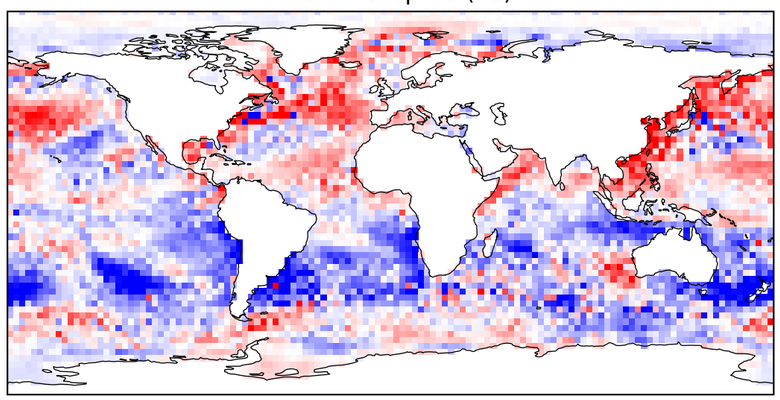
Actual — Step 46 (°C)



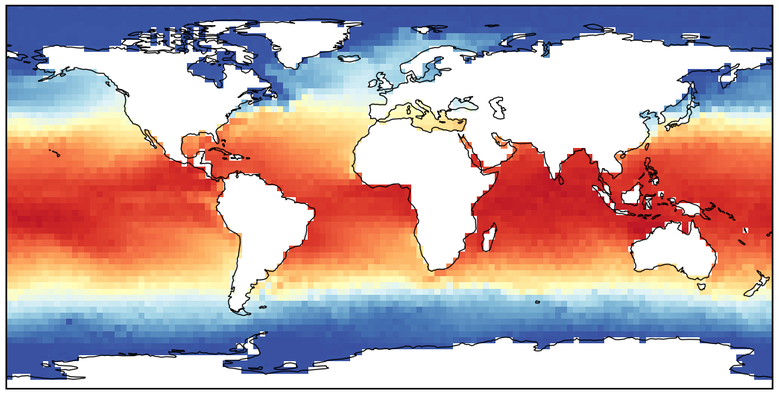
SFNO — Step 46 (°C)



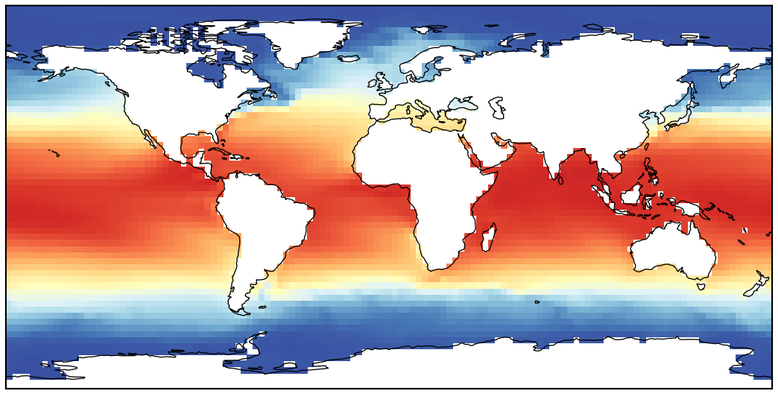
Error — Step 46 (°C)



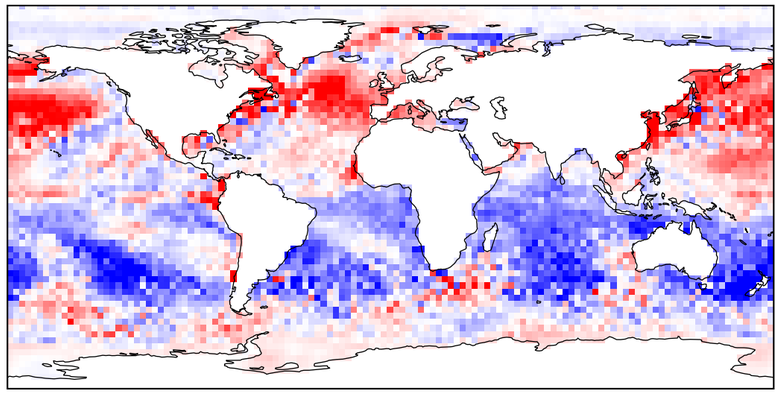
Actual — Step 90 (°C)

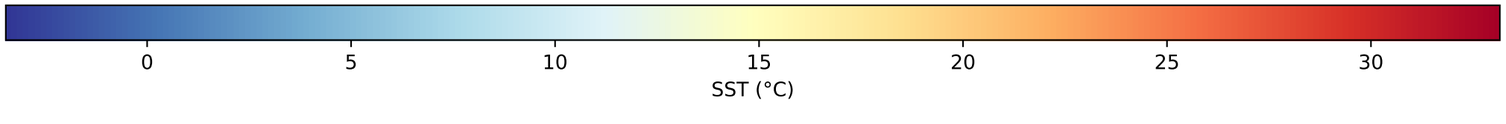


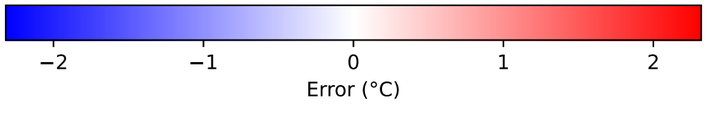
SFNO — Step 90 (°C)



Error — Step 90 (°C)

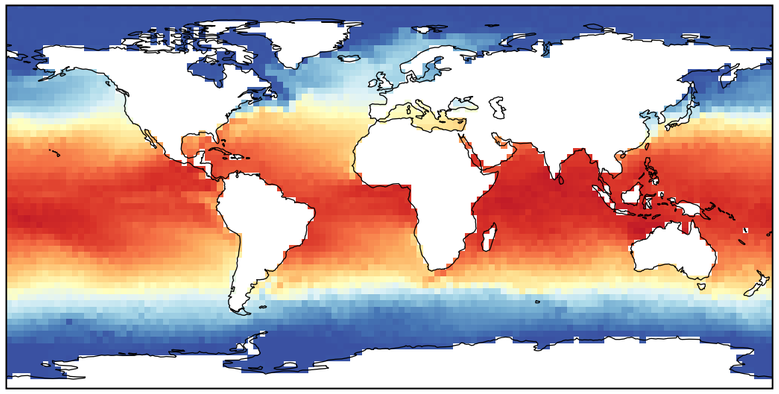




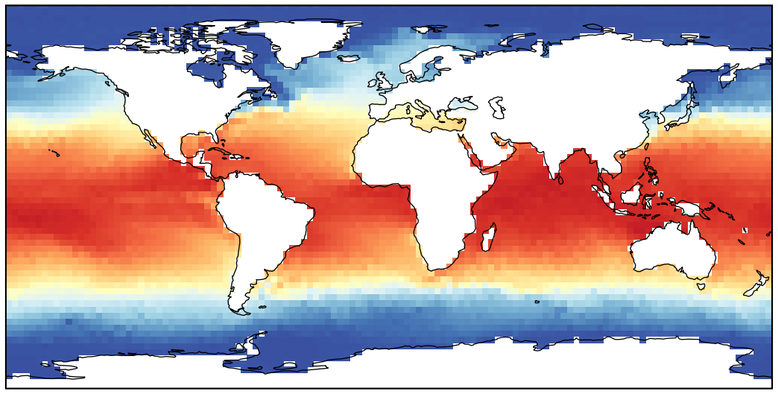



# Chain init: 2016-04-10 — Validation (°C)

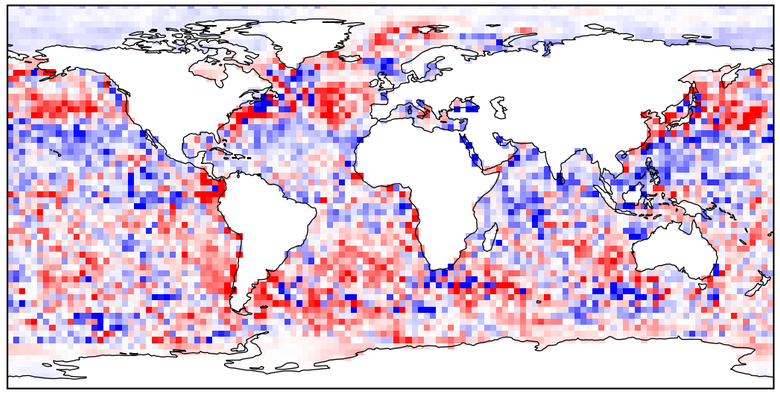
Actual — Step 1 (°C)



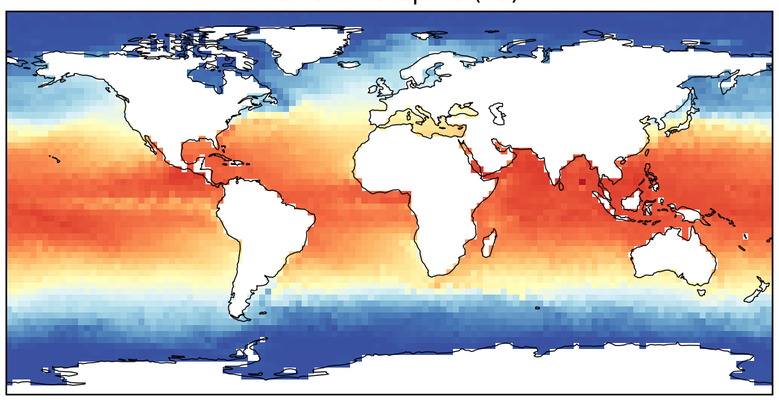
SFNO — Step 1 (°C)



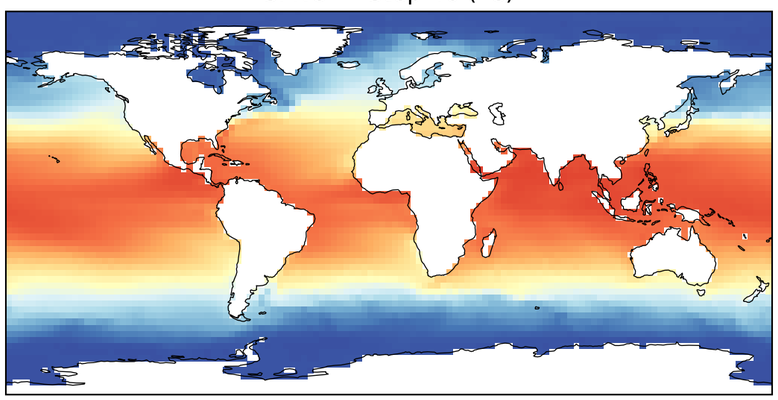
Error — Step 1 (°C)



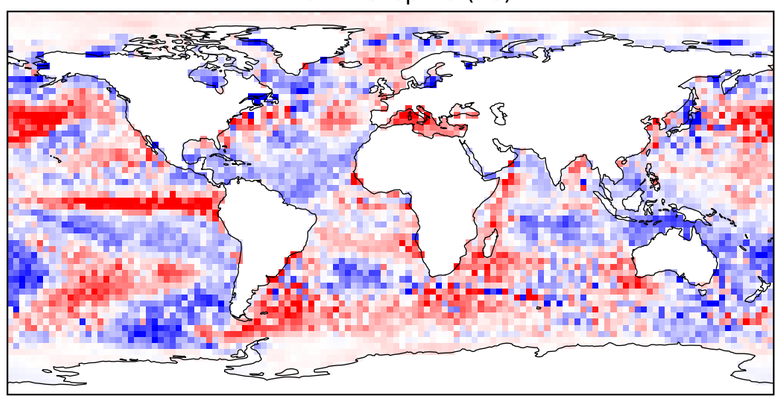
Actual — Step 46 (°C)



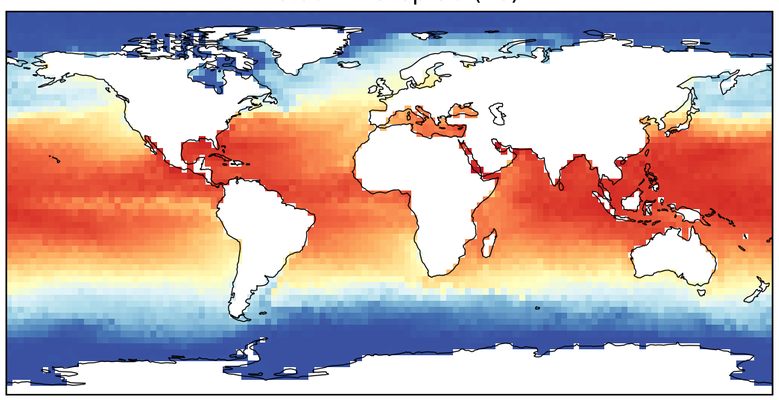
SFNO — Step 46 (°C)



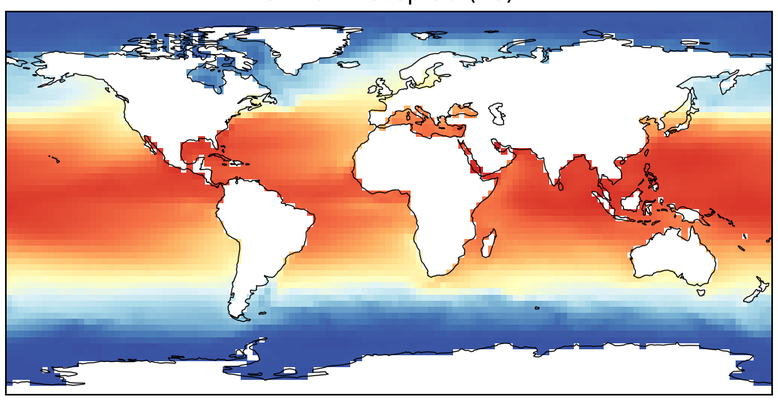
Error — Step 46 (°C)



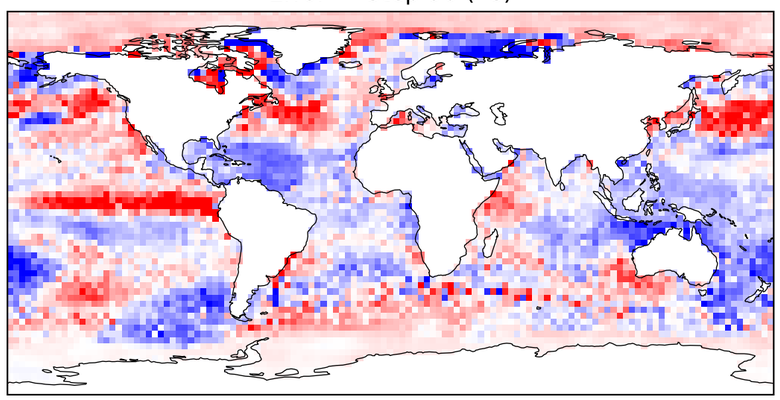
Actual — Step 90 (°C)

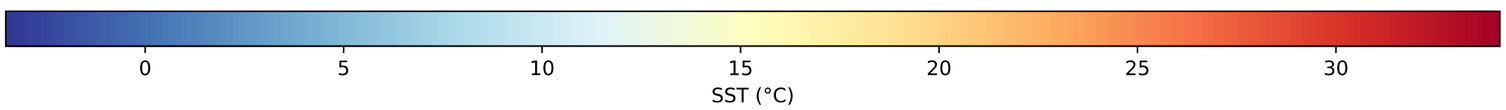


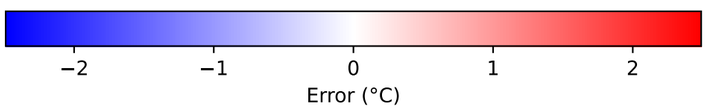
SFNO — Step 90 (°C)



Error — Step 90 (°C)

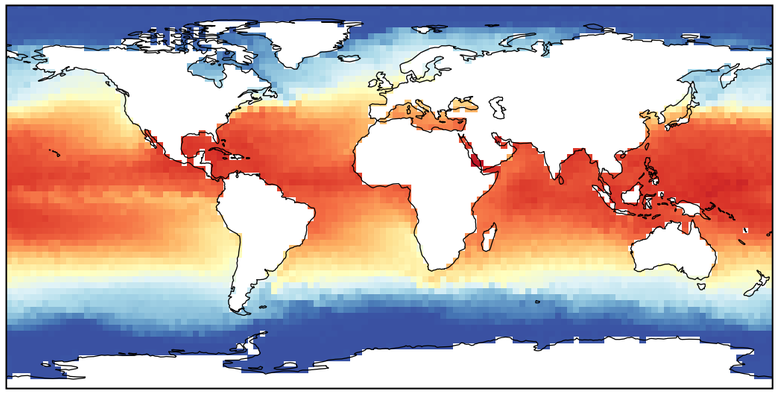




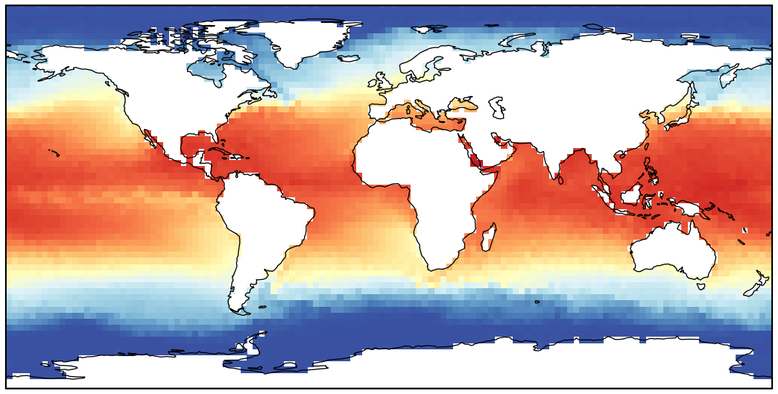



# Chain init: 2016-10-07 — Validation (°C)

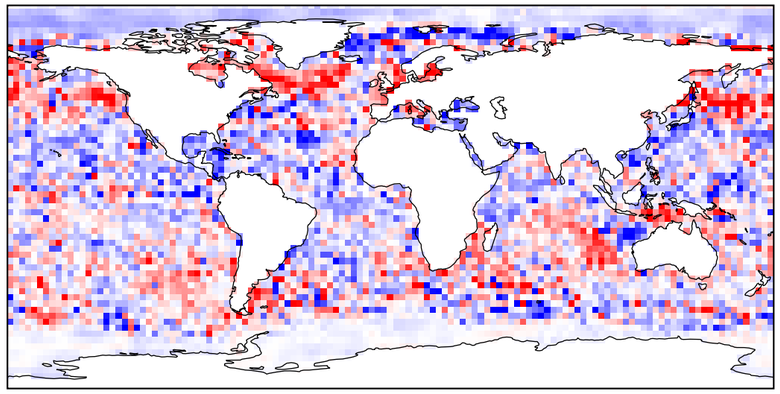
Actual — Step 1 (°C)



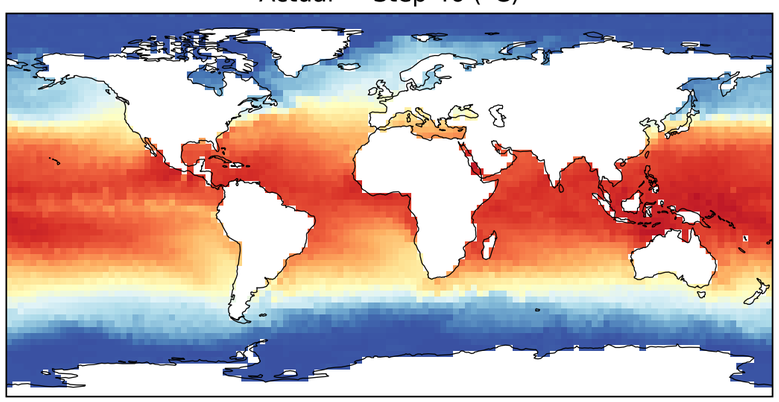
SFNO — Step 1 (°C)



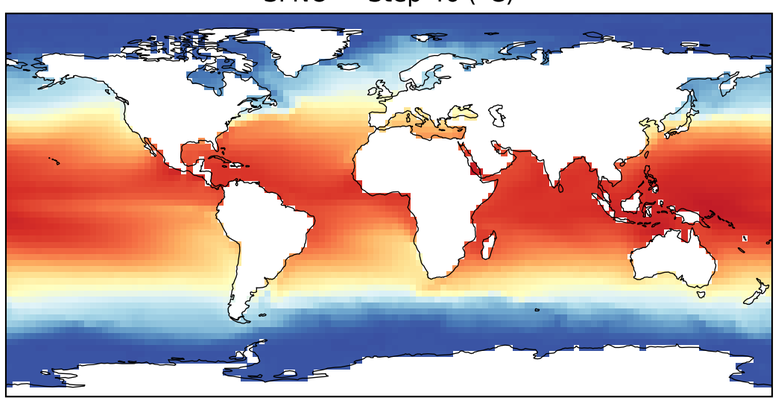
Error — Step 1 (°C)



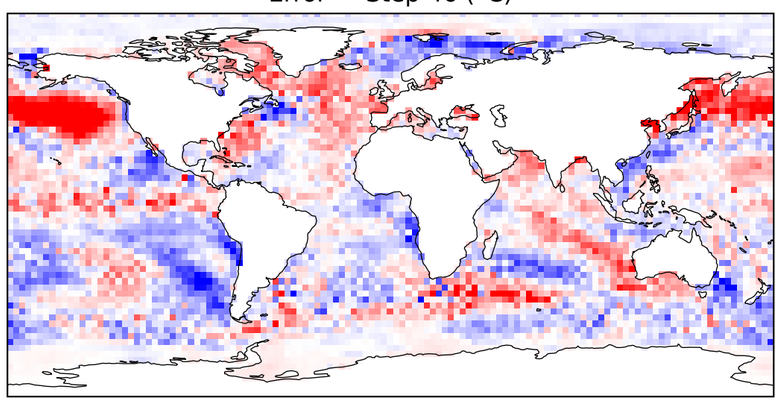
Actual — Step 46 (°C)



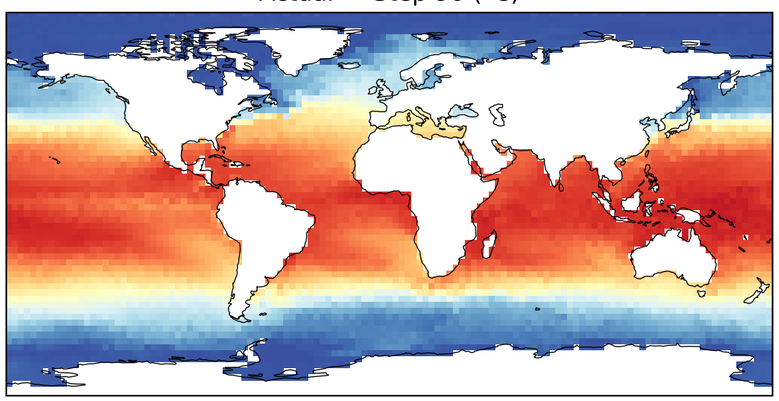
SFNO — Step 46 (°C)



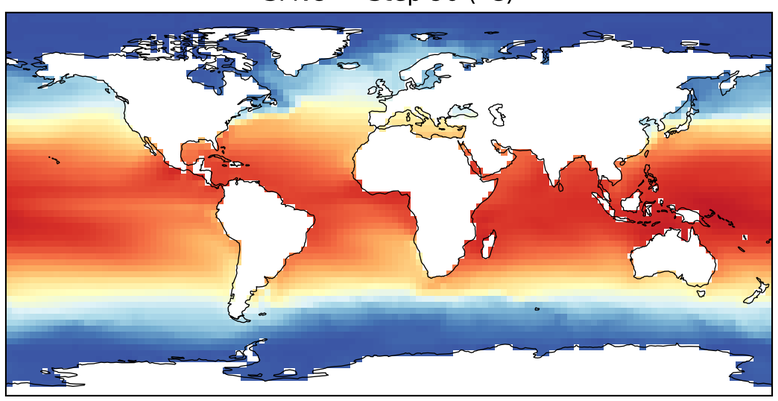
Error — Step 46 (°C)



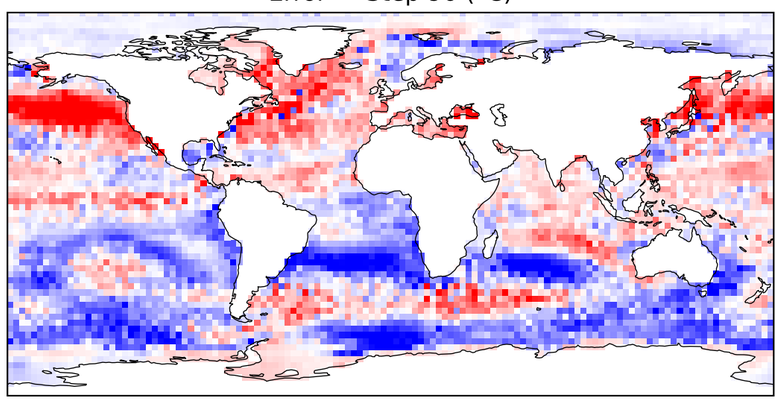
Actual — Step 90 (°C)

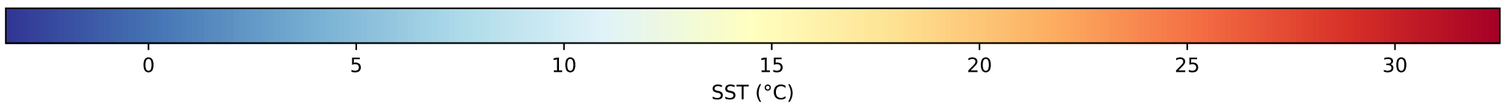


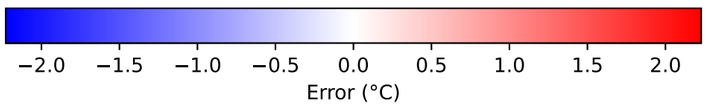
SFNO — Step 90 (°C)



Error — Step 90 (°C)

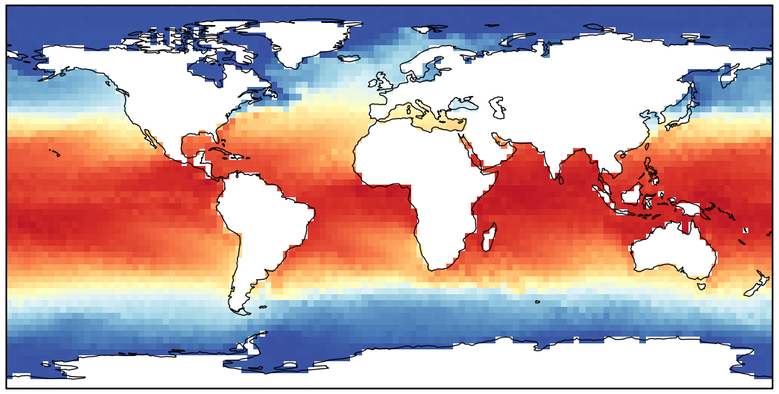




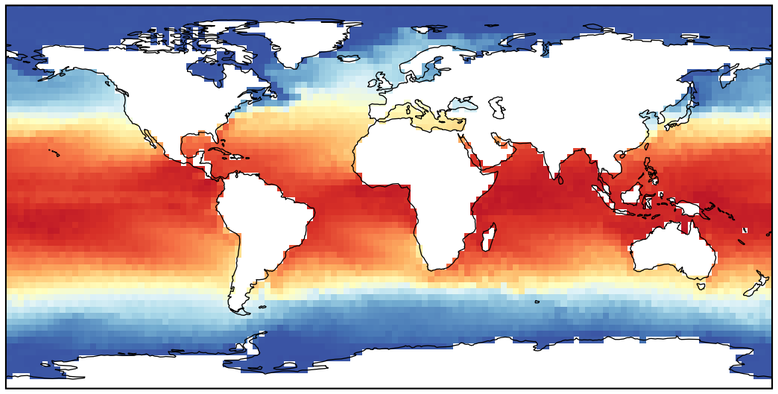



# Chain init: 2017-04-05 — Validation (°C)

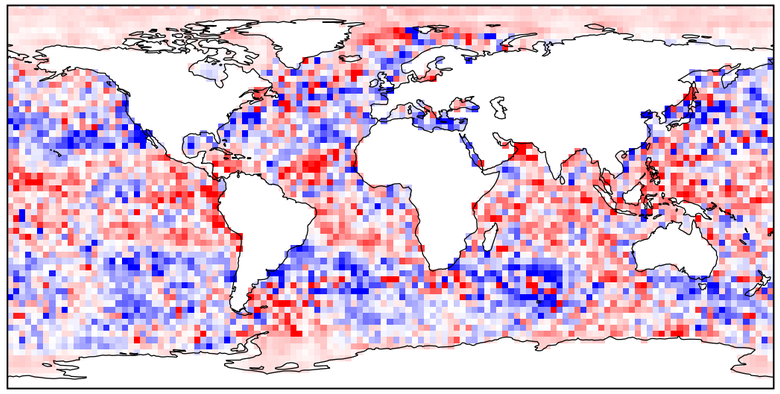
Actual — Step 1 (°C)



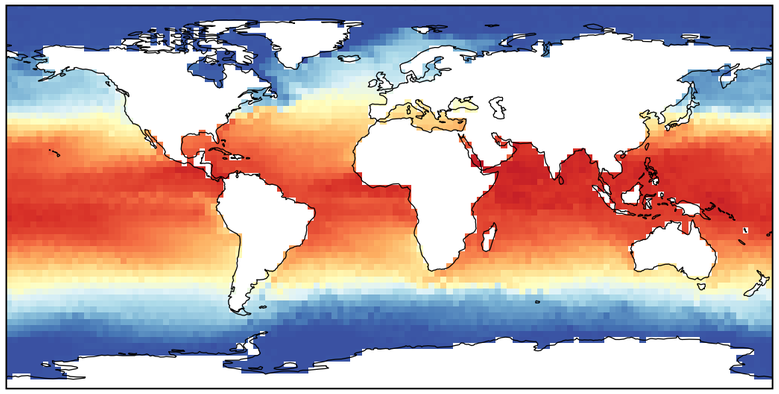
SFNO — Step 1 (°C)



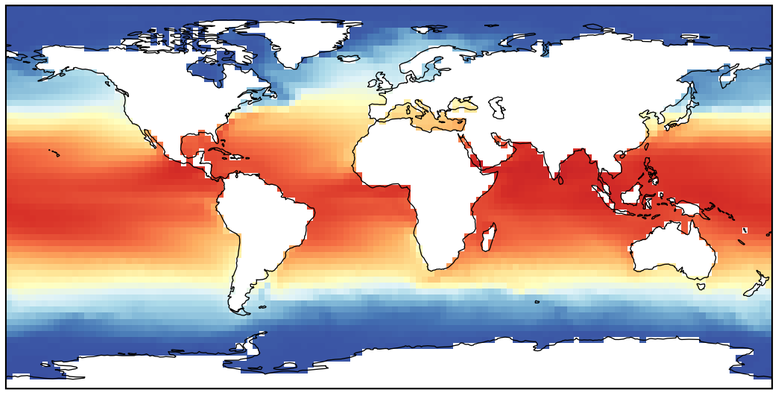
Error — Step 1 (°C)



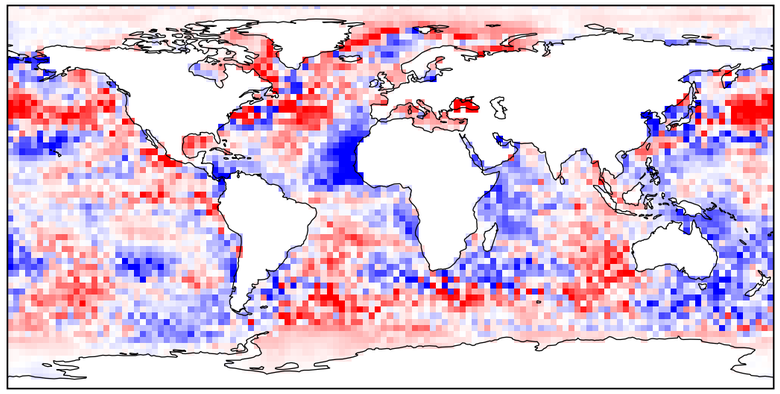
Actual — Step 46 (°C)



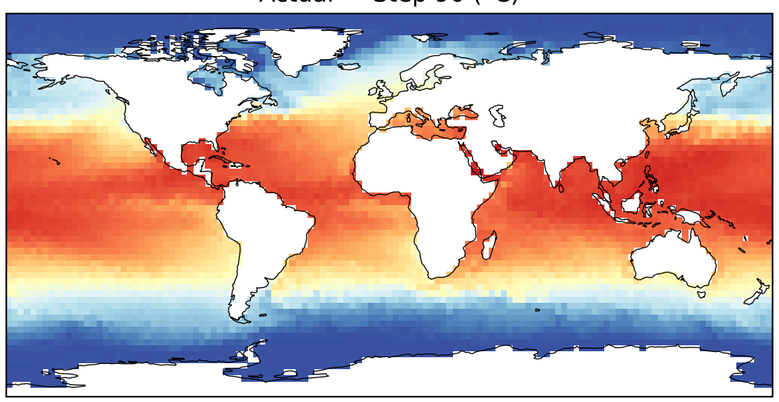
SFNO — Step 46 (°C)



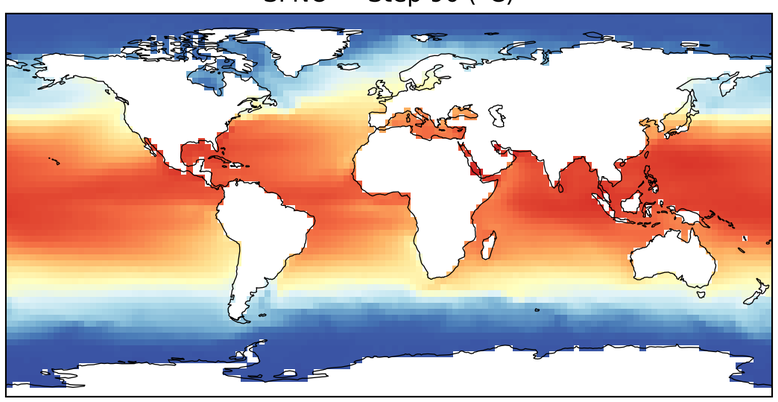
Error — Step 46 (°C)



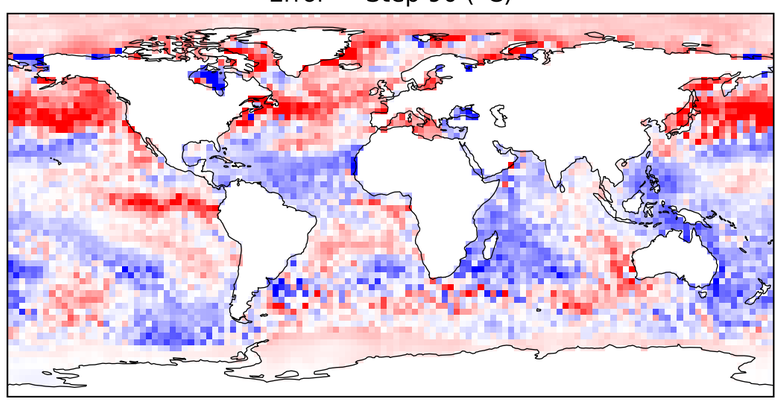
Actual — Step 90 (°C)

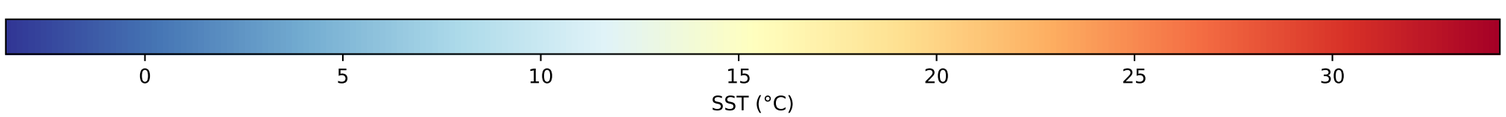


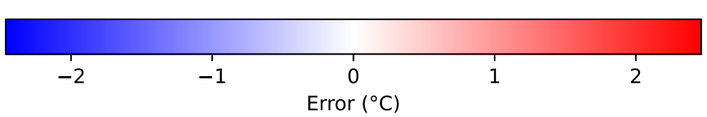
SFNO — Step 90 (°C)



Error — Step 90 (°C)

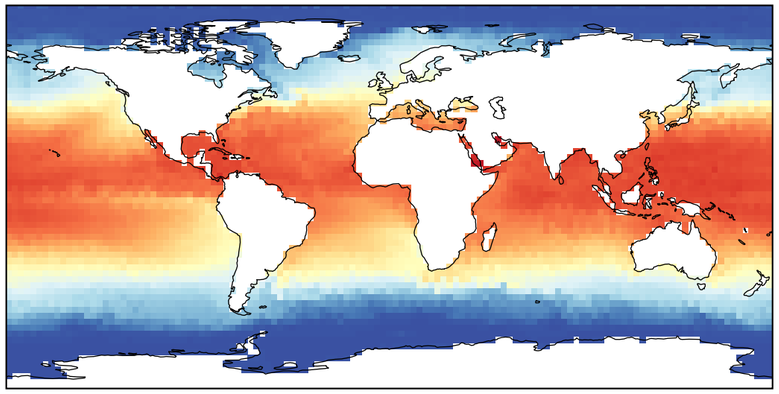




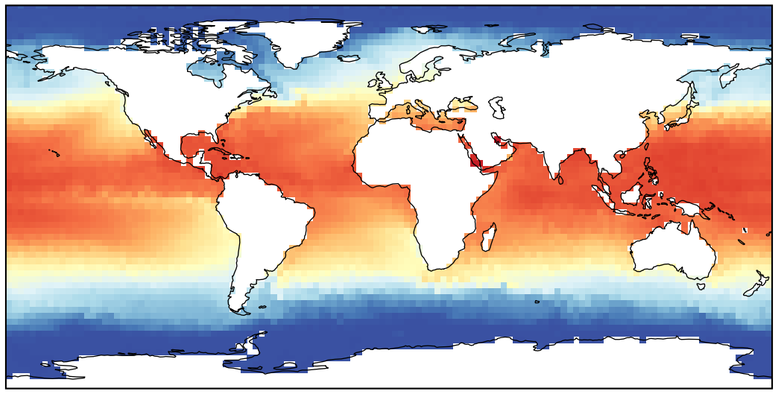



# Chain init: 2017-10-02 — Validation (°C)

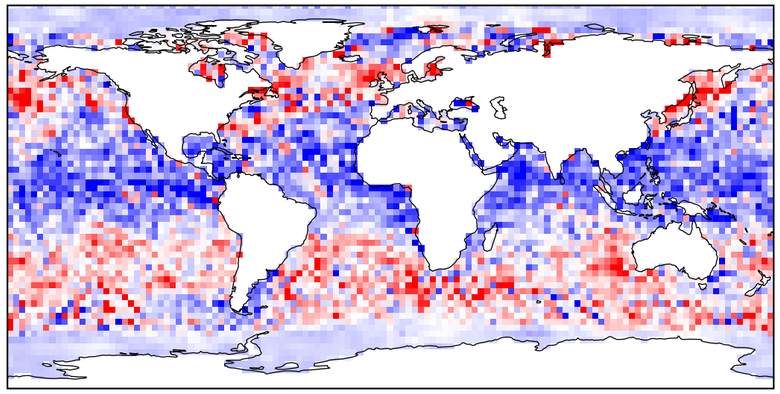
Actual — Step 1 (°C)



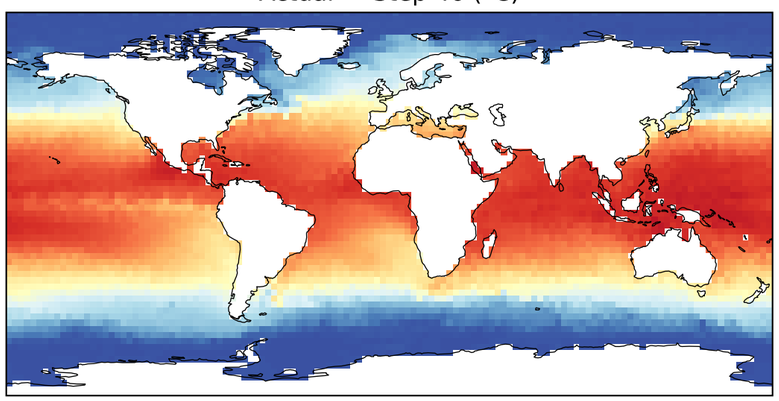
SFNO — Step 1 (°C)



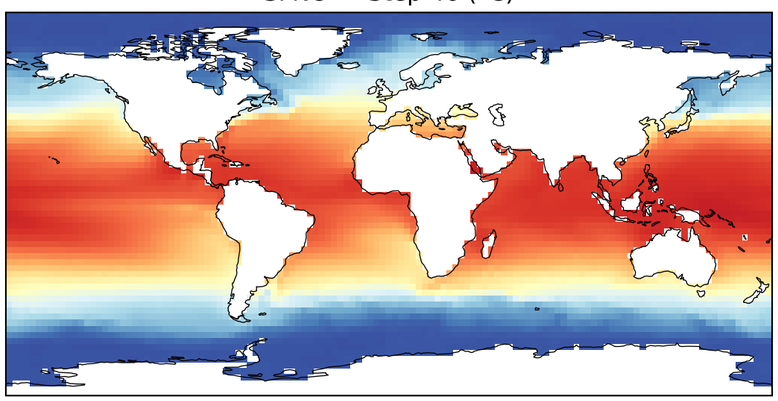
Error — Step 1 (°C)



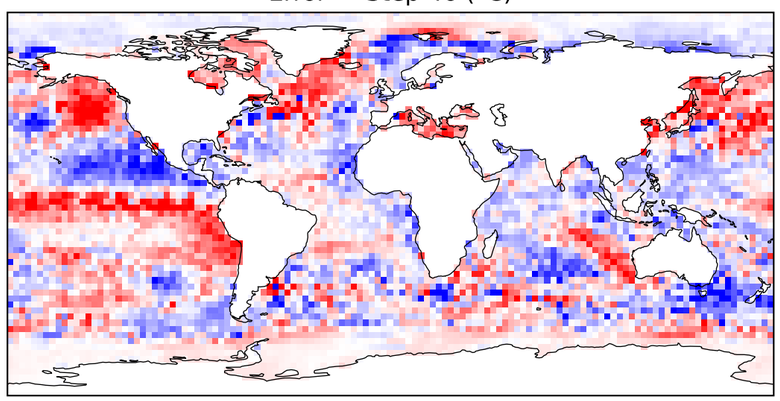
Actual — Step 46 (°C)



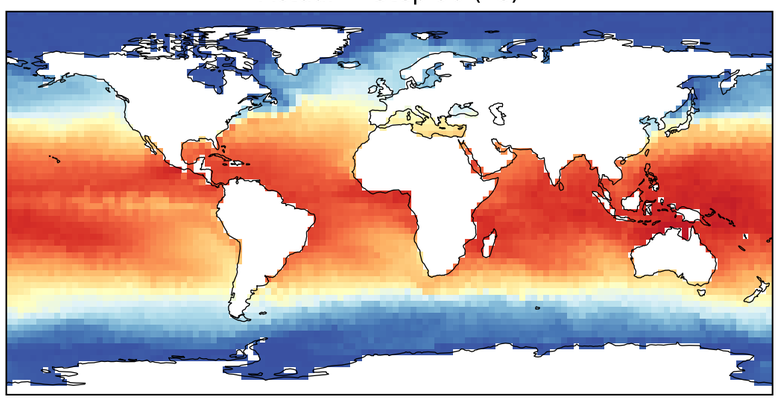
SFNO — Step 46 (°C)



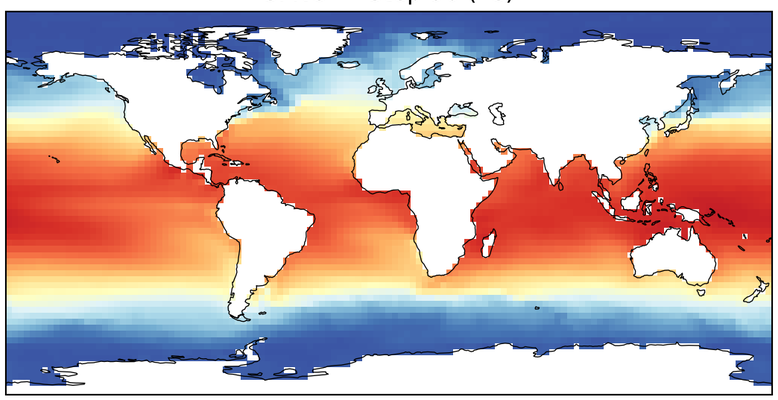
Error — Step 46 (°C)



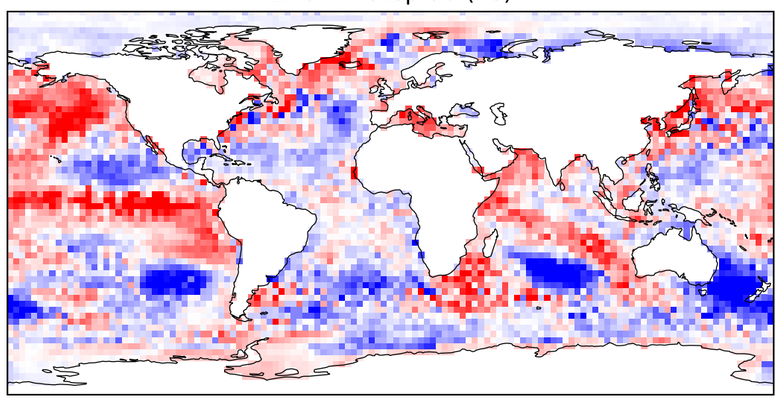
Actual — Step 90 (°C)

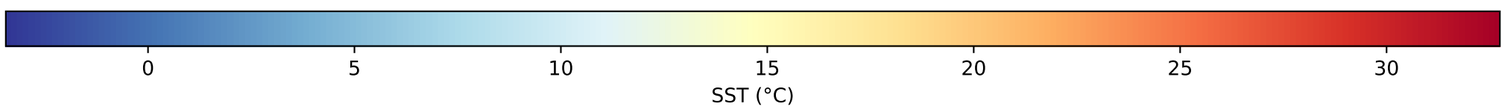


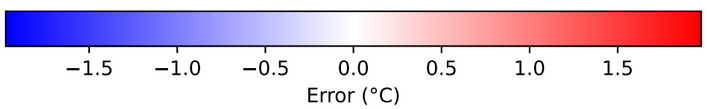
SFNO — Step 90 (°C)



Error — Step 90 (°C)

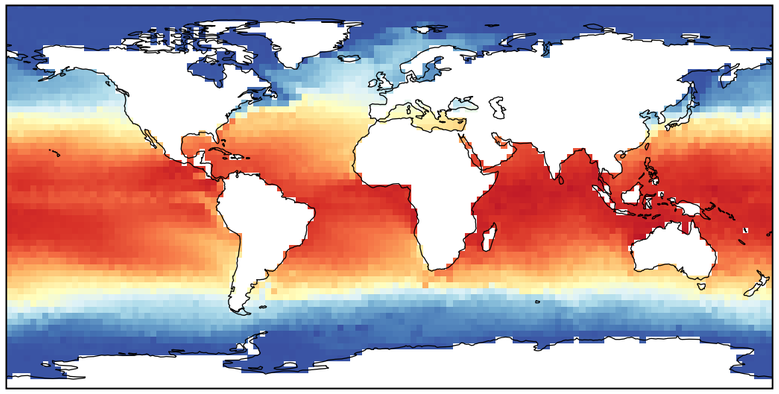




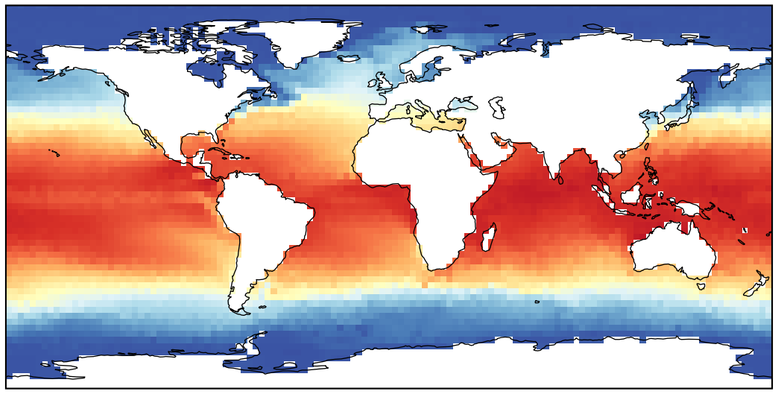



# Chain init: 2018-03-31 — Validation (°C)

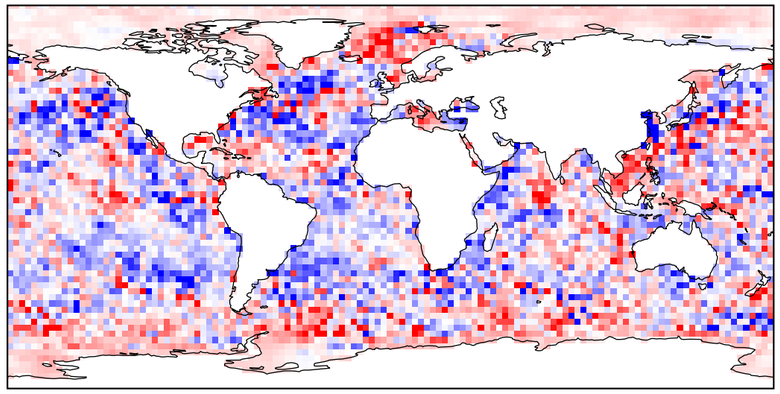
Actual — Step 1 (°C)



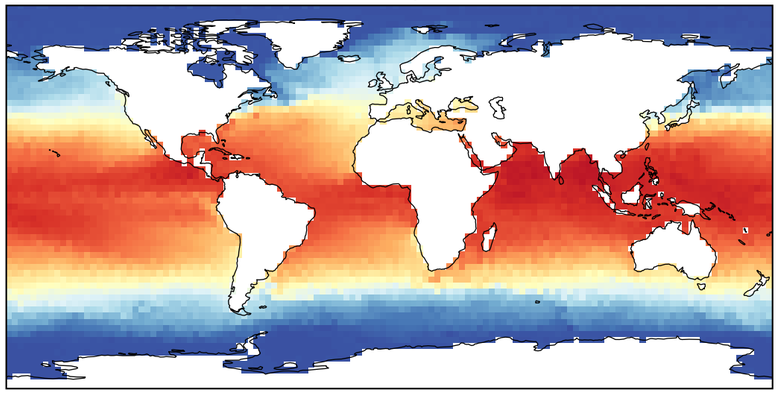
SFNO — Step 1 (°C)



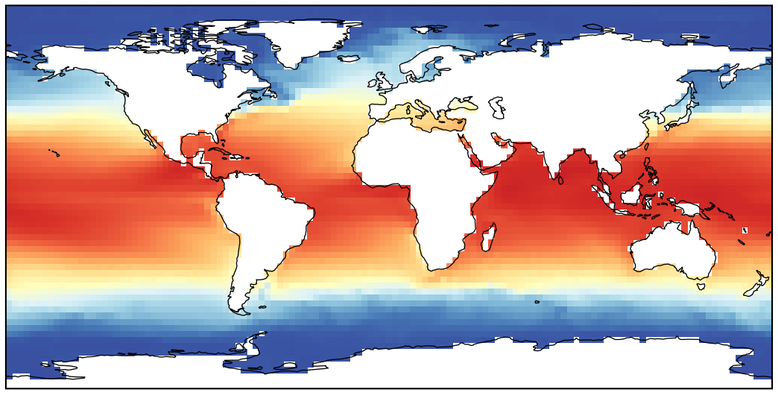
Error — Step 1 (°C)



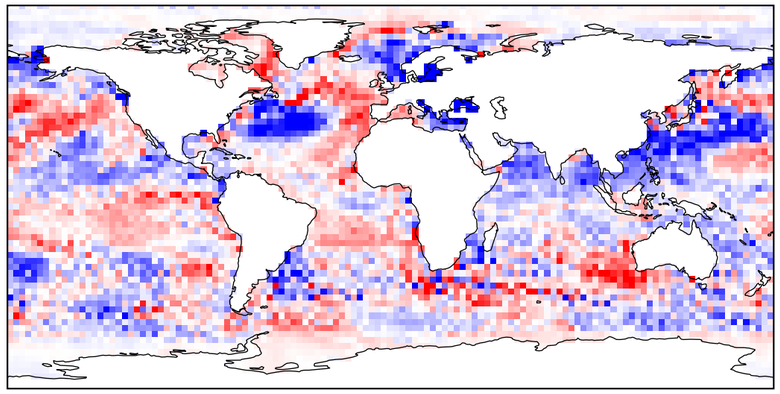
Actual — Step 46 (°C)



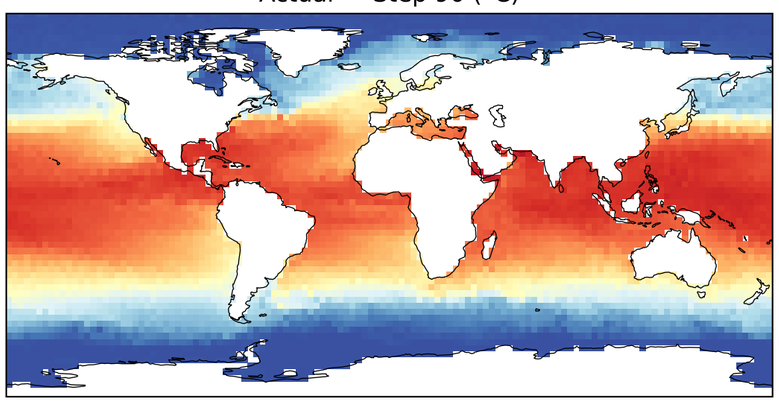
SFNO — Step 46 (°C)



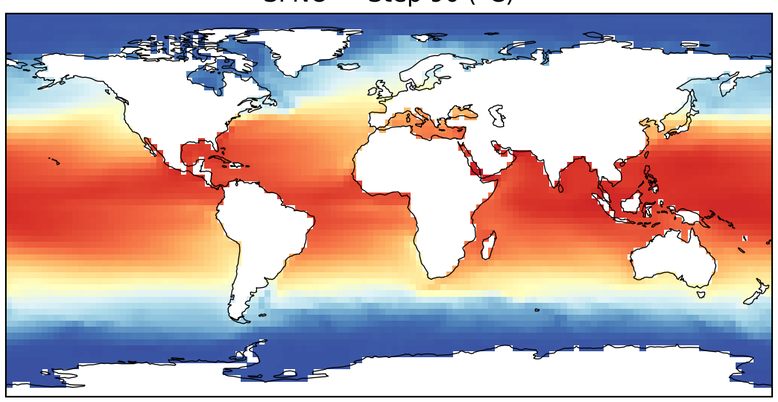
Error — Step 46 (°C)



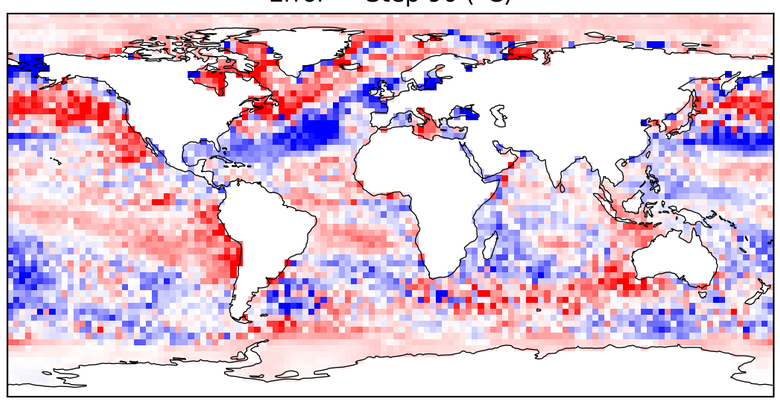
Actual — Step 90 (°C)

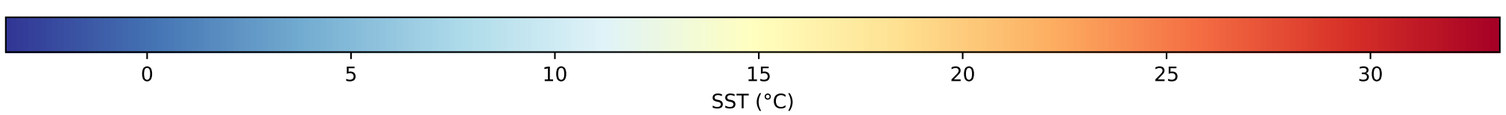


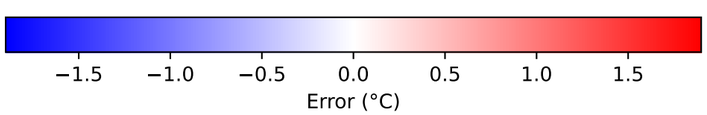
SFNO — Step 90 (°C)



Error — Step 90 (°C)

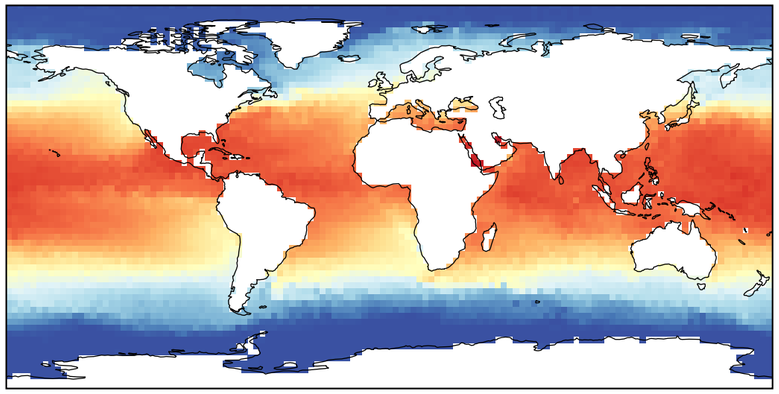




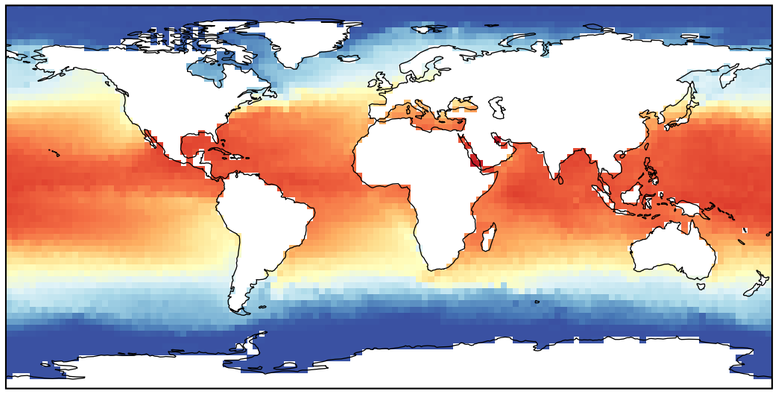



# Chain init: 2018-09-27 — Validation (°C)

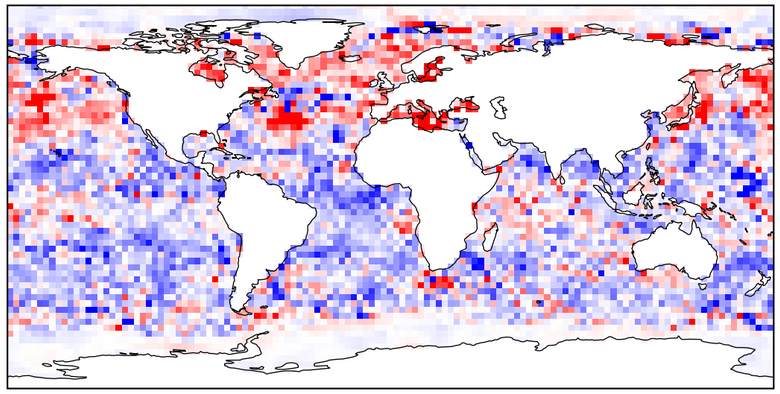
Actual — Step 1 (°C)



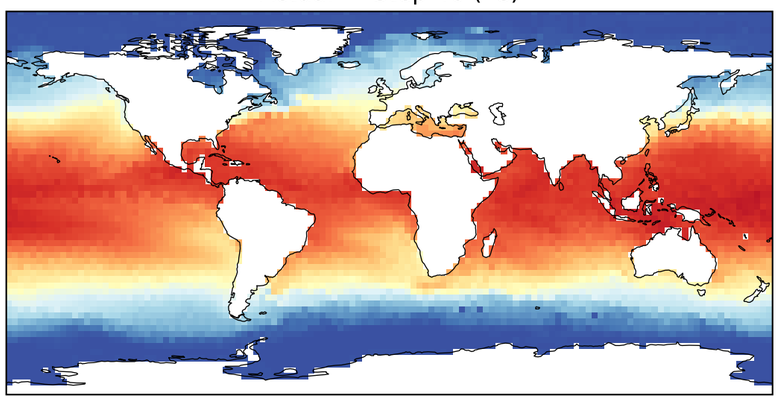
SFNO — Step 1 (°C)



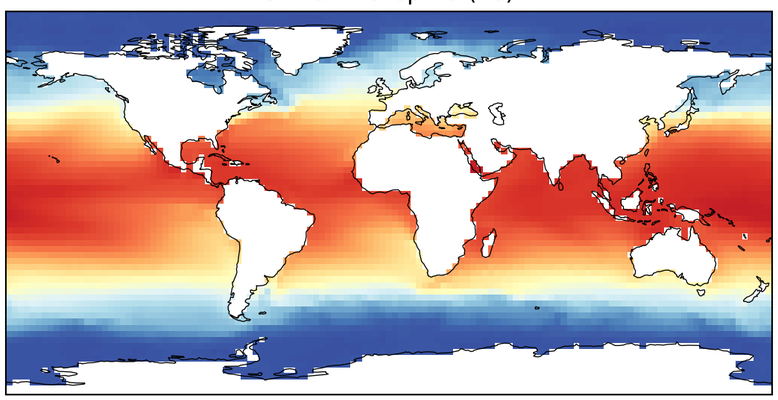
Error — Step 1 (°C)



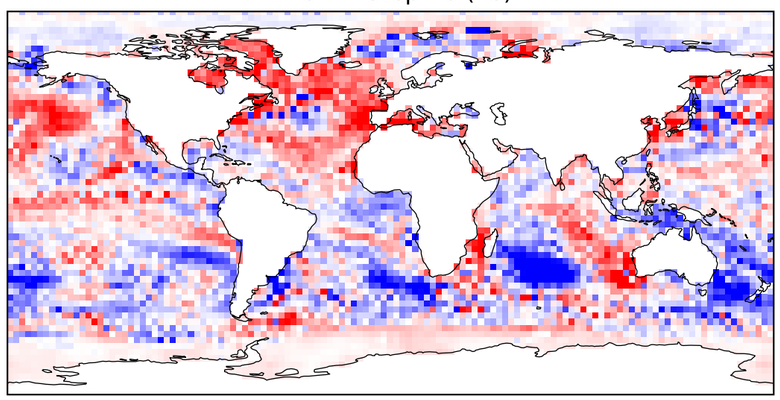
Actual — Step 46 (°C)



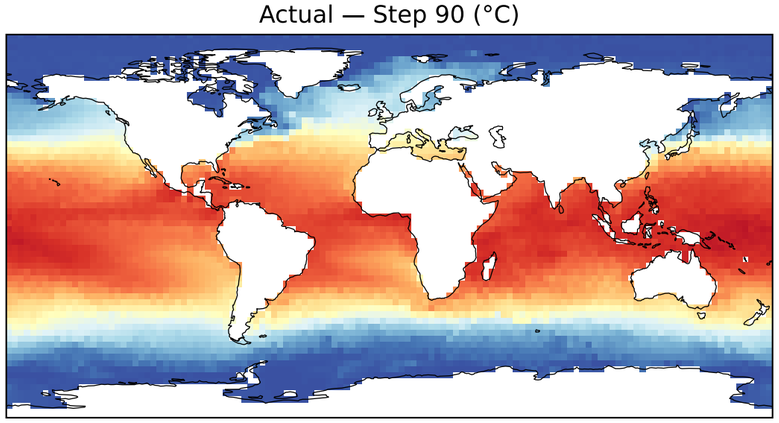
SFNO — Step 46 (°C)



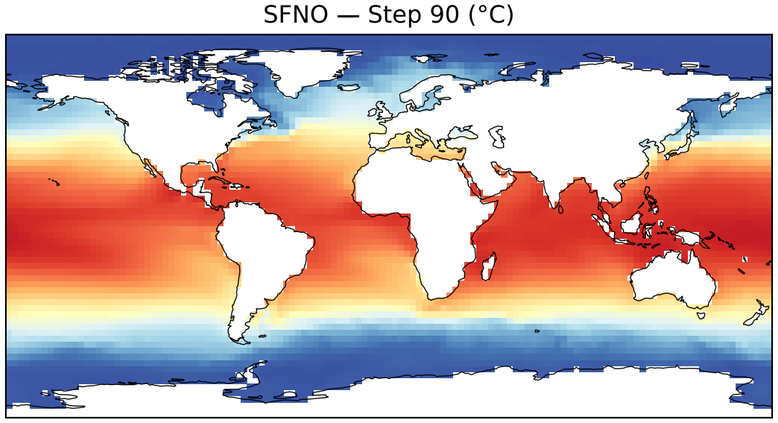
Error — Step 46 (°C)



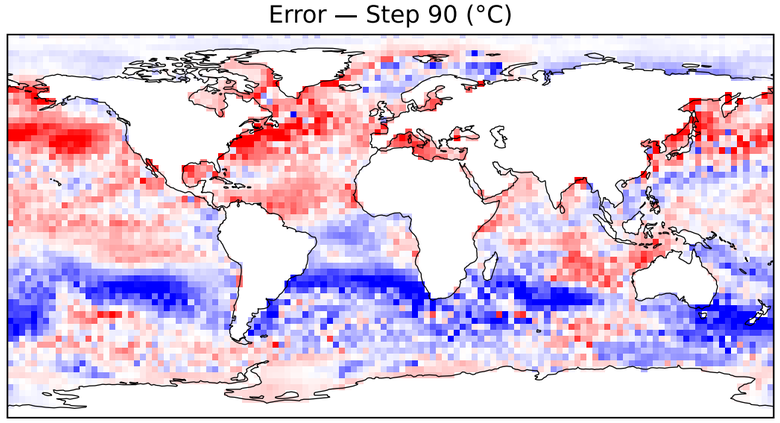
Actual — Step 90 (°C)

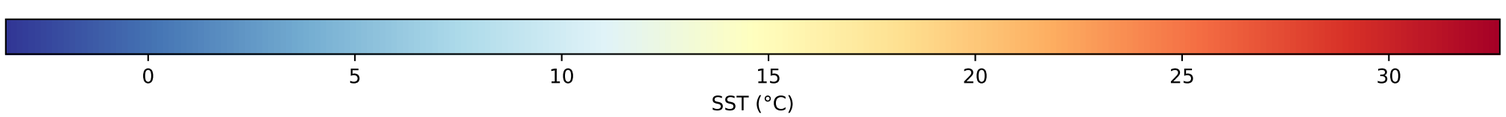


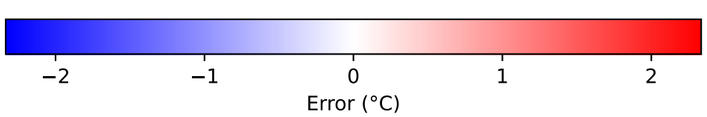
SFNO — Step 90 (°C)



Error — Step 90 (°C)

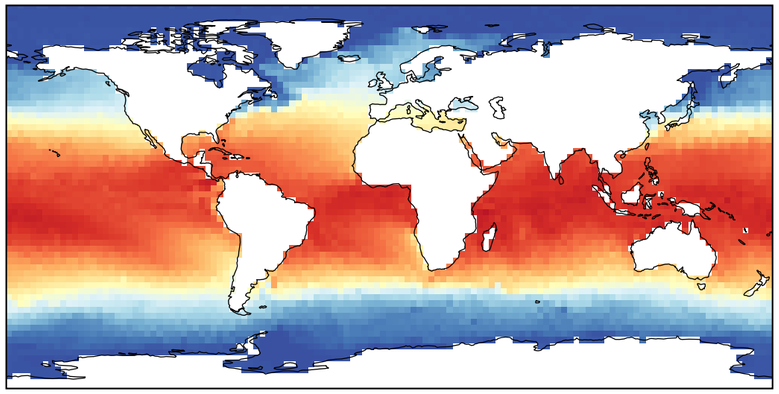




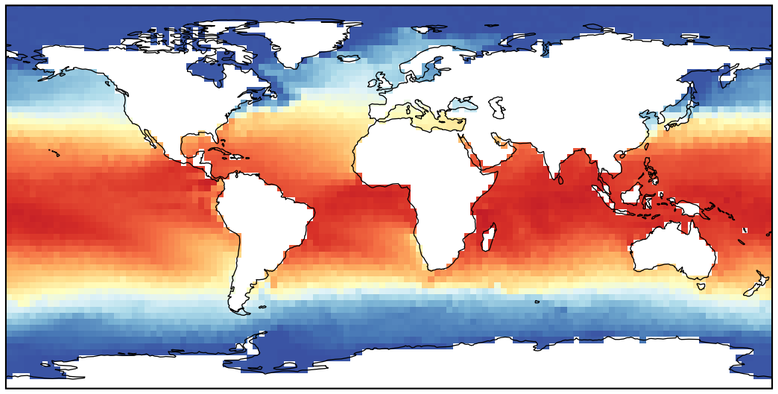



# Chain init: 2019-03-26 — Validation (°C)

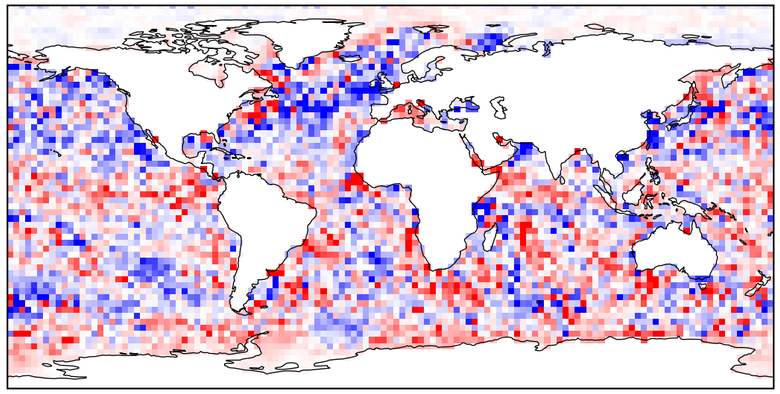
Actual — Step 1 (°C)



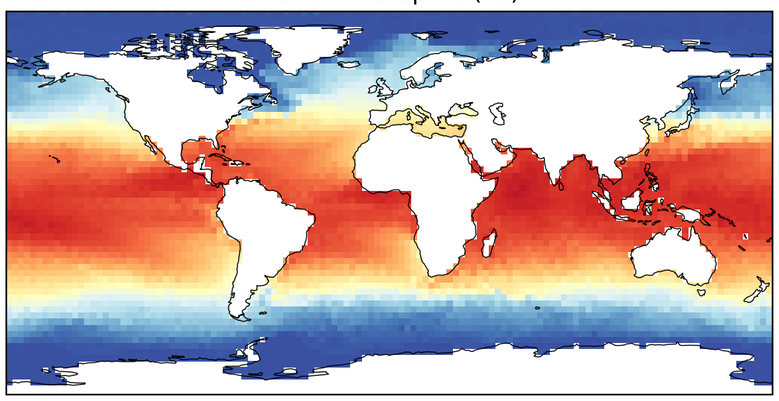
SFNO — Step 1 (°C)



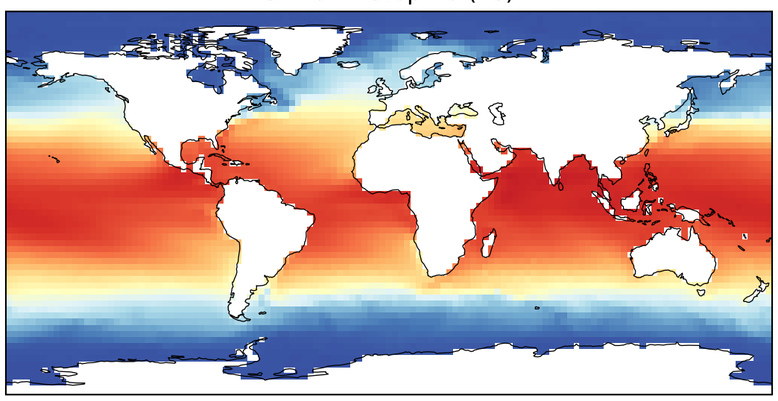
Error — Step 1 (°C)



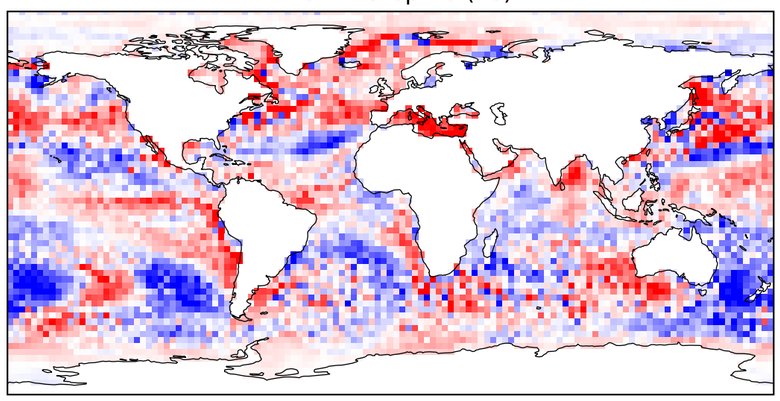
Actual — Step 46 (°C)



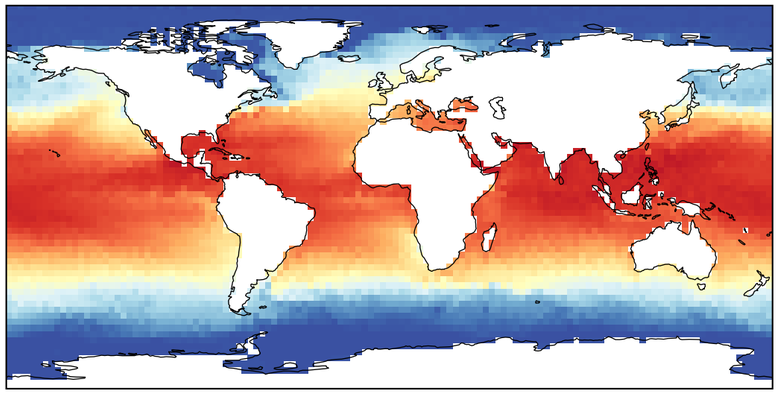
SFNO — Step 46 (°C)



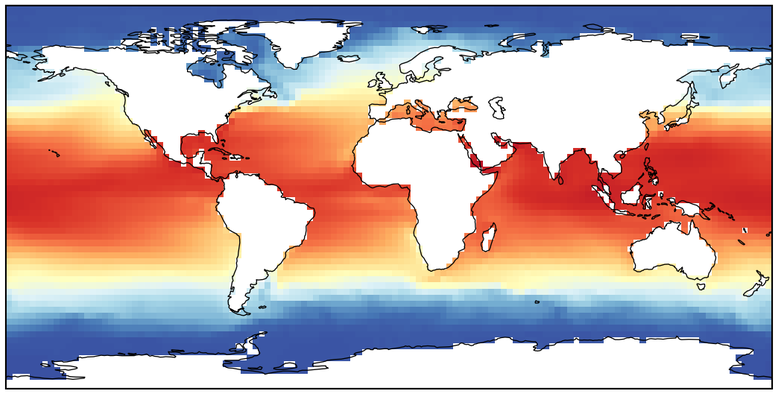
Error — Step 46 (°C)



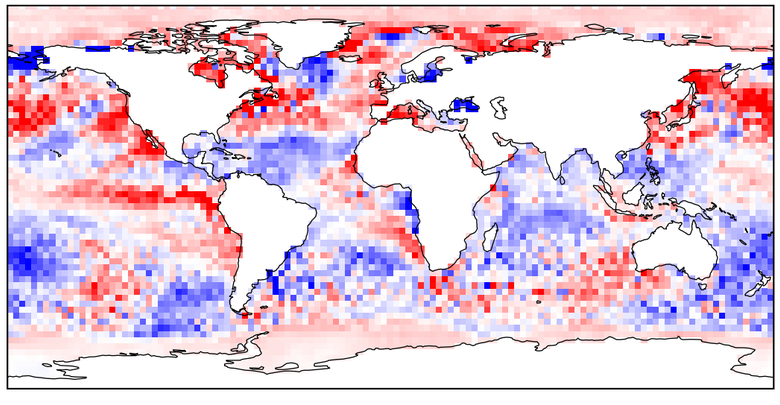
Actual — Step 90 (°C)

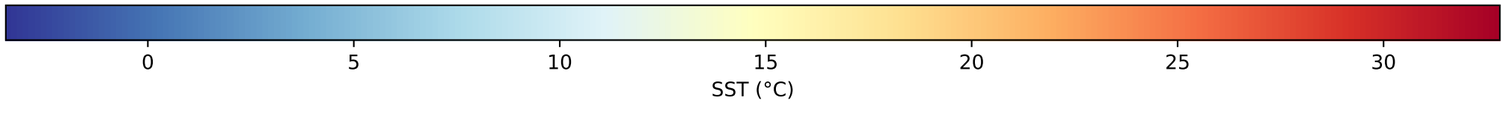


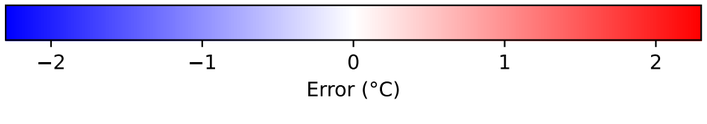
SFNO — Step 90 (°C)



Error — Step 90 (°C)

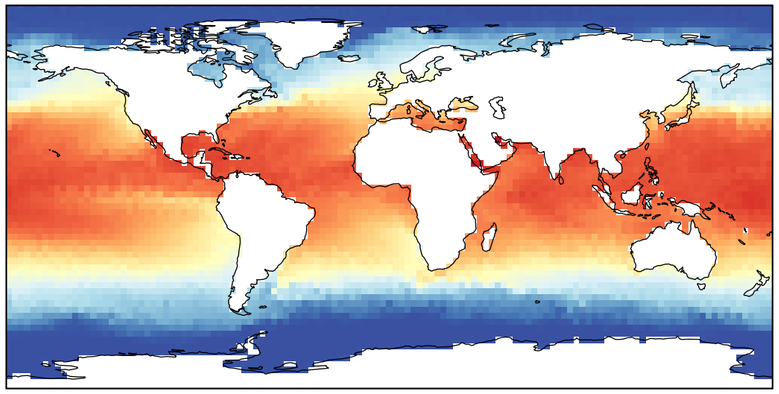




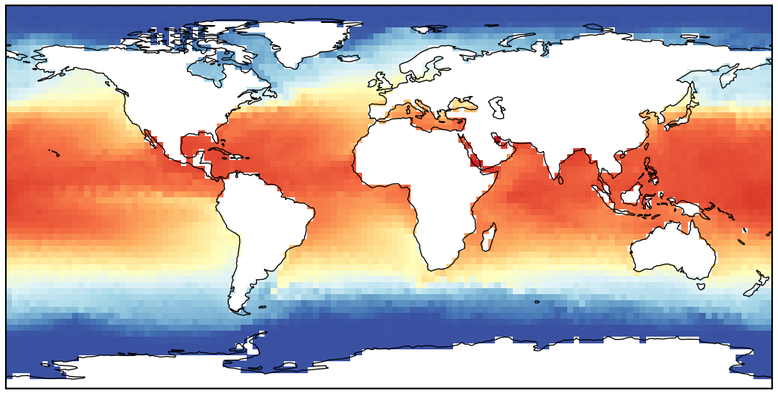



# Chain init: 2019-09-22 — Validation (°C)

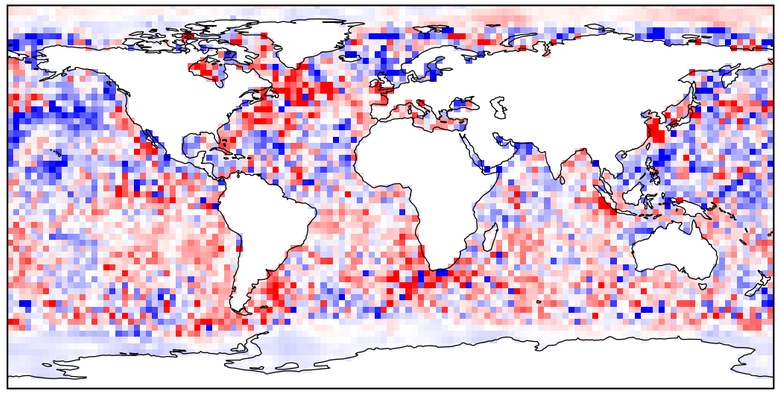
Actual — Step 1 (°C)



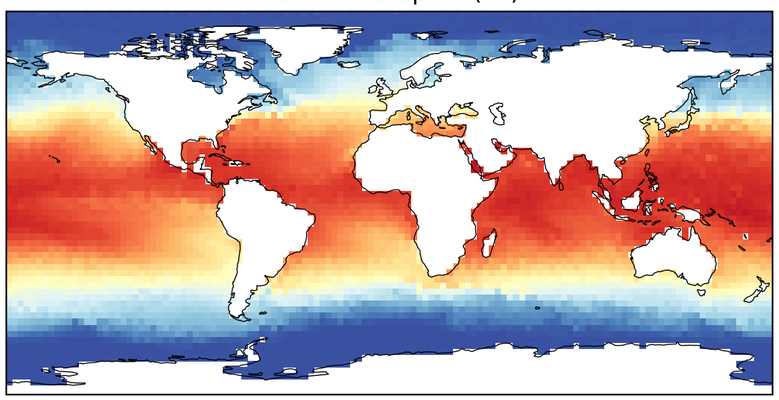
SFNO — Step 1 (°C)



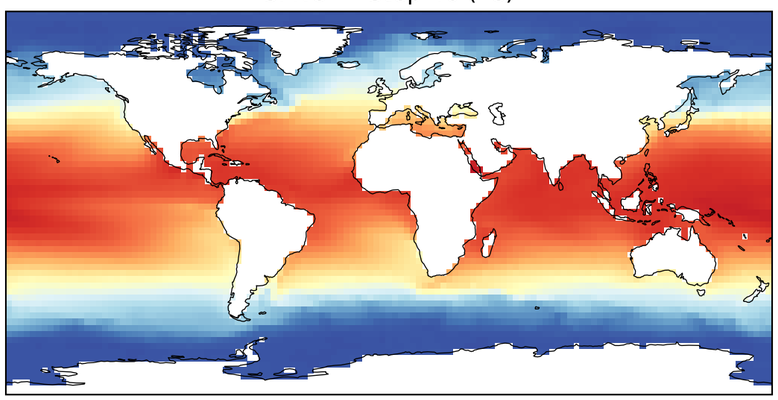
Error — Step 1 (°C)



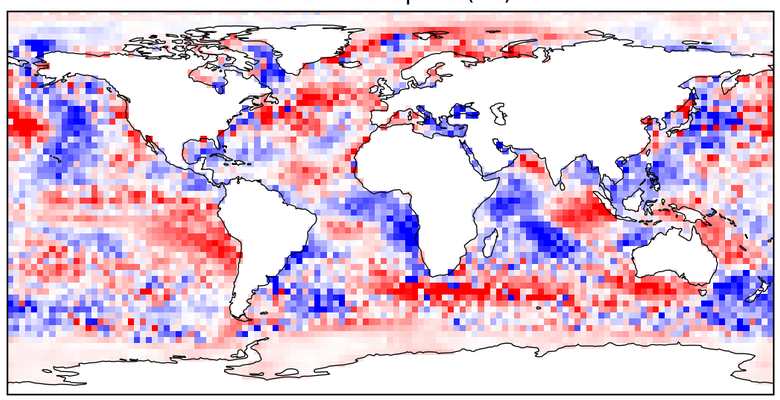
Actual — Step 46 (°C)



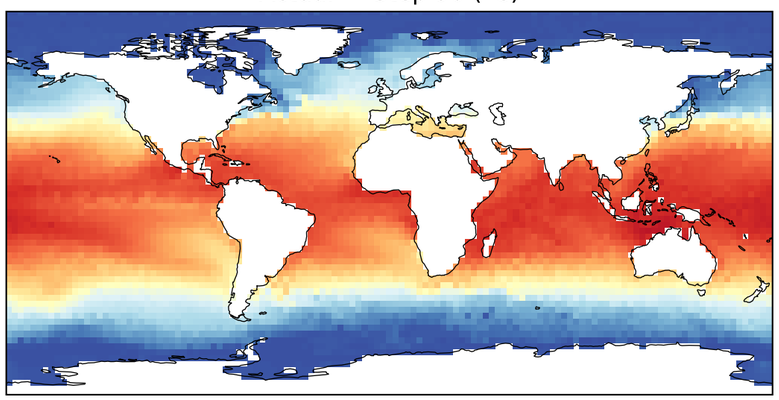
SFNO — Step 46 (°C)



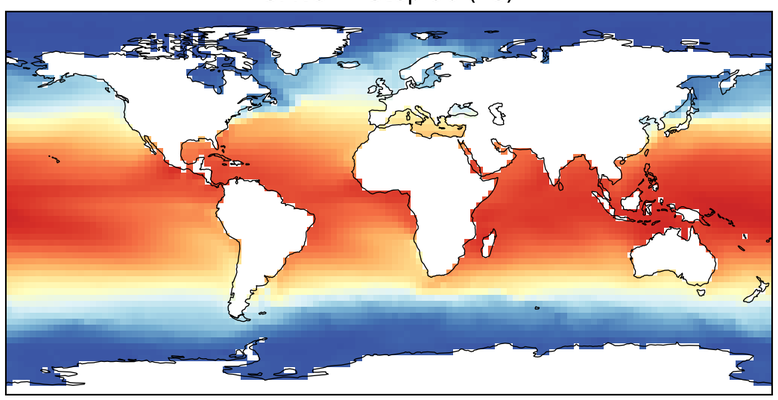
Error — Step 46 (°C)



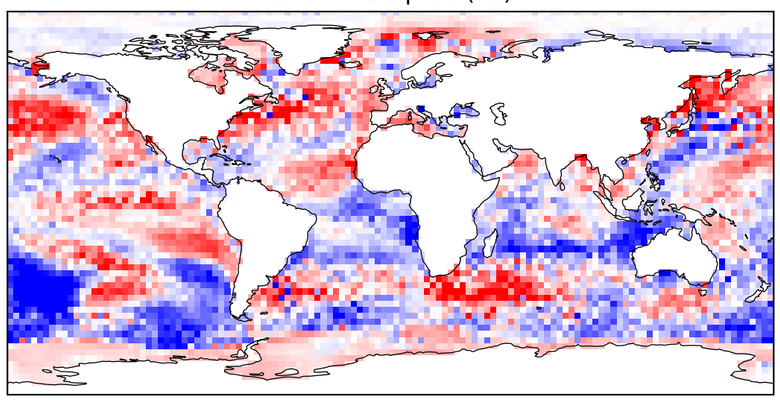
Actual — Step 90 (°C)

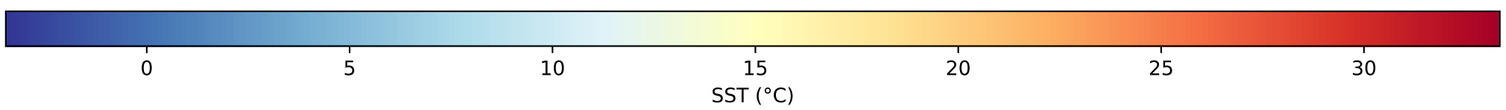


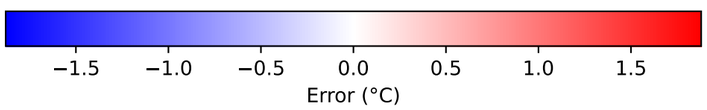
SFNO — Step 90 (°C)



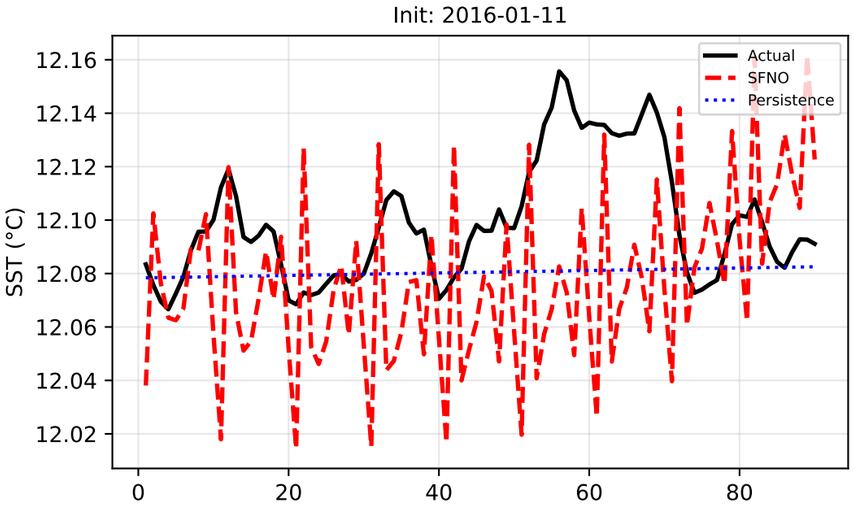
Error — Step 90 (°C)

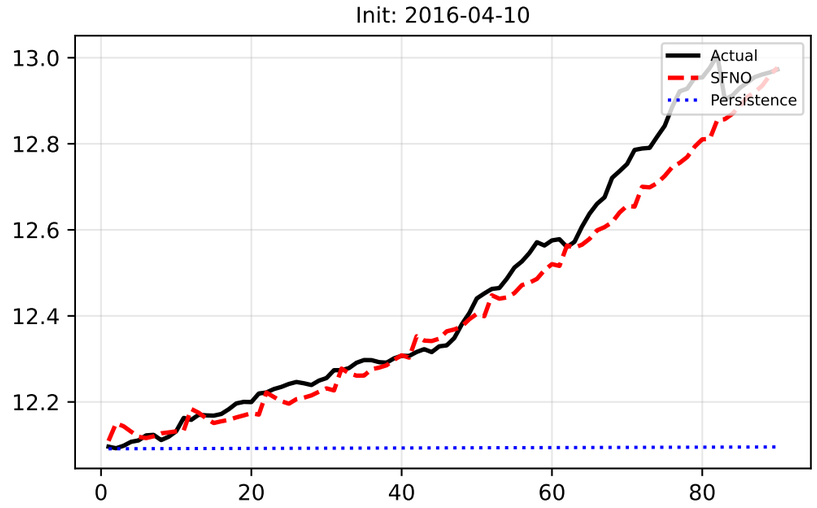


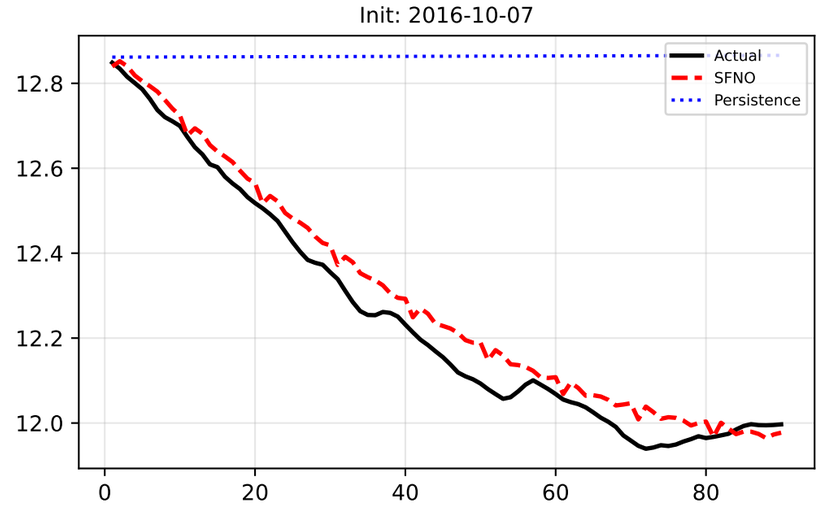


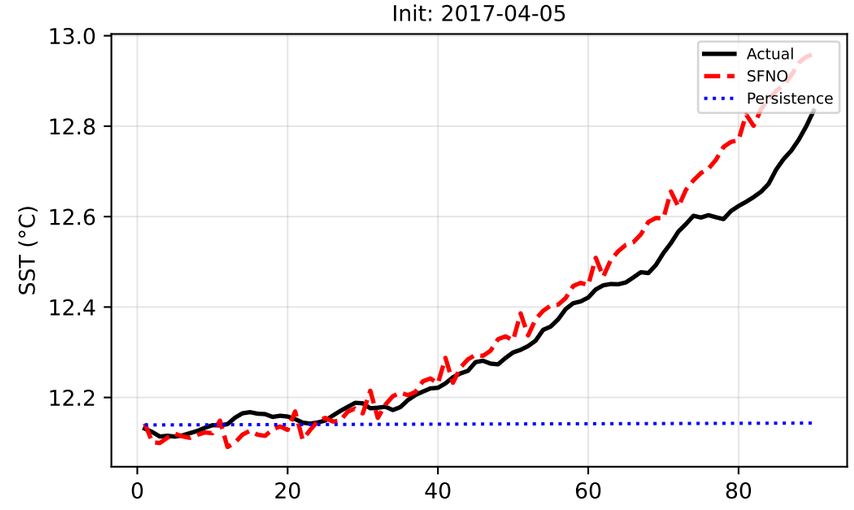



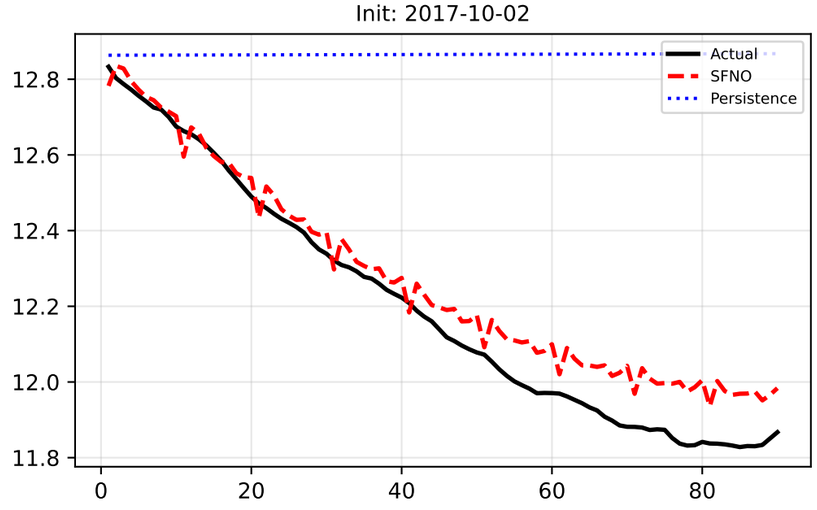
# Global-Mean SST — Validation (°C)

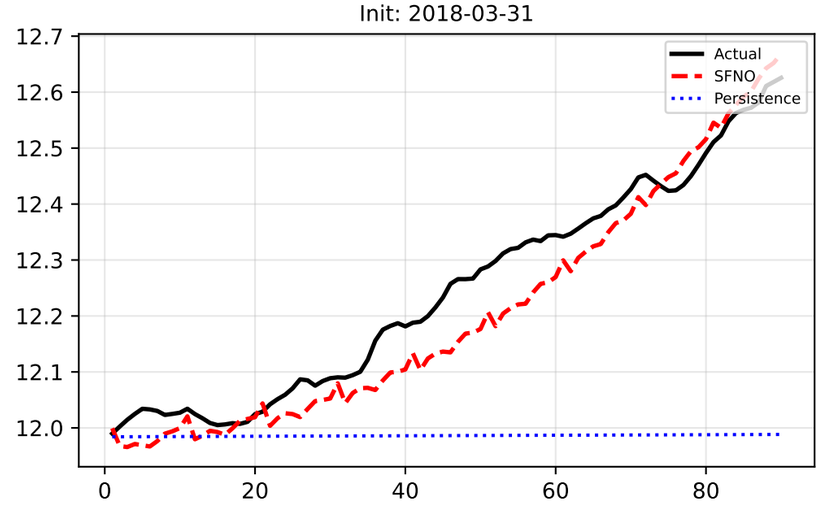


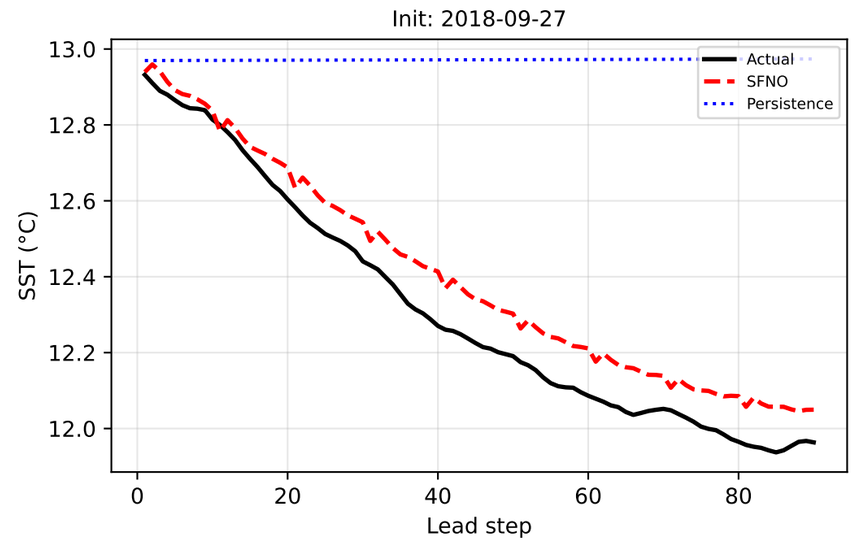



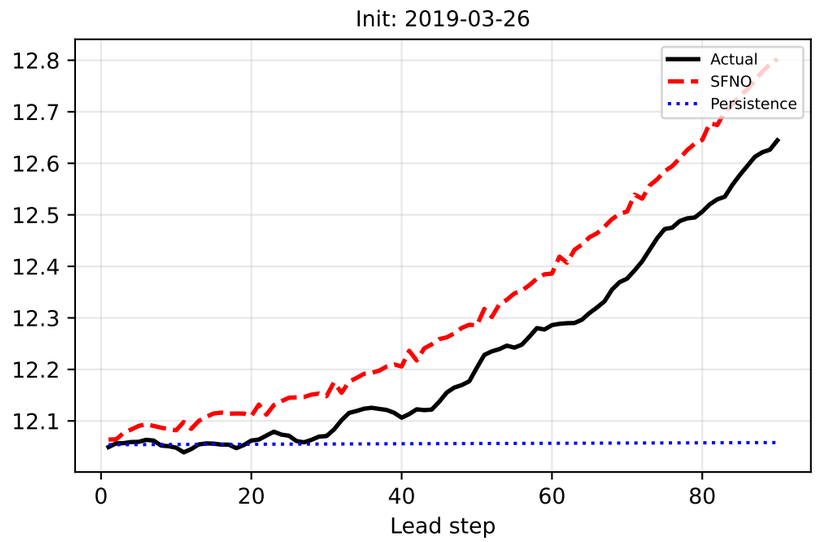



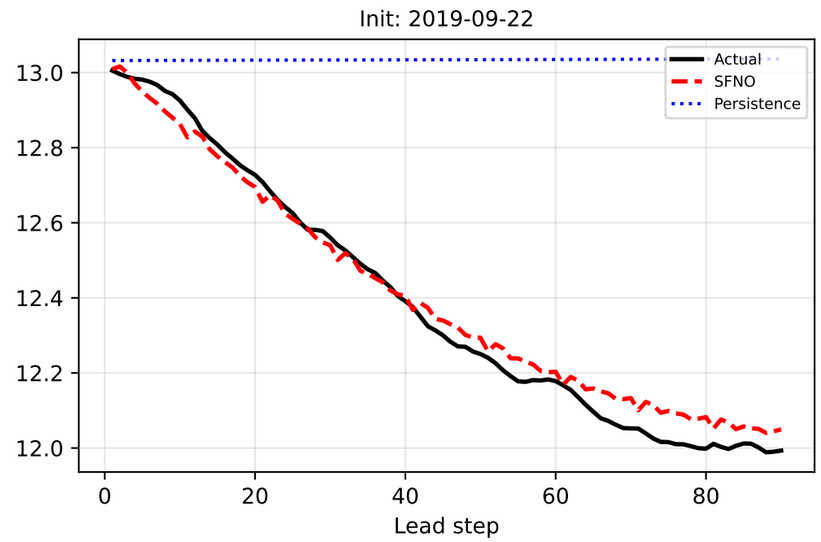



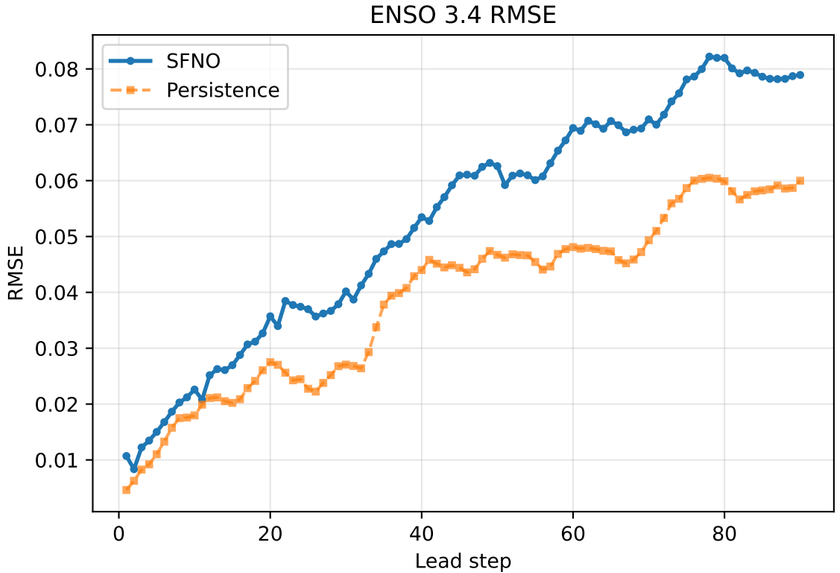



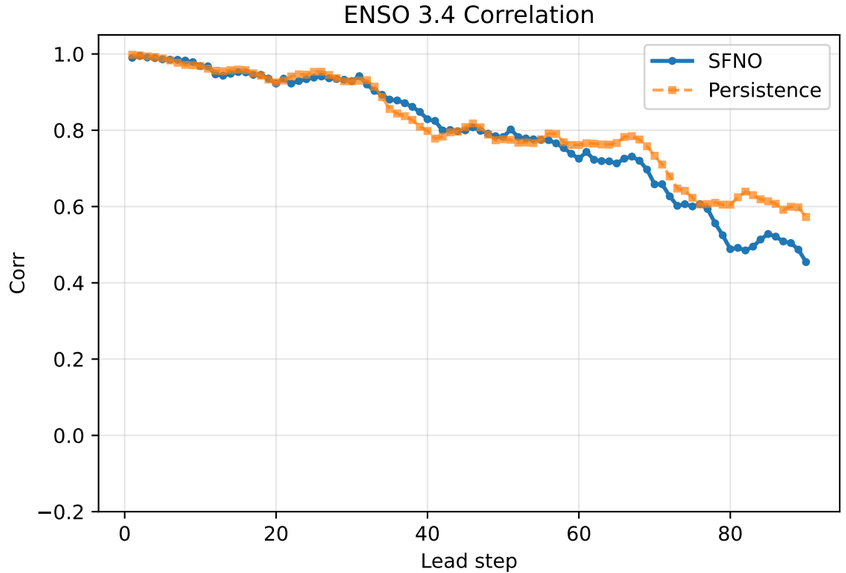



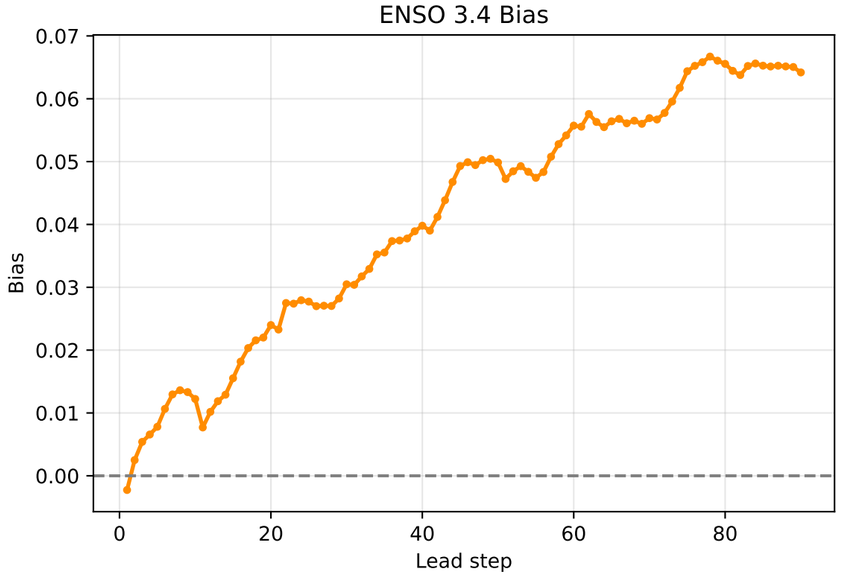
# ENSO 3.4 Metrics — Validation

| Lead Step | SFNO RMSE | Persist RMSE | SFNO Corr | Persist Corr | SFNO Bias |
|---|---|---|---|---|---|
| 1 | 0.0107 | 0.0046 | 0.9899 | 0.9981 | -0.0023 |
| 2 | 0.0083 | 0.0063 | 0.9948 | 0.9964 | 0.0025 |
| 3 | 0.0122 | 0.0083 | 0.9907 | 0.9938 | 0.0054 |
| 4 | 0.0135 | 0.0092 | 0.9893 | 0.9920 | 0.0066 |
| 5 | 0.0150 | 0.0110 | 0.9859 | 0.9881 | 0.0078 |
| 6 | 0.0168 | 0.0133 | 0.9851 | 0.9833 | 0.0106 |
| 7 | 0.0186 | 0.0158 | 0.9847 | 0.9771 | 0.0129 |
| 8 | 0.0203 | 0.0175 | 0.9827 | 0.9714 | 0.0136 |
| 9 | 0.0212 | 0.0176 | 0.9787 | 0.9705 | 0.0133 |
| 10 | 0.0226 | 0.0180 | 0.9682 | 0.9684 | 0.0122 |
| 11 | 0.0208 | 0.0199 | 0.9676 | 0.9615 | 0.0077 |
| 12 | 0.0252 | 0.0211 | 0.9465 | 0.9562 | 0.0102 |
| 13 | 0.0263 | 0.0212 | 0.9432 | 0.9548 | 0.0119 |
| 14 | 0.0261 | 0.0205 | 0.9481 | 0.9576 | 0.0129 |
| 15 | 0.0270 | 0.0202 | 0.9527 | 0.9592 | 0.0155 |
| 16 | 0.0288 | 0.0209 | 0.9518 | 0.9576 | 0.0182 |
| 17 | 0.0307 | 0.0228 | 0.9453 | 0.9494 | 0.0203 |
| 18 | 0.0312 | 0.0241 | 0.9454 | 0.9431 | 0.0216 |
| 19 | 0.0327 | 0.0261 | 0.9356 | 0.9328 | 0.0220 |
| 20 | 0.0357 | 0.0275 | 0.9225 | 0.9257 | 0.0240 |
| 21 | 0.0340 | 0.0270 | 0.9354 | 0.9303 | 0.0233 |
| 22 | 0.0385 | 0.0256 | 0.9224 | 0.9410 | 0.0275 |
| 23 | 0.0377 | 0.0243 | 0.9290 | 0.9466 | 0.0274 |
| 24 | 0.0374 | 0.0245 | 0.9339 | 0.9458 | 0.0279 |
| 25 | 0.0370 | 0.0227 | 0.9380 | 0.9531 | 0.0277 |
| 26 | 0.0357 | 0.0222 | 0.9410 | 0.9533 | 0.0270 |
| 27 | 0.0362 | 0.0238 | 0.9364 | 0.9448 | 0.0271 |
| 28 | 0.0367 | 0.0252 | 0.9344 | 0.9366 | 0.0270 |
| 29 | 0.0379 | 0.0268 | 0.9324 | 0.9284 | 0.0282 |
| 30 | 0.0402 | 0.0271 | 0.9291 | 0.9280 | 0.0305 |
| 31 | 0.0387 | 0.0268 | 0.9419 | 0.9302 | 0.0304 |
| 32 | 0.0412 | 0.0264 | 0.9201 | 0.9312 | 0.0317 |
| 33 | 0.0433 | 0.0293 | 0.9033 | 0.9144 | 0.0329 |
| 34 | 0.0460 | 0.0338 | 0.8931 | 0.8869 | 0.0352 |
| 35 | 0.0473 | 0.0378 | 0.8804 | 0.8557 | 0.0355 |
| 36 | 0.0486 | 0.0394 | 0.8782 | 0.8439 | 0.0374 |
| 37 | 0.0487 | 0.0399 | 0.8710 | 0.8370 | 0.0374 |
| 38 | 0.0495 | 0.0408 | 0.8614 | 0.8272 | 0.0378 |
| 39 | 0.0515 | 0.0429 | 0.8481 | 0.8091 | 0.0389 |
| 40 | 0.0535 | 0.0440 | 0.8289 | 0.7981 | 0.0398 |
| 41 | 0.0528 | 0.0458 | 0.8245 | 0.7782 | 0.0390 |
| 42 | 0.0553 | 0.0451 | 0.7996 | 0.7835 | 0.0412 |
| 43 | 0.0571 | 0.0445 | 0.8007 | 0.7945 | 0.0439 |
| 44 | 0.0592 | 0.0449 | 0.7973 | 0.7965 | 0.0468 |
| 45 | 0.0609 | 0.0444 | 0.7997 | 0.8085 | 0.0493 |
| 46 | 0.0611 | 0.0436 | 0.8078 | 0.8178 | 0.0499 |
| 47 | 0.0609 | 0.0441 | 0.7985 | 0.8079 | 0.0494 |
| 48 | 0.0625 | 0.0460 | 0.7910 | 0.7888 | 0.0502 |
| 49 | 0.0632 | 0.0474 | 0.7844 | 0.7738 | 0.0505 |
| 50 | 0.0626 | 0.0467 | 0.7824 | 0.7763 | 0.0499 |
| 51 | 0.0592 | 0.0462 | 0.8021 | 0.7751 | 0.0472 |
| 52 | 0.0609 | 0.0468 | 0.7821 | 0.7672 | 0.0485 |
| 53 | 0.0613 | 0.0467 | 0.7786 | 0.7680 | 0.0493 |
| 54 | 0.0610 | 0.0466 | 0.7763 | 0.7665 | 0.0484 |
| 55 | 0.0601 | 0.0454 | 0.7747 | 0.7763 | 0.0474 |
| 56 | 0.0608 | 0.0441 | 0.7741 | 0.7919 | 0.0483 |
| 57 | 0.0631 | 0.0446 | 0.7657 | 0.7902 | 0.0508 |
| 58 | 0.0654 | 0.0469 | 0.7535 | 0.7685 | 0.0528 |
| 59 | 0.0672 | 0.0477 | 0.7385 | 0.7616 | 0.0542 |
| 60 | 0.0694 | 0.0481 | 0.7254 | 0.7614 | 0.0557 |
| 61 | 0.0689 | 0.0478 | 0.7433 | 0.7659 | 0.0556 |
| 62 | 0.0707 | 0.0480 | 0.7226 | 0.7649 | 0.0576 |
| 63 | 0.0701 | 0.0477 | 0.7192 | 0.7628 | 0.0563 |
| 64 | 0.0693 | 0.0474 | 0.7186 | 0.7625 | 0.0555 |
| 65 | 0.0707 | 0.0473 | 0.7130 | 0.7666 | 0.0564 |
| 66 | 0.0699 | 0.0458 | 0.7256 | 0.7823 | 0.0568 |
| 67 | 0.0687 | 0.0452 | 0.7310 | 0.7845 | 0.0561 |
| 68 | 0.0691 | 0.0459 | 0.7201 | 0.7758 | 0.0565 |
| 69 | 0.0693 | 0.0472 | 0.6969 | 0.7581 | 0.0560 |
| 70 | 0.0710 | 0.0493 | 0.6583 | 0.7330 | 0.0569 |
| 71 | 0.0700 | 0.0510 | 0.6586 | 0.7100 | 0.0567 |
| 72 | 0.0718 | 0.0533 | 0.6267 | 0.6791 | 0.0577 |
| 73 | 0.0742 | 0.0559 | 0.6020 | 0.6473 | 0.0596 |
| 74 | 0.0756 | 0.0567 | 0.6063 | 0.6412 | 0.0617 |
| 75 | 0.0781 | 0.0587 | 0.6002 | 0.6229 | 0.0644 |
| 76 | 0.0786 | 0.0600 | 0.6062 | 0.6055 | 0.0652 |
| 77 | 0.0800 | 0.0603 | 0.5936 | 0.6065 | 0.0658 |
| 78 | 0.0822 | 0.0605 | 0.5559 | 0.6097 | 0.0667 |
| 79 | 0.0820 | 0.0603 | 0.5246 | 0.6050 | 0.0661 |
| 80 | 0.0820 | 0.0599 | 0.4884 | 0.6048 | 0.0656 |
| 81 | 0.0801 | 0.0581 | 0.4918 | 0.6243 | 0.0644 |
| 82 | 0.0792 | 0.0566 | 0.4849 | 0.6391 | 0.0638 |
| 83 | 0.0797 | 0.0574 | 0.4949 | 0.6298 | 0.0652 |
| 84 | 0.0793 | 0.0581 | 0.5136 | 0.6190 | 0.0656 |
| 85 | 0.0786 | 0.0583 | 0.5280 | 0.6136 | 0.0653 |
| 86 | 0.0782 | 0.0585 | 0.5212 | 0.6072 | 0.0651 |
| 87 | 0.0782 | 0.0591 | 0.5084 | 0.5917 | 0.0653 |
| 88 | 0.0782 | 0.0586 | 0.5044 | 0.5994 | 0.0651 |
| 89 | 0.0787 | 0.0587 | 0.4869 | 0.5977 | 0.0651 |
| 90 | 0.0789 | 0.0600 | 0.4543 | 0.5725 | 0.0642 |

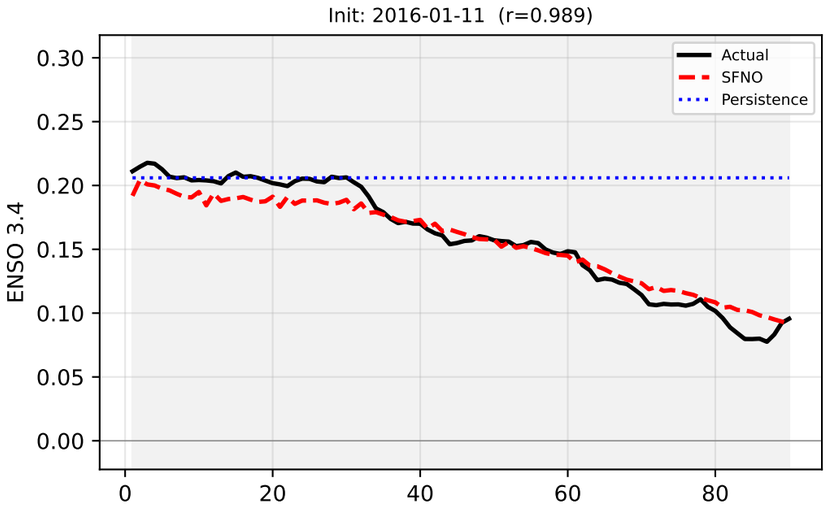
# ENSO 3.4 Forecast Skill — Validation

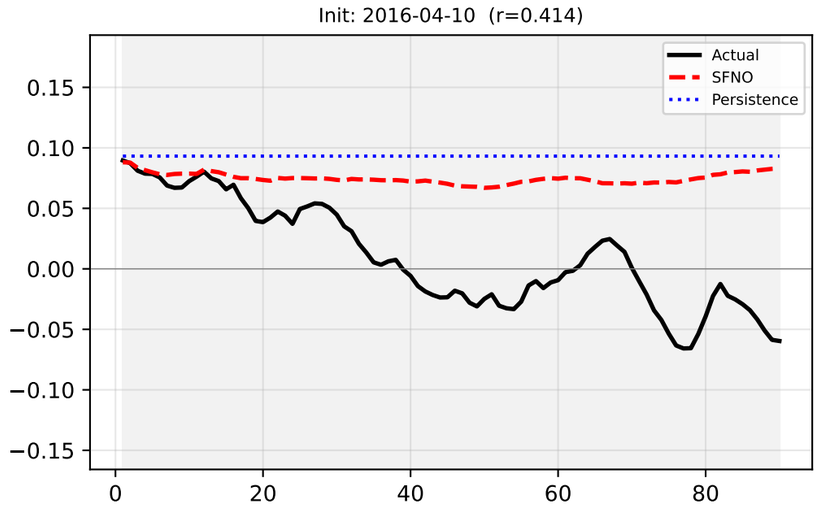


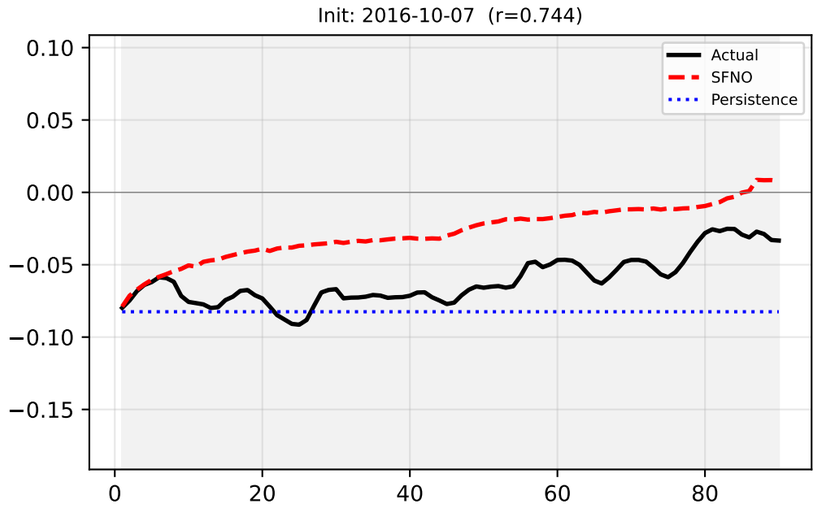



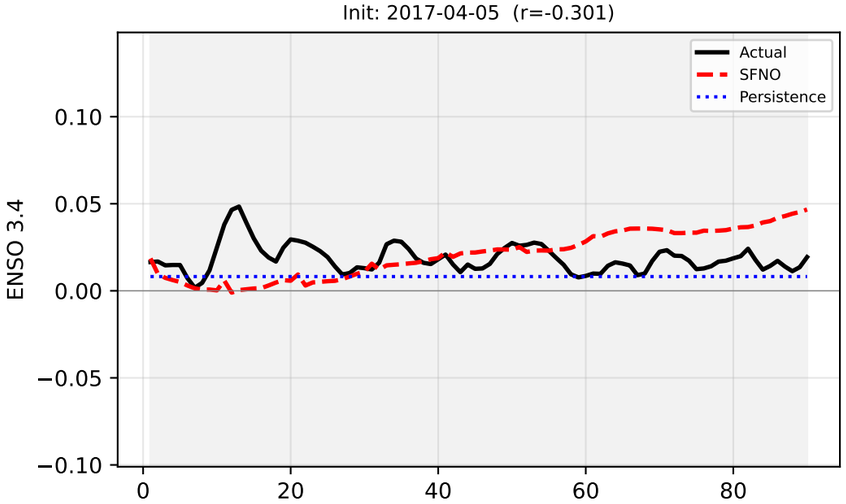



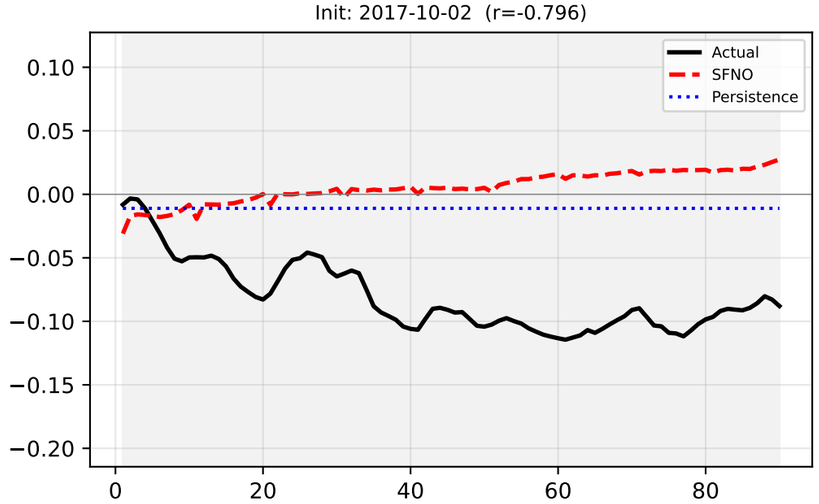
# ENSO 3.4 Index — Validation

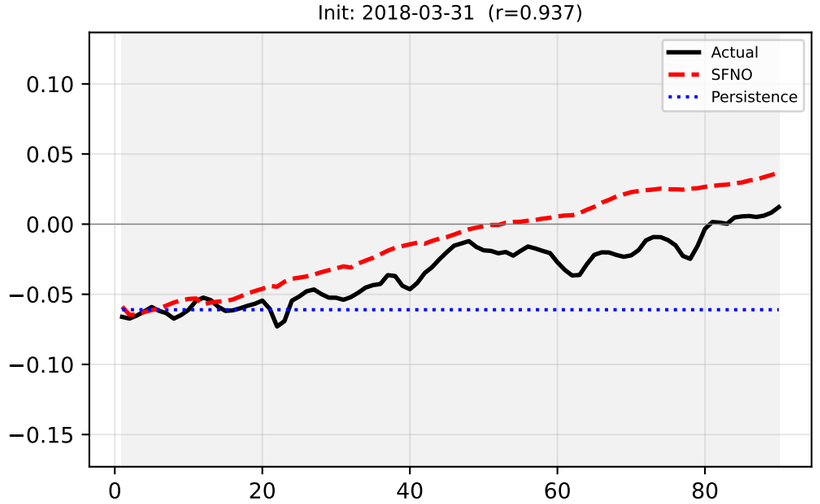


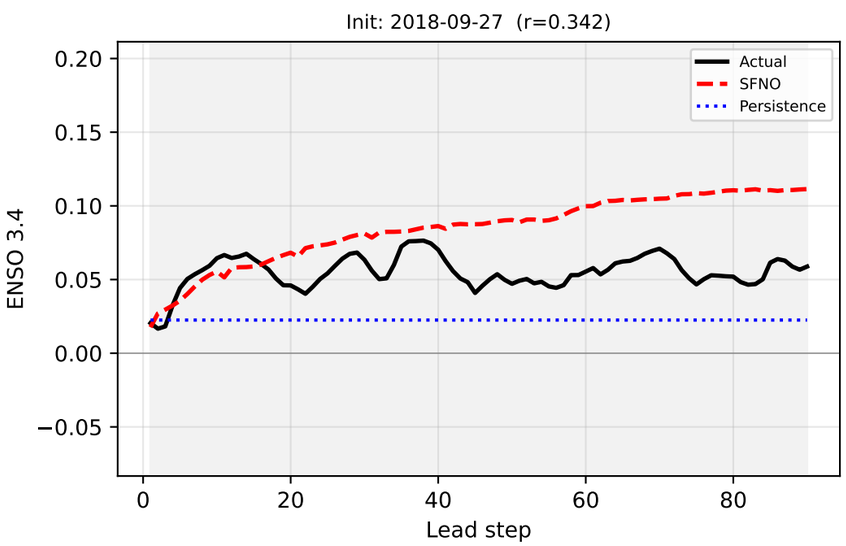



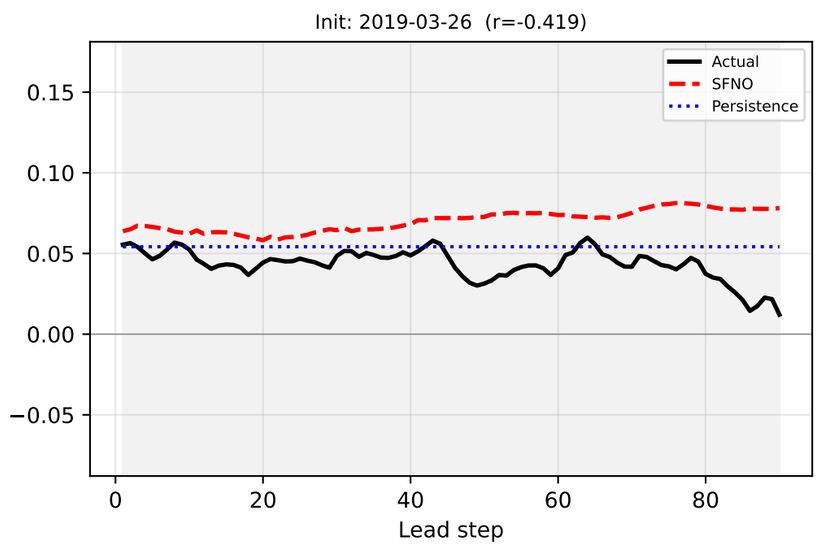



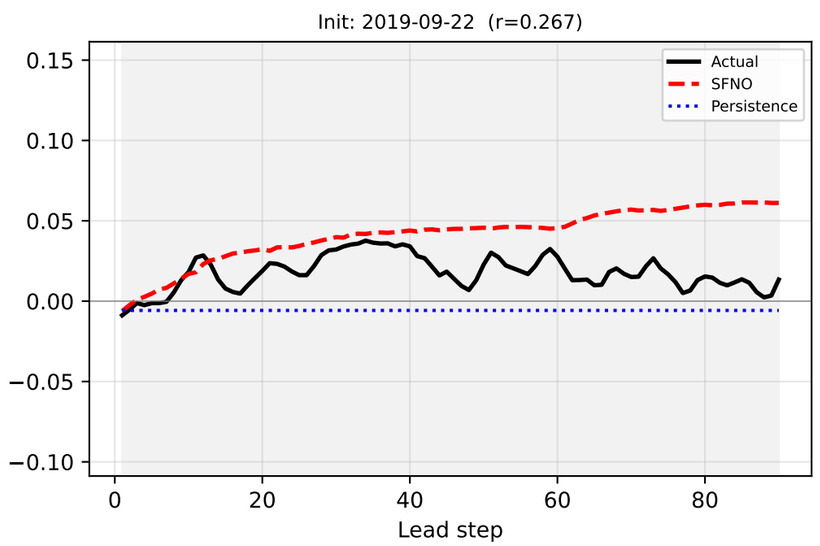



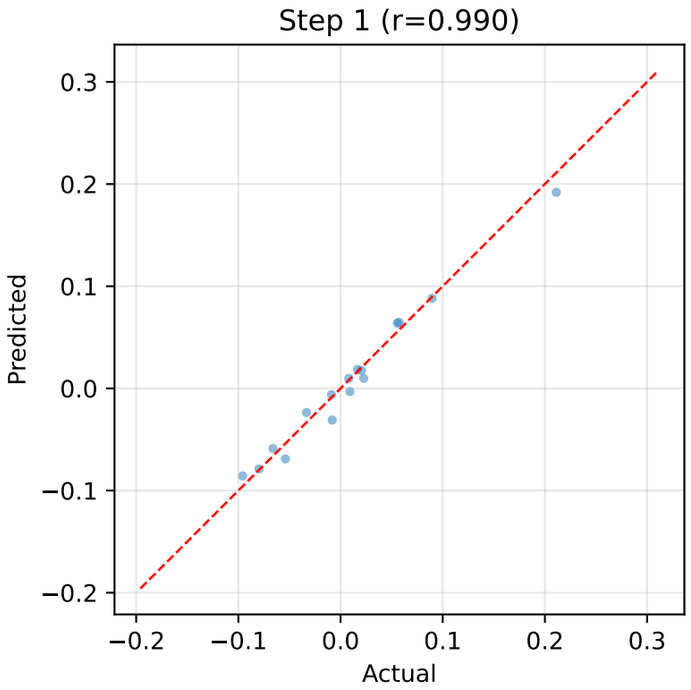



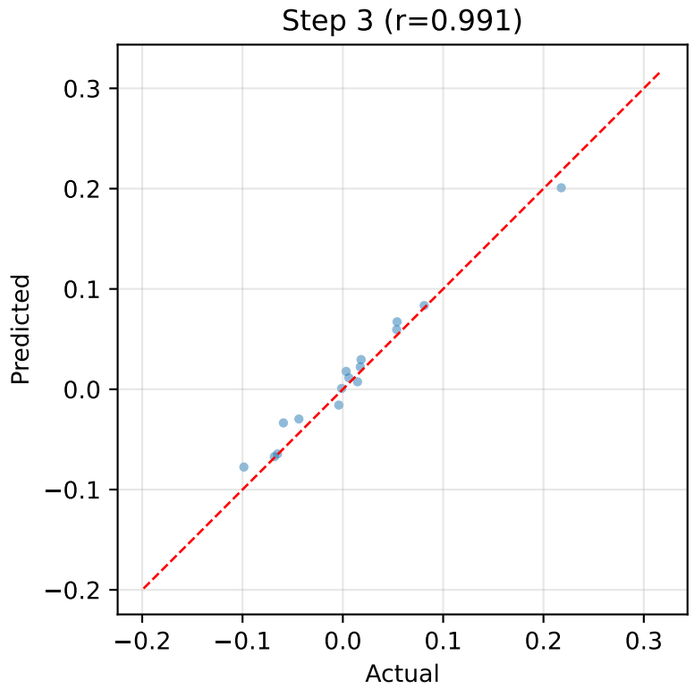



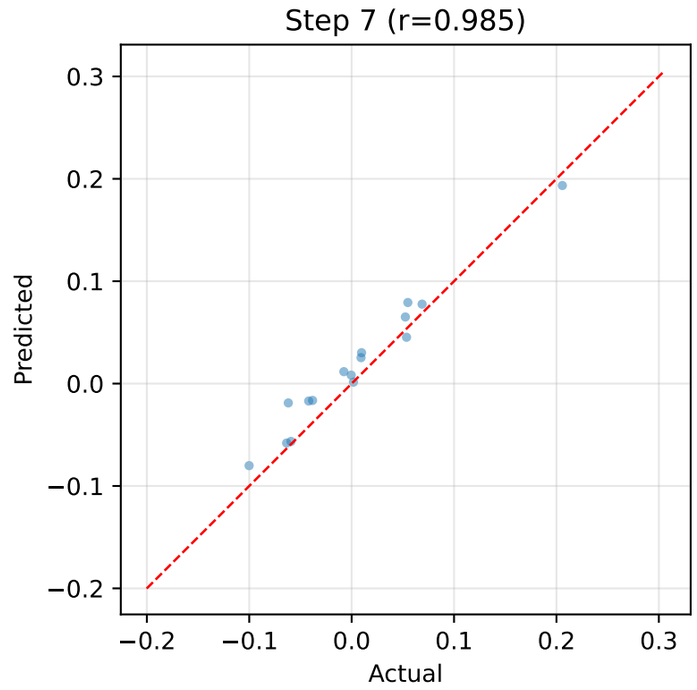



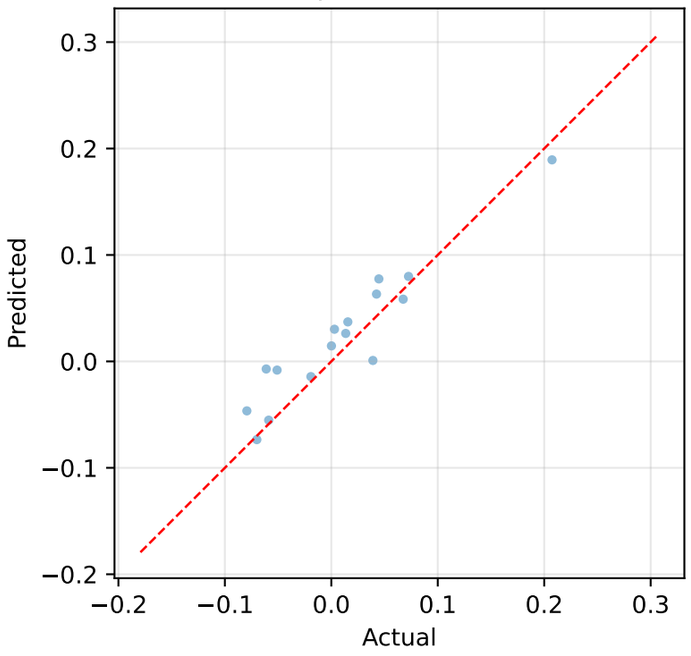



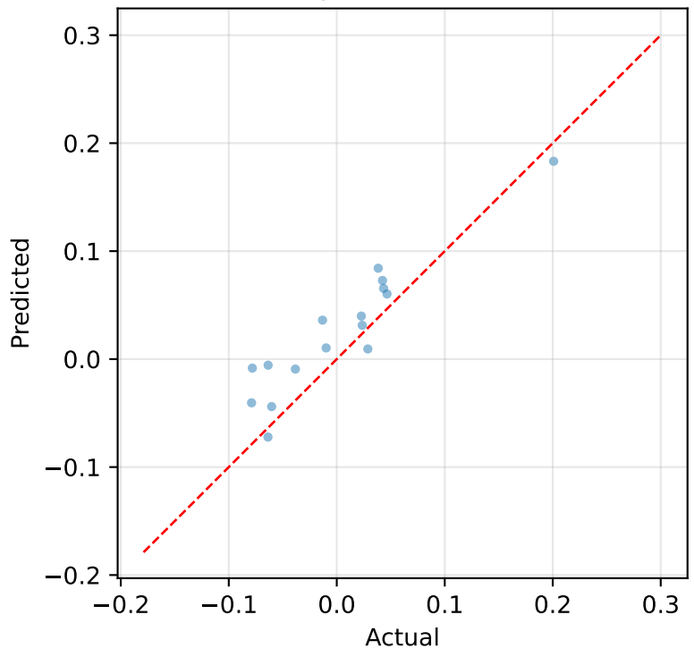
# ENSO 3.4 Scatter — Validation

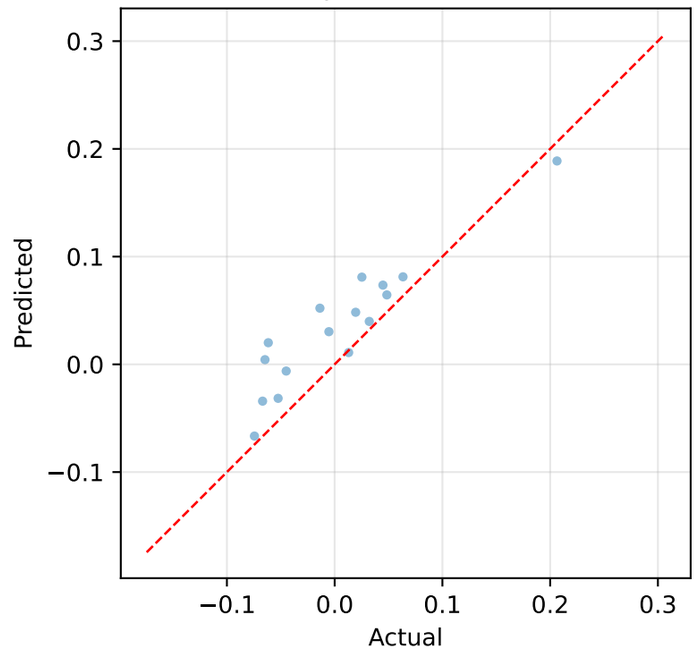


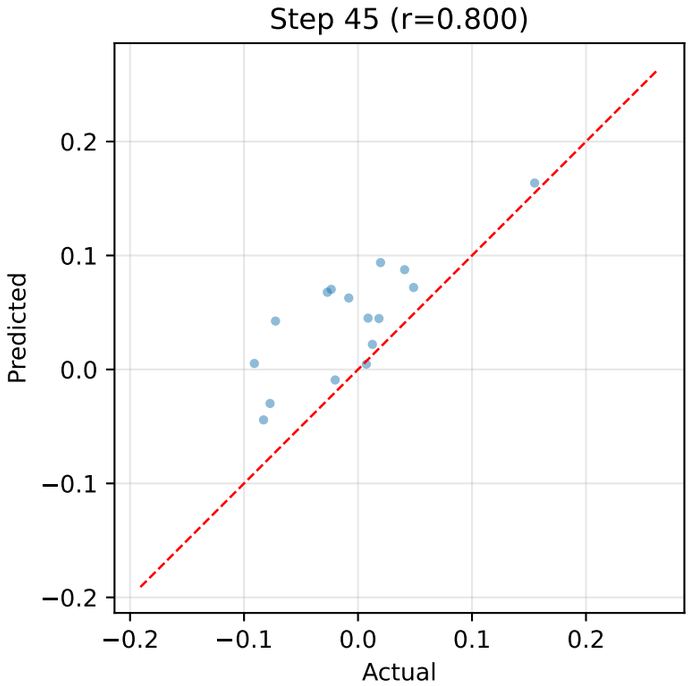



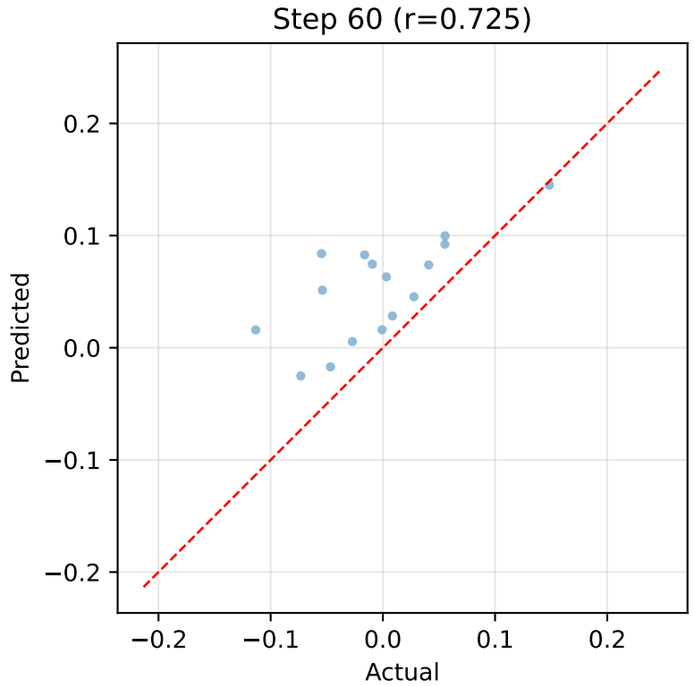



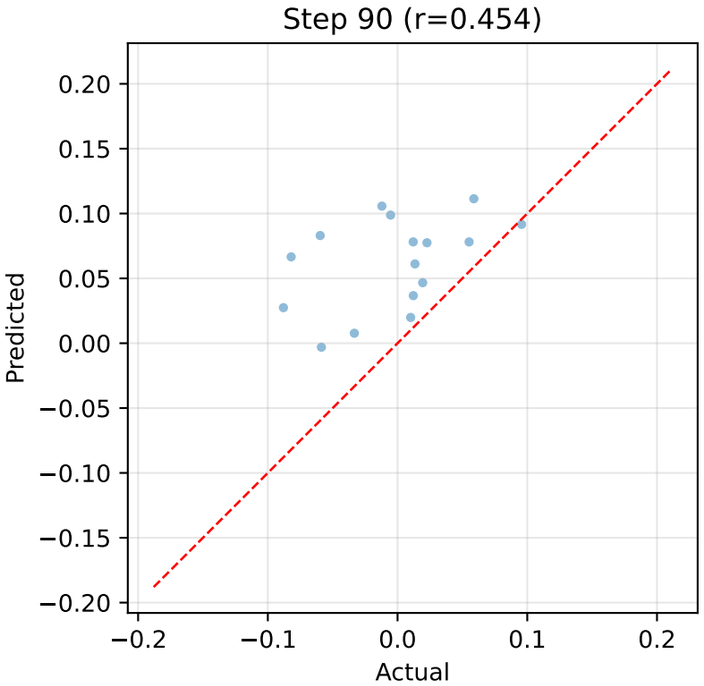



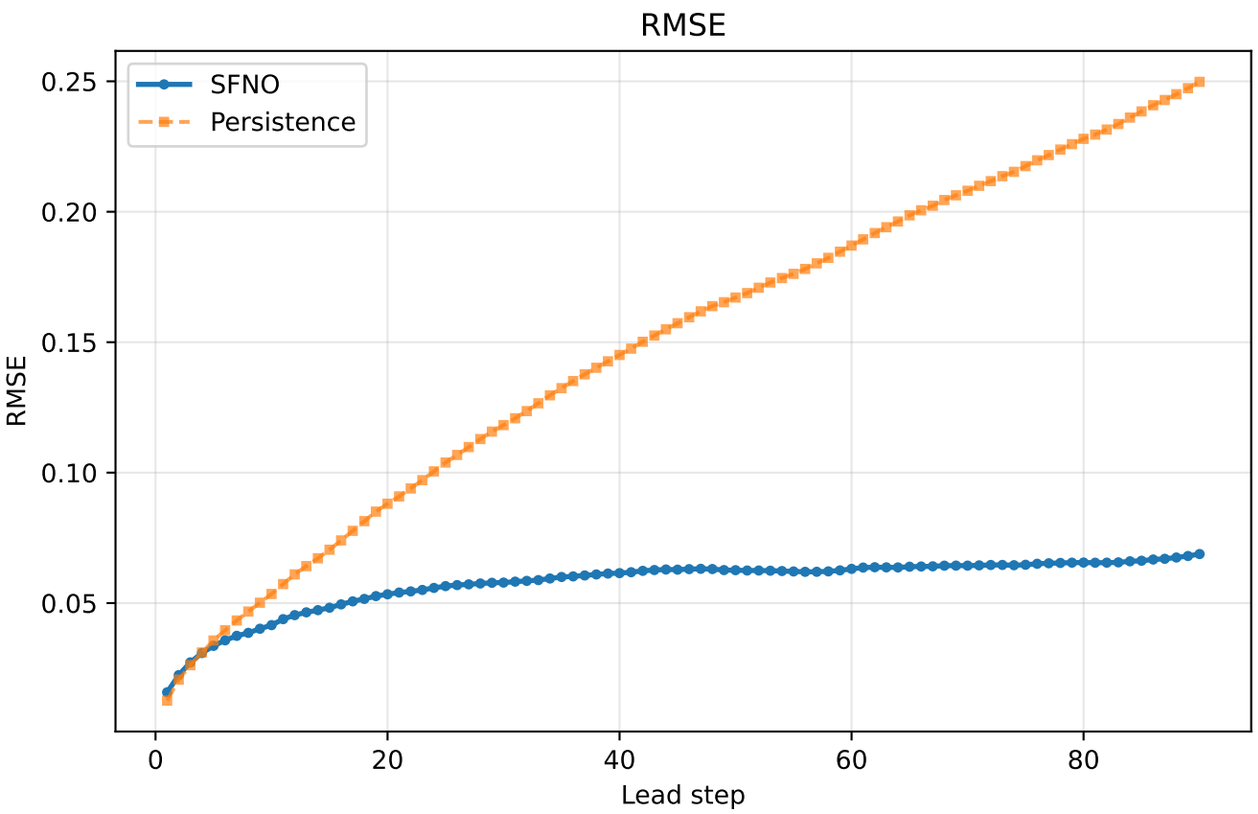



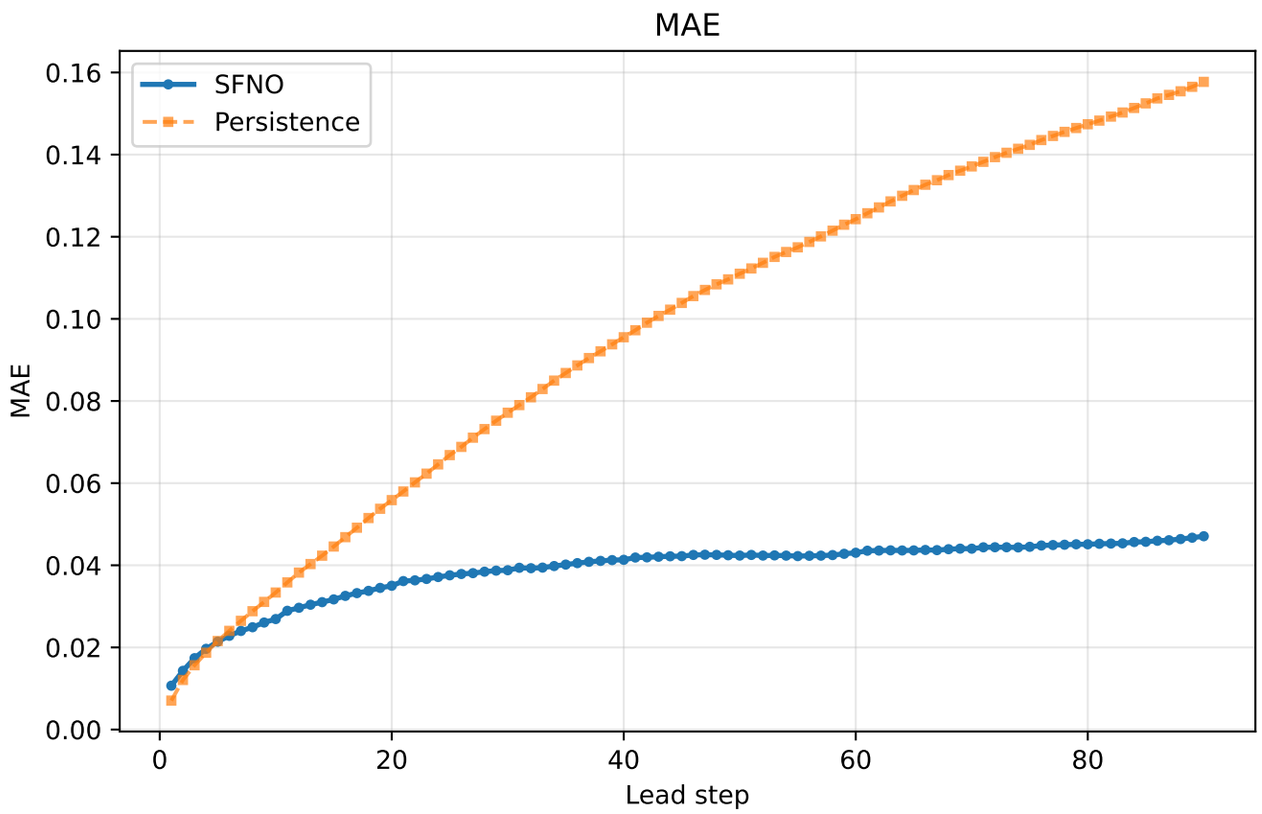



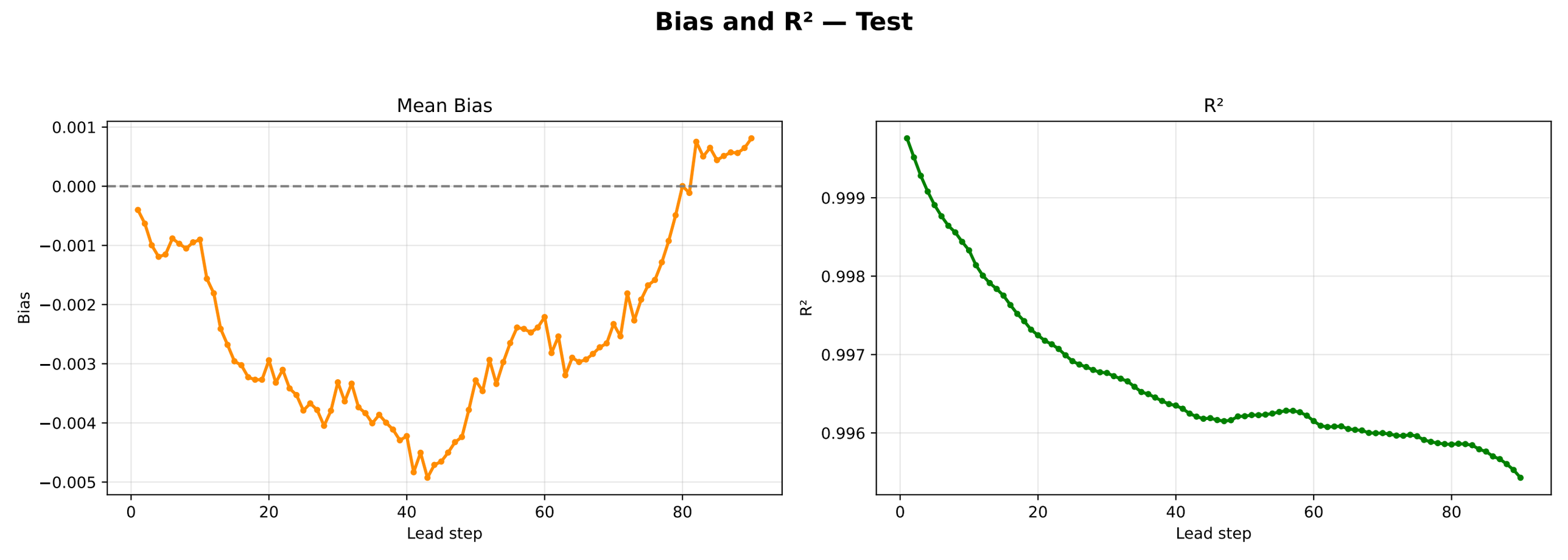
## Metrics by Lead Step — Test

| Lead Step | SFNO RMSE | Persist RMSE | Skill % | SFNO MAE | SFNO Bias | SFNO ACC | Persist ACC | SFNO $R^2$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.0158 | 0.0126 | -25.1% | 0.0107 | -0.0004 | 0.9719 | 0.9796 | 0.9998 |
| 2 | 0.0224 | 0.0206 | -8.4% | 0.0143 | -0.0006 | 0.9407 | 0.9462 | 0.9995 |
| 3 | 0.0273 | 0.0262 | -4.1% | 0.0174 | -0.0010 | 0.9078 | 0.9132 | 0.9993 |
| 4 | 0.0308 | 0.0311 | 0.7% | 0.0197 | -0.0012 | 0.8778 | 0.8780 | 0.9991 |
| 5 | 0.0336 | 0.0357 | 5.7% | 0.0214 | -0.0012 | 0.8506 | 0.8411 | 0.9989 |
| 6 | 0.0357 | 0.0396 | 9.8% | 0.0228 | -0.0009 | 0.8265 | 0.8072 | 0.9988 |
| 7 | 0.0374 | 0.0433 | 13.6% | 0.0240 | -0.0010 | 0.8051 | 0.7728 | 0.9986 |
| 8 | 0.0386 | 0.0467 | 17.5% | 0.0249 | -0.0011 | 0.7868 | 0.7389 | 0.9986 |
| 9 | 0.0402 | 0.0501 | 19.9% | 0.0261 | -0.0009 | 0.7667 | 0.7063 | 0.9984 |
| 10 | 0.0415 | 0.0535 | 22.3% | 0.0269 | -0.0009 | 0.7491 | 0.6743 | 0.9983 |
| 11 | 0.0438 | 0.0573 | 23.5% | 0.0289 | -0.0016 | 0.7237 | 0.6410 | 0.9981 |
| 12 | 0.0454 | 0.0610 | 25.6% | 0.0297 | -0.0018 | 0.7034 | 0.6068 | 0.9980 |
| 13 | 0.0465 | 0.0642 | 27.6% | 0.0304 | -0.0024 | 0.6873 | 0.5769 | 0.9979 |
| 14 | 0.0473 | 0.0672 | 29.6% | 0.0310 | -0.0027 | 0.6741 | 0.5500 | 0.9978 |
| 15 | 0.0482 | 0.0705 | 31.6% | 0.0317 | -0.0030 | 0.6604 | 0.5219 | 0.9978 |
| 16 | 0.0495 | 0.0740 | 33.1% | 0.0325 | -0.0030 | 0.6439 | 0.4938 | 0.9976 |
| 17 | 0.0507 | 0.0777 | 34.8% | 0.0332 | -0.0032 | 0.6288 | 0.4629 | 0.9975 |
| 18 | 0.0516 | 0.0814 | 36.6% | 0.0338 | -0.0033 | 0.6157 | 0.4320 | 0.9974 |
| 19 | 0.0527 | 0.0851 | 38.1% | 0.0345 | -0.0033 | 0.6004 | 0.4030 | 0.9973 |
| 20 | 0.0534 | 0.0881 | 39.4% | 0.0350 | -0.0029 | 0.5857 | 0.3801 | 0.9972 |
| 21 | 0.0540 | 0.0909 | 40.5% | 0.0362 | -0.0033 | 0.5676 | 0.3592 | 0.9972 |
| 22 | 0.0545 | 0.0939 | 42.0% | 0.0363 | -0.0031 | 0.5585 | 0.3356 | 0.9971 |
| 23 | 0.0550 | 0.0971 | 43.3% | 0.0367 | -0.0034 | 0.5496 | 0.3131 | 0.9971 |
| 24 | 0.0558 | 0.1005 | 44.5% | 0.0371 | -0.0035 | 0.5397 | 0.2889 | 0.9970 |
| 25 | 0.0565 | 0.1039 | 45.6% | 0.0376 | -0.0038 | 0.5295 | 0.2670 | 0.9969 |
| 26 | 0.0569 | 0.1068 | 46.7% | 0.0379 | -0.0037 | 0.5199 | 0.2487 | 0.9969 |
| 27 | 0.0572 | 0.1098 | 47.9% | 0.0381 | -0.0038 | 0.5105 | 0.2287 | 0.9968 |
| 28 | 0.0575 | 0.1129 | 49.0% | 0.0384 | -0.0040 | 0.5028 | 0.2138 | 0.9968 |
| 29 | 0.0578 | 0.1157 | 50.1% | 0.0387 | -0.0038 | 0.4929 | 0.2032 | 0.9968 |
| 30 | 0.0579 | 0.1182 | 51.1% | 0.0388 | -0.0033 | 0.4828 | 0.1947 | 0.9968 |
| 31 | 0.0582 | 0.1208 | 51.8% | 0.0394 | -0.0036 | 0.4688 | 0.1854 | 0.9967 |
| 32 | 0.0585 | 0.1236 | 52.7% | 0.0393 | -0.0033 | 0.4605 | 0.1758 | 0.9967 |
| 33 | 0.0588 | 0.1266 | 53.5% | 0.0394 | -0.0037 | 0.4513 | 0.1635 | 0.9967 |
| 34 | 0.0594 | 0.1296 | 54.2% | 0.0398 | -0.0038 | 0.4418 | 0.1532 | 0.9966 |
| 35 | 0.0600 | 0.1324 | 54.7% | 0.0402 | -0.0040 | 0.4297 | 0.1445 | 0.9965 |
| 36 | 0.0602 | 0.1351 | 55.4% | 0.0405 | -0.0039 | 0.4180 | 0.1347 | 0.9965 |
| 37 | 0.0606 | 0.1377 | 56.0% | 0.0408 | -0.0040 | 0.4072 | 0.1282 | 0.9965 |
| 38 | 0.0610 | 0.1402 | 56.5% | 0.0411 | -0.0041 | 0.4000 | 0.1243 | 0.9964 |
| 39 | 0.0613 | 0.1427 | 57.0% | 0.0413 | -0.0043 | 0.3945 | 0.1209 | 0.9964 |
| 40 | 0.0615 | 0.1451 | 57.6% | 0.0413 | -0.0042 | 0.3889 | 0.1145 | 0.9963 |
| 41 | 0.0618 | 0.1475 | 58.1% | 0.0419 | -0.0048 | 0.3787 | 0.1073 | 0.9963 |
| 42 | 0.0624 | 0.1502 | 58.5% | 0.0419 | -0.0045 | 0.3709 | 0.1002 | 0.9962 |
| 43 | 0.0627 | 0.1526 | 58.9% | 0.0421 | -0.0049 | 0.3638 | 0.0960 | 0.9962 |
| 44 | 0.0629 | 0.1549 | 59.4% | 0.0422 | -0.0047 | 0.3562 | 0.0912 | 0.9962 |
| 45 | 0.0628 | 0.1573 | 60.1% | 0.0422 | -0.0047 | 0.3478 | 0.0831 | 0.9962 |
| 46 | 0.0630 | 0.1596 | 60.5% | 0.0425 | -0.0045 | 0.3404 | 0.0757 | 0.9962 |
| 47 | 0.0631 | 0.1618 | 61.0% | 0.0426 | -0.0043 | 0.3362 | 0.0700 | 0.9962 |
| 48 | 0.0630 | 0.1638 | 61.5% | 0.0425 | -0.0042 | 0.3336 | 0.0685 | 0.9962 |
| 49 | 0.0626 | 0.1653 | 62.1% | 0.0424 | -0.0038 | 0.3311 | 0.0718 | 0.9962 |
| 50 | 0.0626 | 0.1671 | 62.5% | 0.0424 | -0.0033 | 0.3254 | 0.0747 | 0.9962 |
| 51 | 0.0625 | 0.1688 | 63.0% | 0.0425 | -0.0035 | 0.3184 | 0.0768 | 0.9962 |
| 52 | 0.0625 | 0.1709 | 63.4% | 0.0424 | -0.0029 | 0.3132 | 0.0753 | 0.9962 |
| 53 | 0.0624 | 0.1729 | 63.9% | 0.0424 | -0.0033 | 0.3081 | 0.0730 | 0.9962 |
| 54 | 0.0623 | 0.1746 | 64.3% | 0.0424 | -0.0030 | 0.3049 | 0.0736 | 0.9962 |
| 55 | 0.0621 | 0.1762 | 64.7% | 0.0422 | -0.0027 | 0.3021 | 0.0758 | 0.9963 |
| 56 | 0.0620 | 0.1781 | 65.2% | 0.0423 | -0.0024 | 0.3002 | 0.0736 | 0.9963 |
| 57 | 0.0620 | 0.1802 | 65.6% | 0.0423 | -0.0024 | 0.2976 | 0.0694 | 0.9963 |
| 58 | 0.0622 | 0.1823 | 65.9% | 0.0425 | -0.0025 | 0.2951 | 0.0654 | 0.9963 |
| 59 | 0.0625 | 0.1847 | 66.1% | 0.0428 | -0.0024 | 0.2900 | 0.0584 | 0.9962 |
| 60 | 0.0631 | 0.1870 | 66.3% | 0.0431 | -0.0022 | 0.2842 | 0.0535 | 0.9962 |
| 61 | 0.0636 | 0.1894 | 66.4% | 0.0436 | -0.0028 | 0.2778 | 0.0470 | 0.9961 |
| 62 | 0.0637 | 0.1918 | 66.8% | 0.0436 | -0.0025 | 0.2785 | 0.0385 | 0.9961 |
| 63 | 0.0637 | 0.1941 | 67.2% | 0.0436 | -0.0032 | 0.2804 | 0.0316 | 0.9961 |
| 64 | 0.0637 | 0.1963 | 67.6% | 0.0436 | -0.0029 | 0.2795 | 0.0248 | 0.9961 |
| 65 | 0.0639 | 0.1986 | 67.8% | 0.0436 | -0.0030 | 0.2775 | 0.0192 | 0.9961 |
| 66 | 0.0640 | 0.2006 | 68.1% | 0.0437 | -0.0029 | 0.2759 | 0.0171 | 0.9960 |
| 67 | 0.0641 | 0.2023 | 68.3% | 0.0437 | -0.0028 | 0.2750 | 0.0183 | 0.9960 |
| 68 | 0.0643 | 0.2045 | 68.5% | 0.0439 | -0.0027 | 0.2711 | 0.0135 | 0.9960 |
| 69 | 0.0644 | 0.2063 | 68.8% | 0.0441 | -0.0027 | 0.2684 | 0.0125 | 0.9960 |
| 70 | 0.0644 | 0.2080 | 69.1% | 0.0440 | -0.0023 | 0.2627 | 0.0122 | 0.9960 |
| 71 | 0.0645 | 0.2099 | 69.3% | 0.0444 | -0.0025 | 0.2554 | 0.0110 | 0.9960 |
| 72 | 0.0646 | 0.2117 | 69.5% | 0.0444 | -0.0018 | 0.2505 | 0.0120 | 0.9960 |
| 73 | 0.0646 | 0.2136 | 69.8% | 0.0444 | -0.0023 | 0.2497 | 0.0109 | 0.9960 |
| 74 | 0.0645 | 0.2153 | 70.0% | 0.0443 | -0.0019 | 0.2499 | 0.0133 | 0.9960 |
| 75 | 0.0647 | 0.2175 | 70.3% | 0.0445 | -0.0017 | 0.2493 | 0.0086 | 0.9960 |
| 76 | 0.0651 | 0.2197 | 70.4% | 0.0448 | -0.0016 | 0.2473 | 0.0050 | 0.9959 |
| 77 | 0.0653 | 0.2217 | 70.6% | 0.0449 | -0.0013 | 0.2455 | 0.0049 | 0.9959 |
| 78 | 0.0654 | 0.2238 | 70.8% | 0.0450 | -0.0009 | 0.2439 | 0.0037 | 0.9959 |
| 79 | 0.0655 | 0.2259 | 71.0% | 0.0451 | -0.0005 | 0.2413 | 0.0019 | 0.9959 |
| 80 | 0.0655 | 0.2279 | 71.2% | 0.0451 | 0.0000 | 0.2376 | -0.0015 | 0.9959 |
| 81 | 0.0655 | 0.2296 | 71.5% | 0.0452 | -0.0001 | 0.2334 | -0.0020 | 0.9959 |
| 82 | 0.0655 | 0.2315 | 71.7% | 0.0453 | 0.0008 | 0.2302 | -0.0056 | 0.9959 |
| 83 | 0.0656 | 0.2336 | 71.9% | 0.0453 | 0.0005 | 0.2308 | -0.0076 | 0.9958 |
| 84 | 0.0660 | 0.2361 | 72.0% | 0.0457 | 0.0006 | 0.2319 | -0.0090 | 0.9958 |
| 85 | 0.0662 | 0.2385 | 72.2% | 0.0457 | 0.0004 | 0.2342 | -0.0119 | 0.9958 |
| 86 | 0.0667 | 0.2408 | 72.3% | 0.0460 | 0.0005 | 0.2328 | -0.0147 | 0.9957 |
| 87 | 0.0670 | 0.2428 | 72.4% | 0.0461 | 0.0006 | 0.2302 | -0.0142 | 0.9957 |
| 88 | 0.0675 | 0.2451 | 72.5% | 0.0464 | 0.0006 | 0.2271 | -0.0133 | 0.9956 |
| 89 | 0.0680 | 0.2474 | 72.5% | 0.0467 | 0.0006 | 0.2248 | -0.0151 | 0.9955 |
| 90 | 0.0688 | 0.2498 | 72.5% | 0.0471 | 0.0008 | 0.2190 | -0.0180 | 0.9954 |

# Forecast Skill vs Lead Step — Test

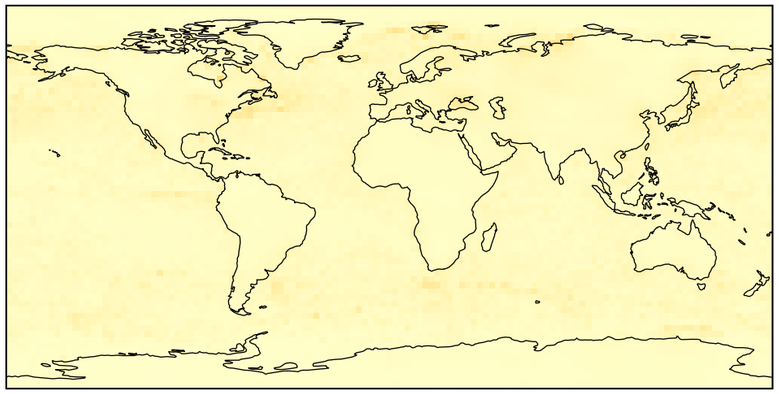


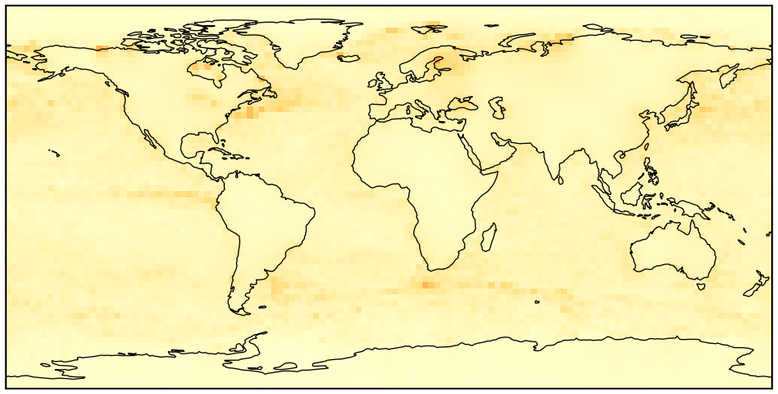

Bias and R² — Test
Mean Bias
Bias
0.001
0.000
−0.001
−0.002
−0.003
−0.004
−0.005
0
20
40
60
80
Lead step
R²
R²
0.999
0.998
0.997
0.996
0
20
40
60
80
Lead step

# Spatial RMSE (all grid) — Test

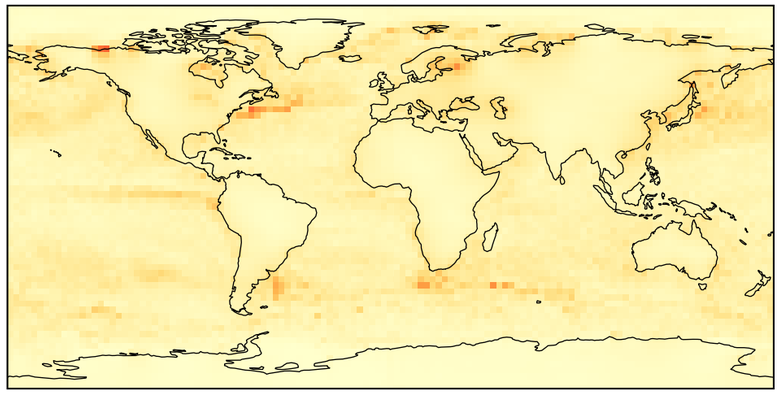
Step 1



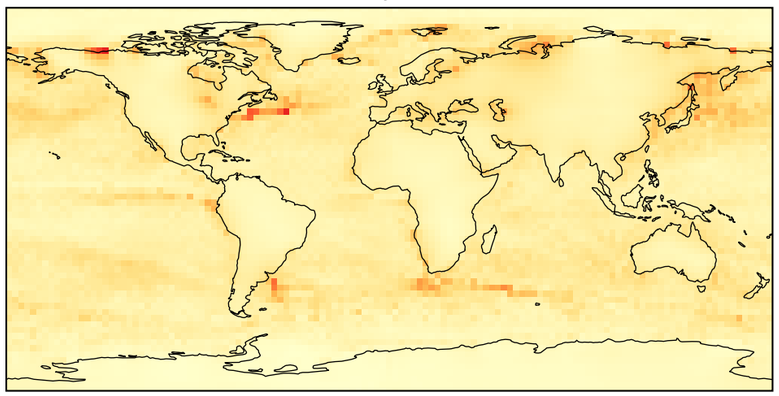
Step 3



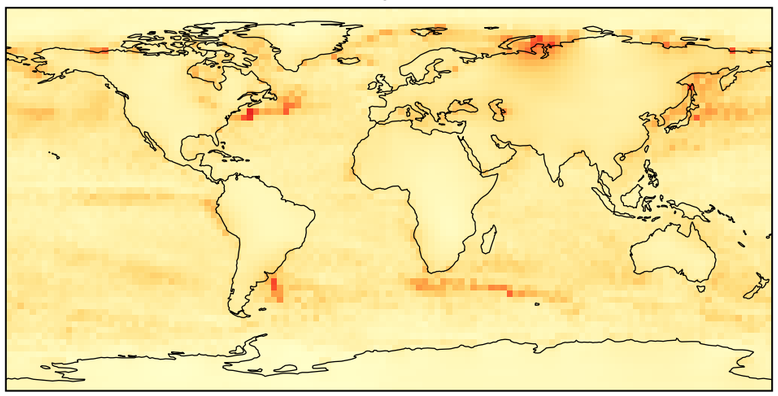
Step 7



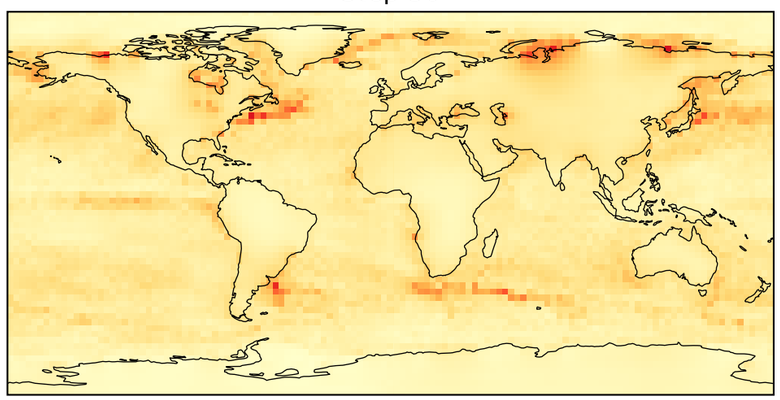
Step 14



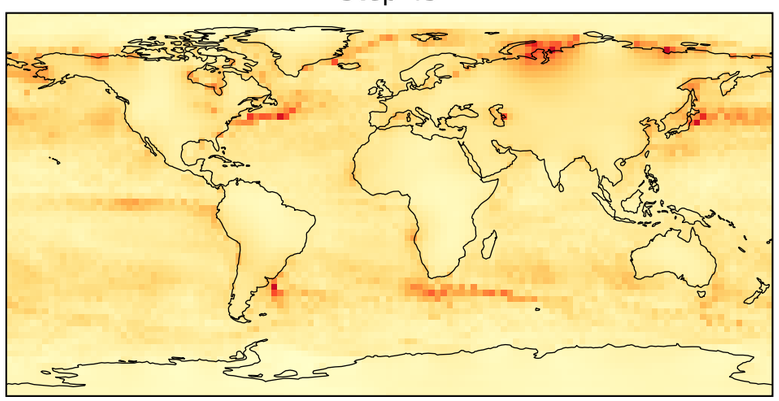
Step 21



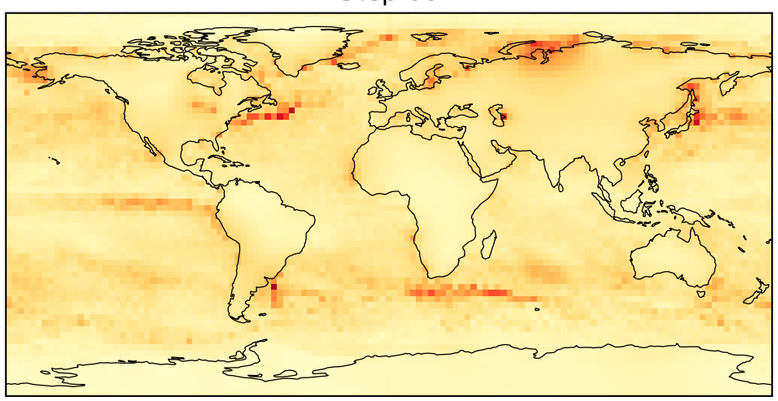
Step 30



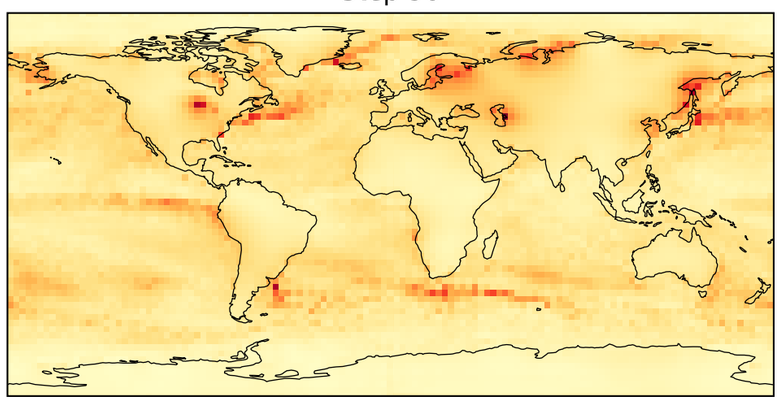
Step 45



Step 60

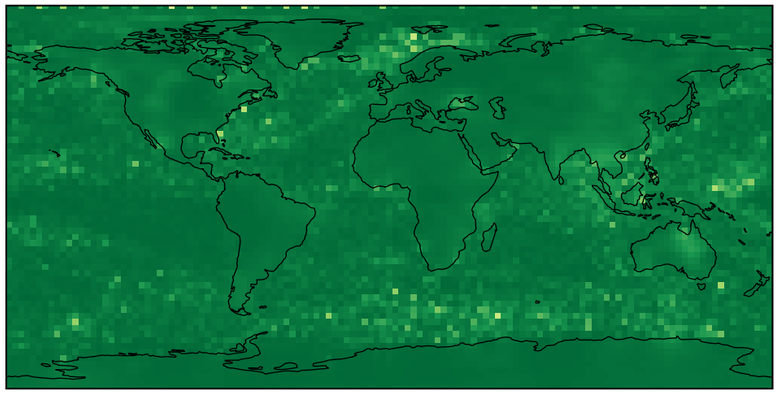


Step 90

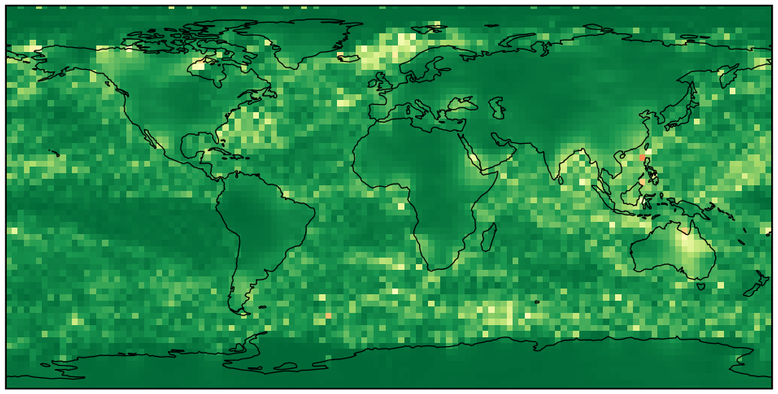


# Spatial ACC (all grid) — Test

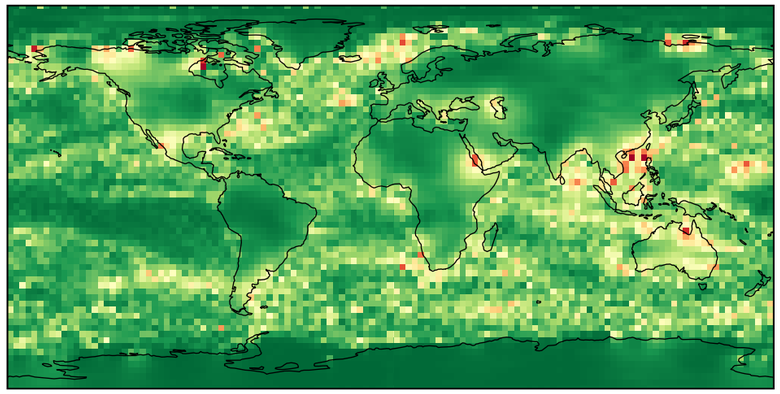
Step 1



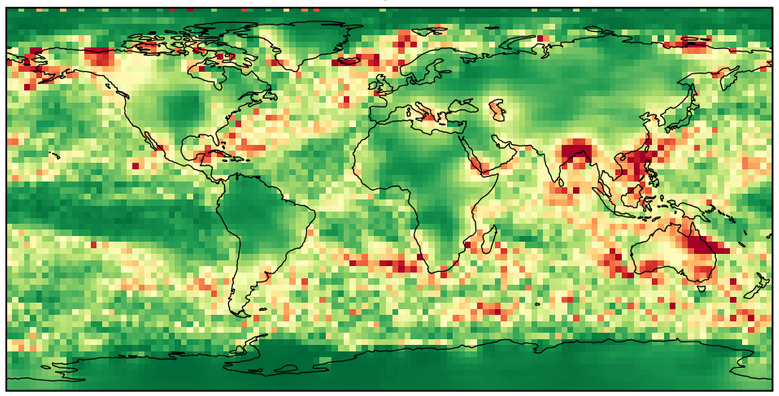
Step 3



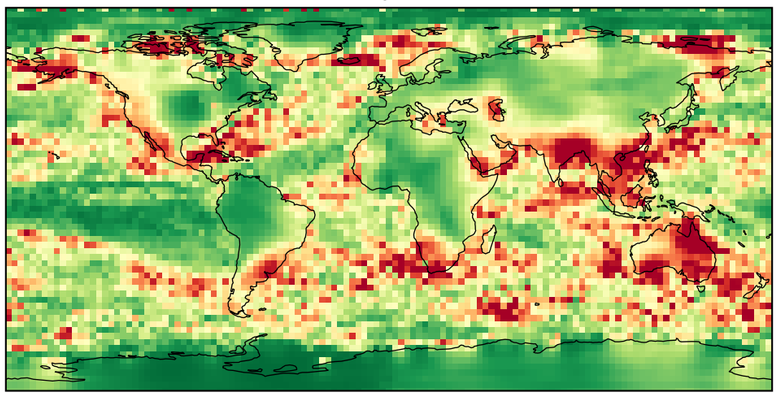
Step 7



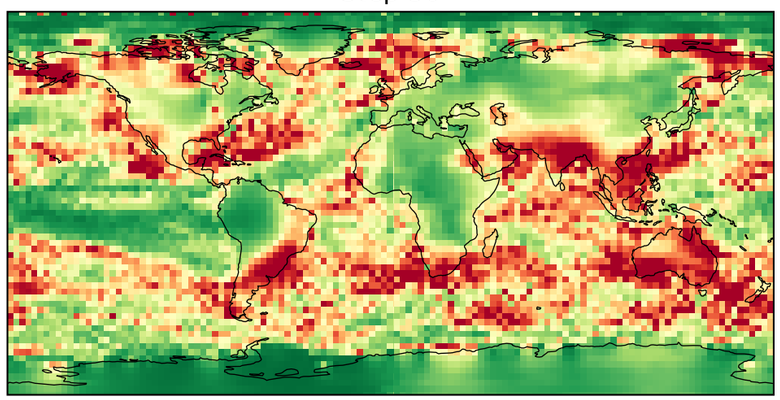
Step 14



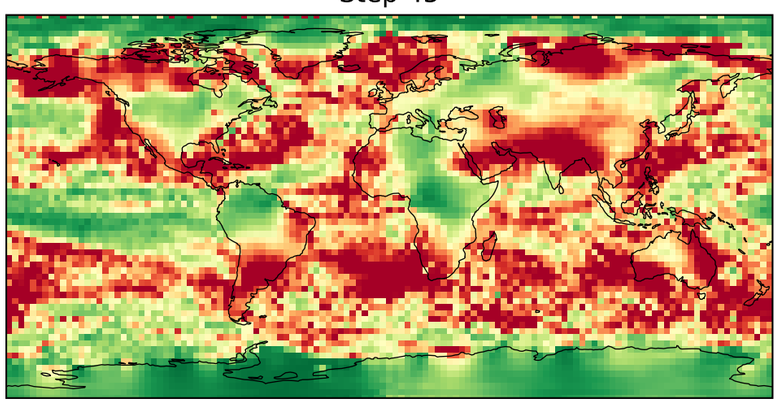
Step 21



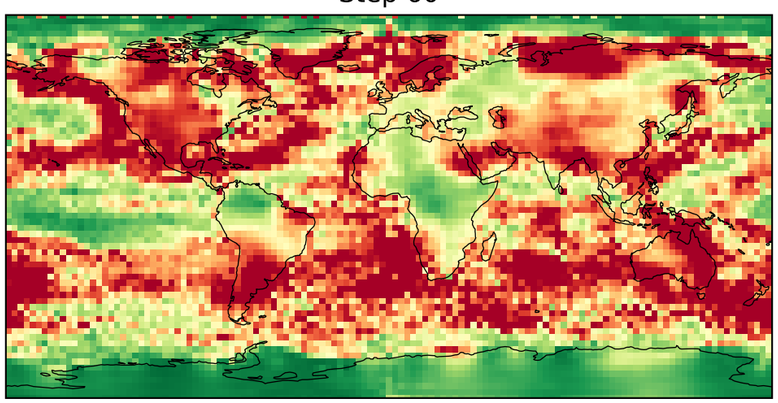
Step 30



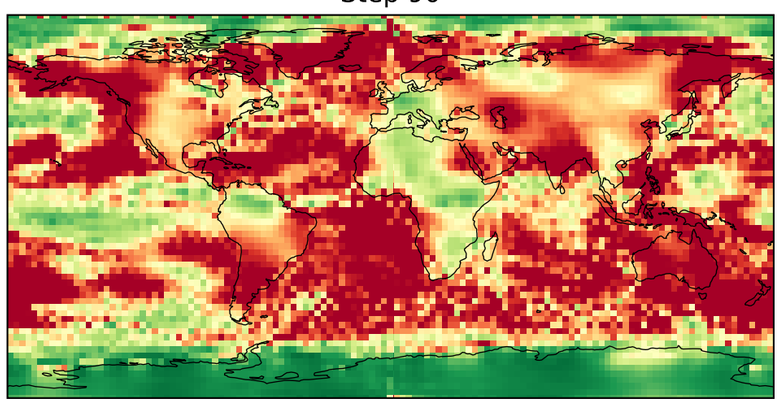
Step 45

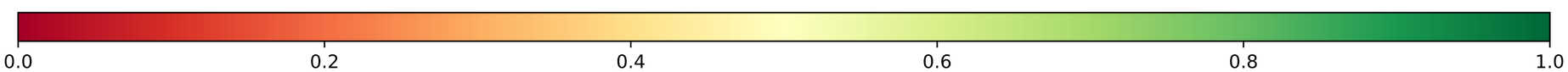


Step 60



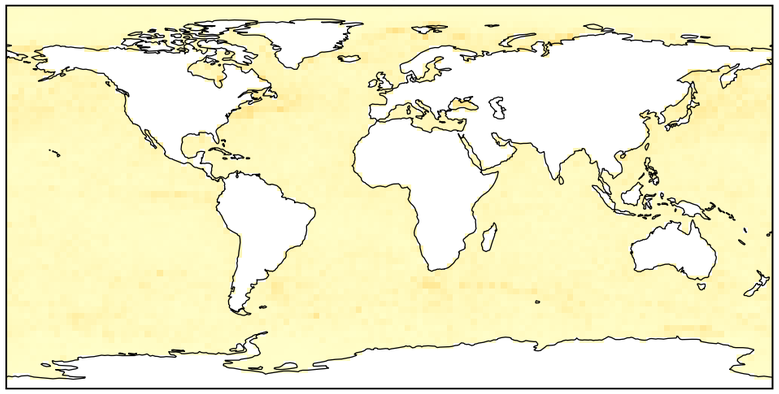
Step 90



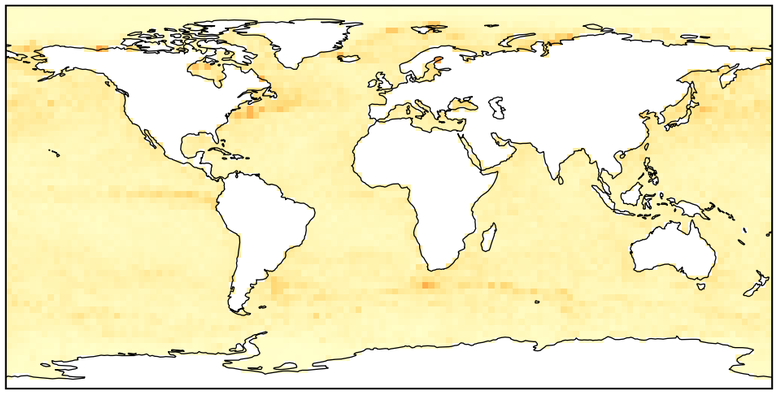


# Spatial RMSE (ocean only) — Test

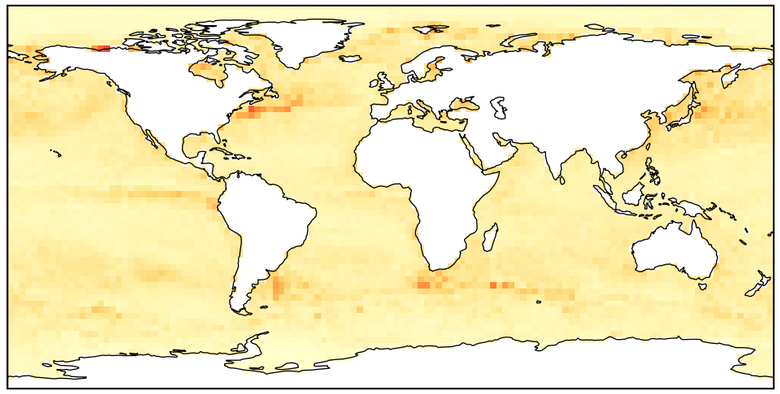
Step 1



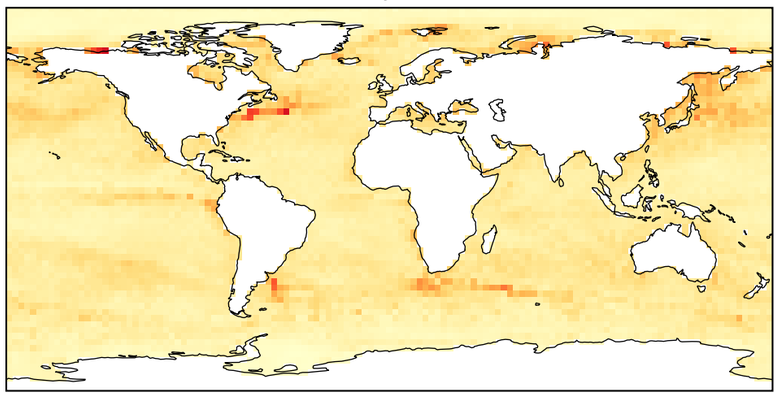
Step 3



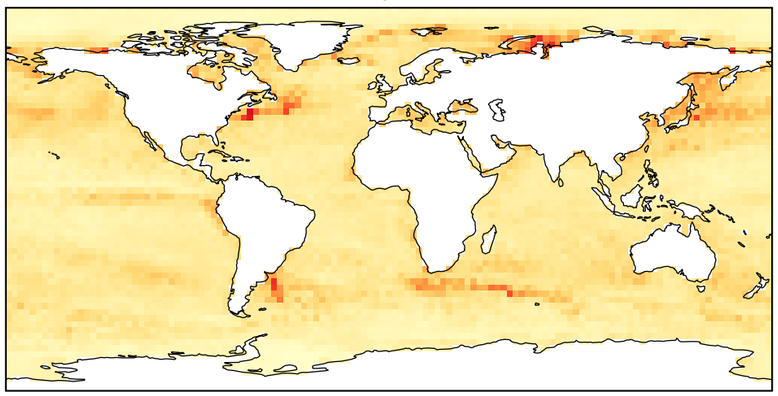
Step 7



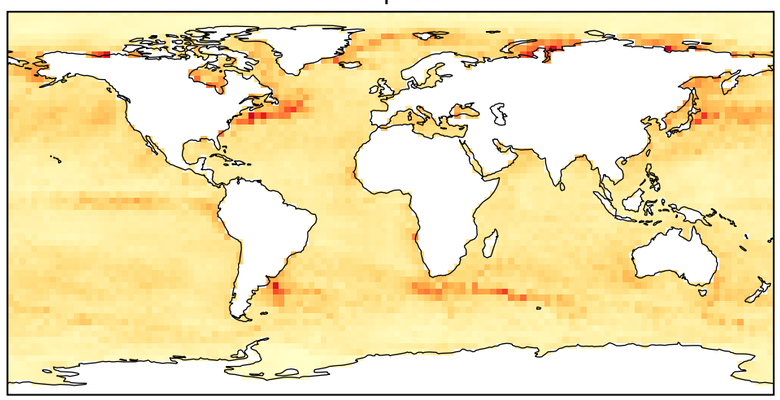
Step 14



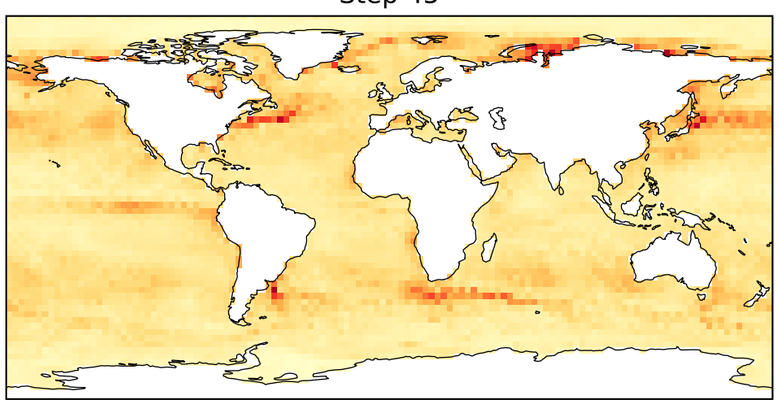
Step 21



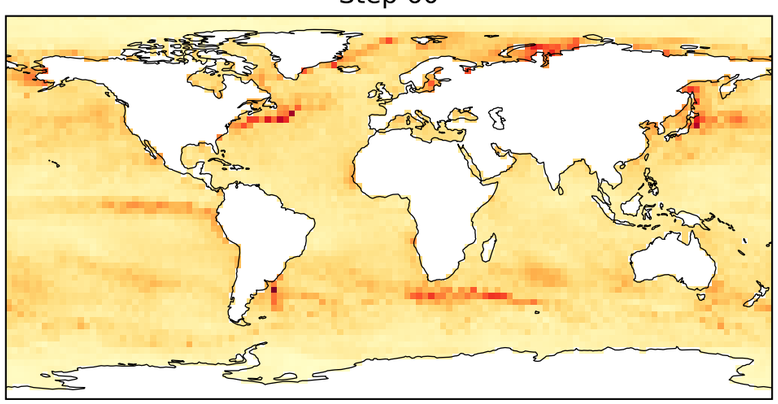
Step 30



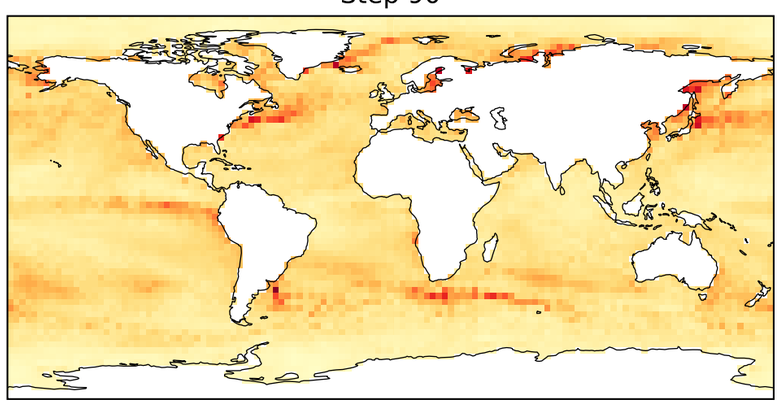
Step 45



Step 60

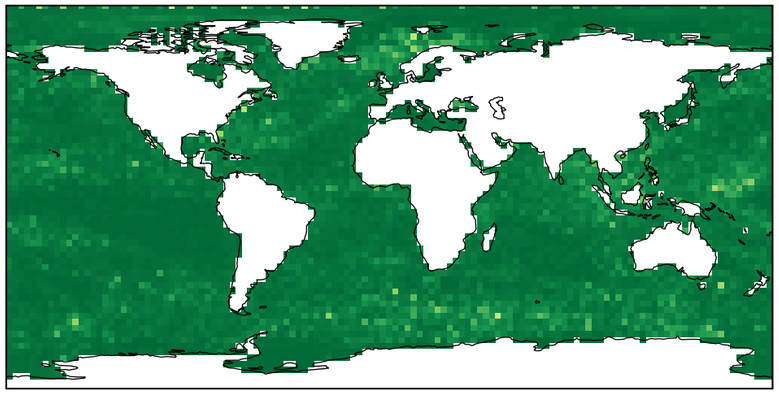


Step 90

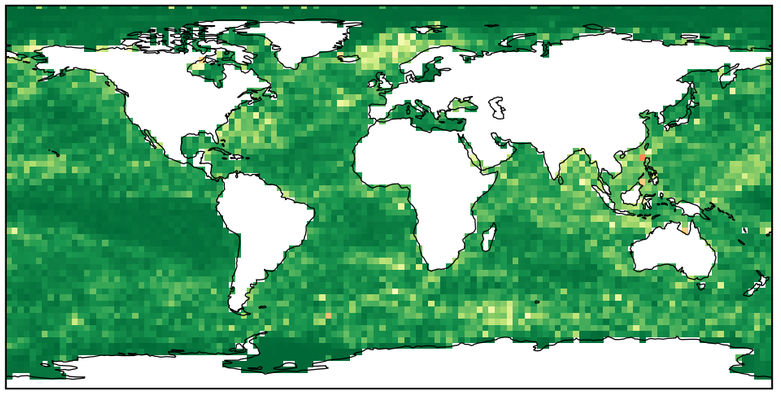


# Spatial ACC (ocean only) — Test

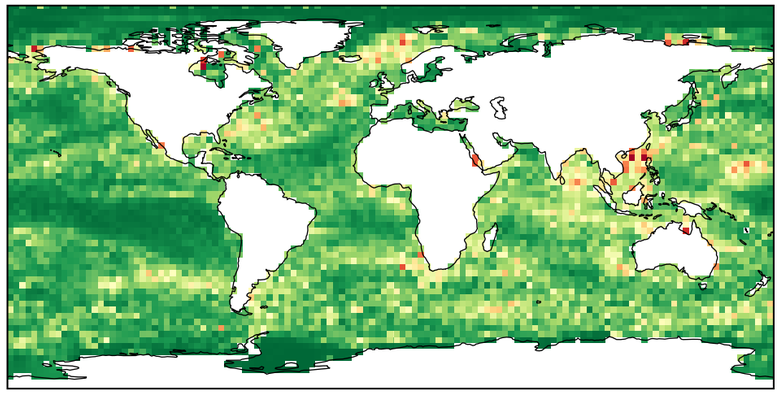
Step 1



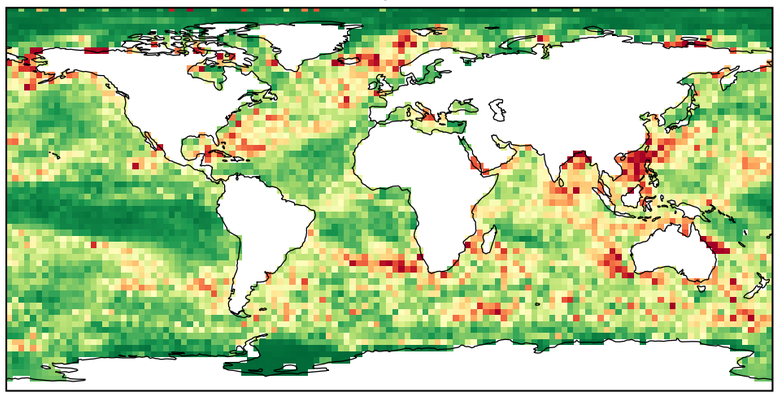
Step 3



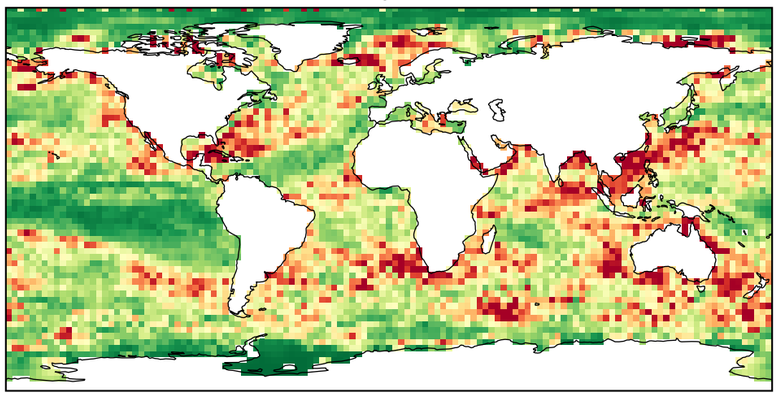
Step 7



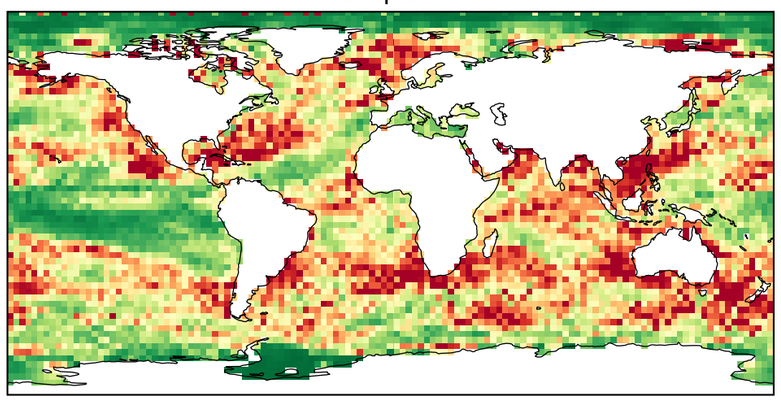
Step 14



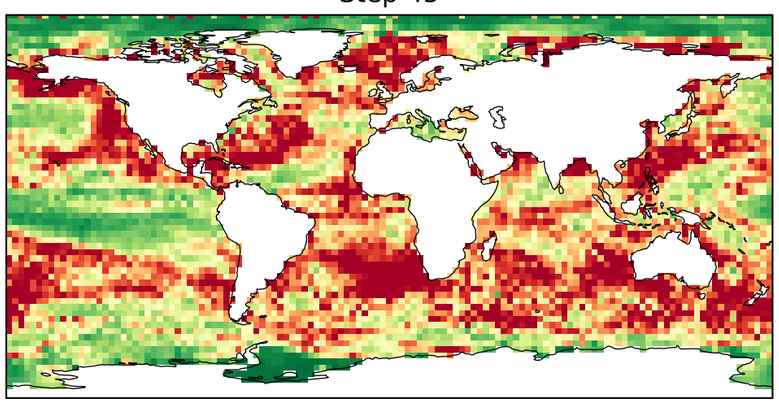
Step 21



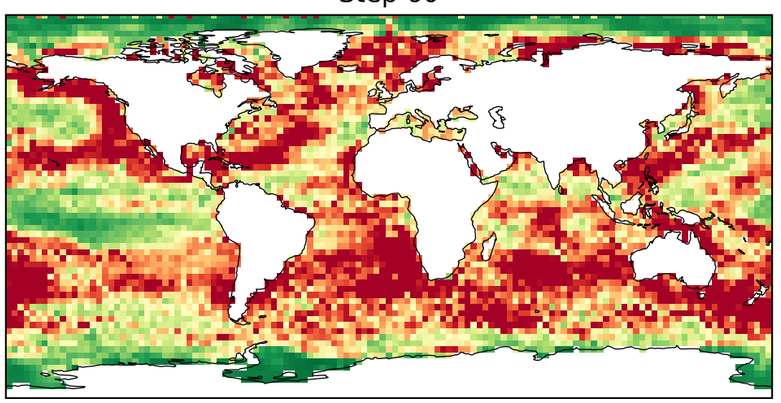
Step 30



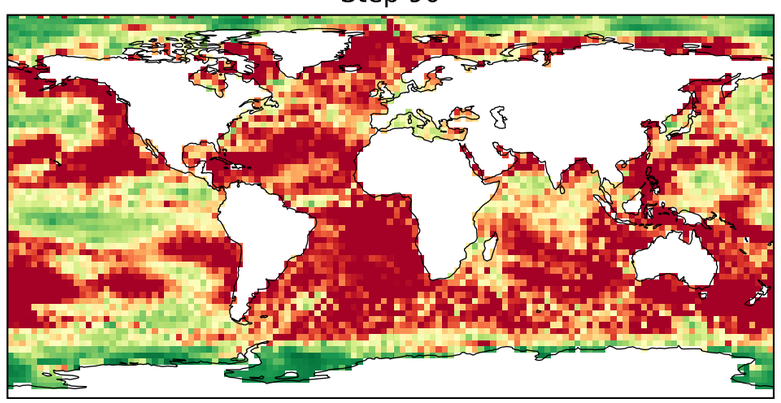
Step 45

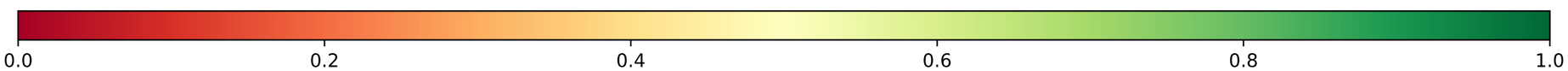


Step 60



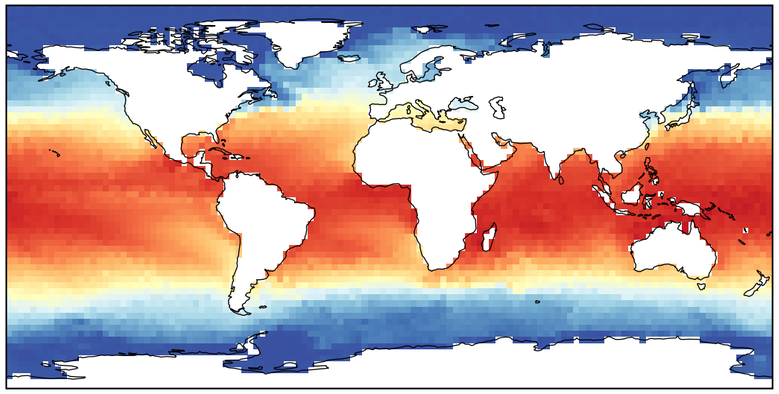
Step 90



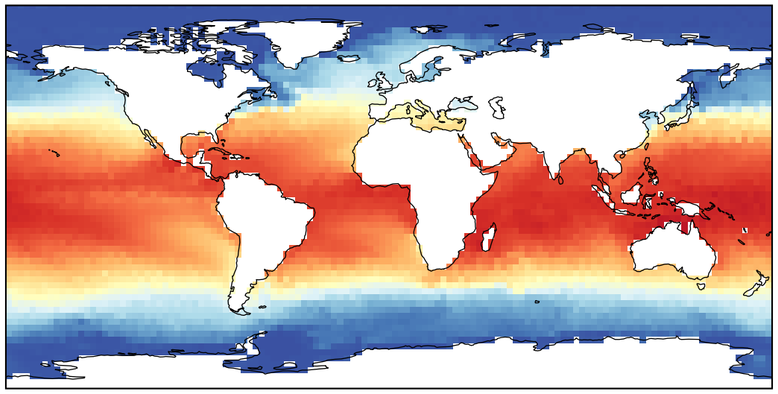


# Chain init: 2020-01-11 — Test (°C)

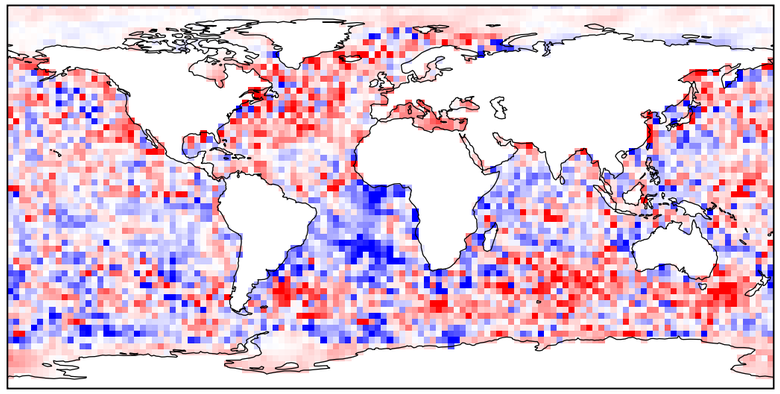
Actual — Step 1 (°C)



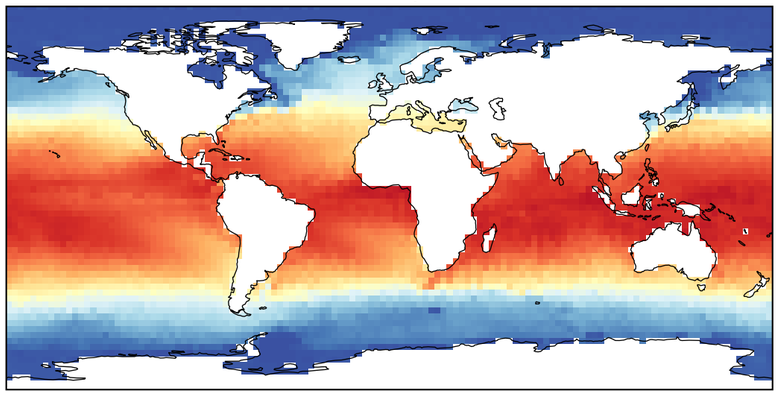
SFNO — Step 1 (°C)



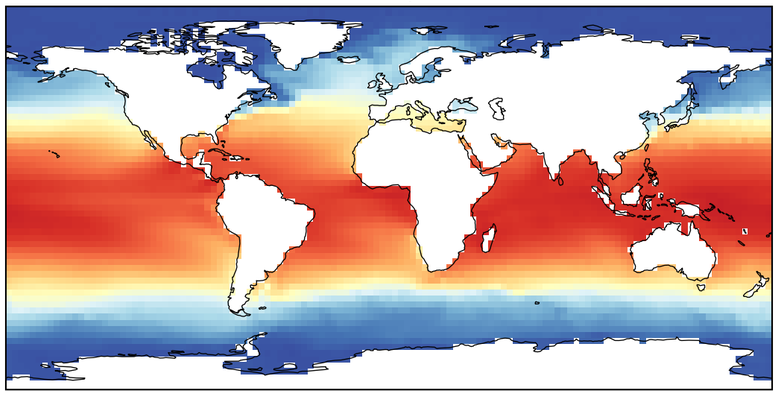
Error — Step 1 (°C)



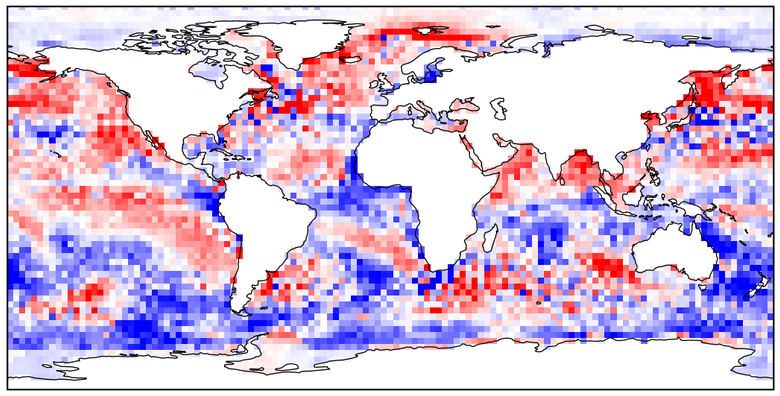
Actual — Step 46 (°C)



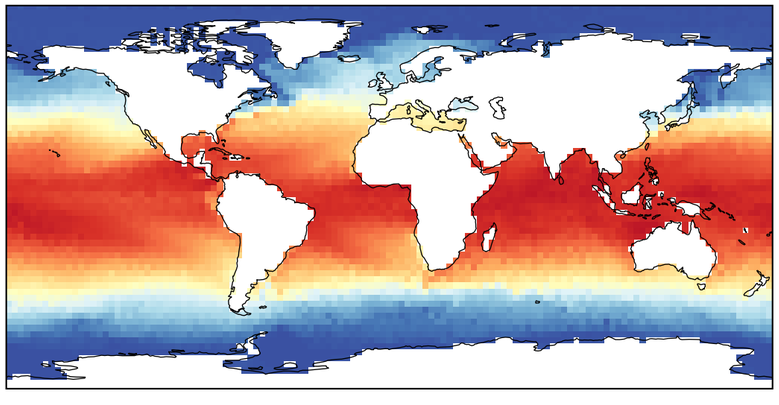
SFNO — Step 46 (°C)



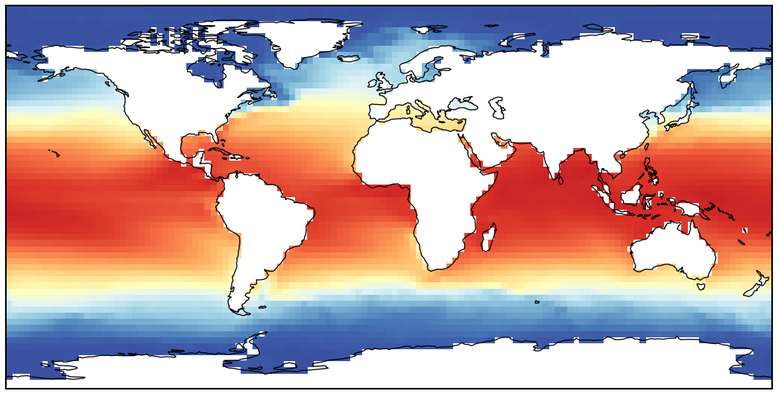
Error — Step 46 (°C)



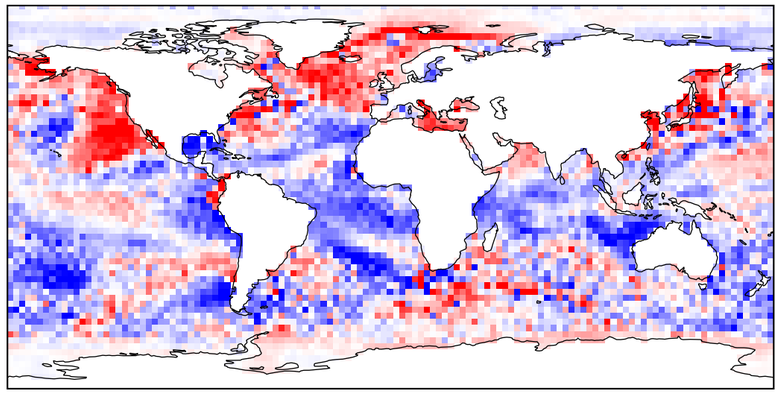
Actual — Step 90 (°C)

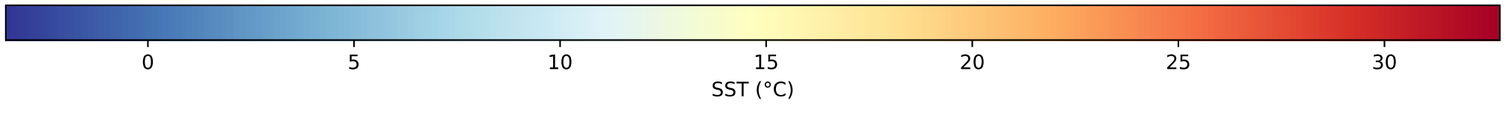


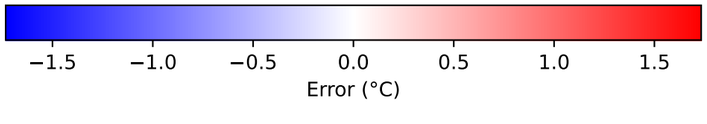
SFNO — Step 90 (°C)



Error — Step 90 (°C)

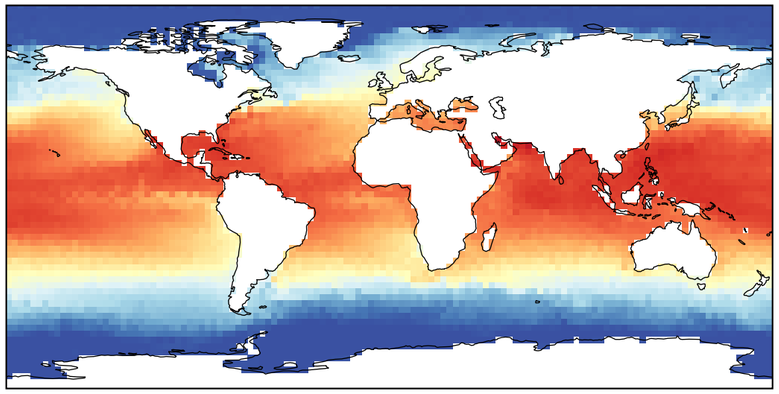




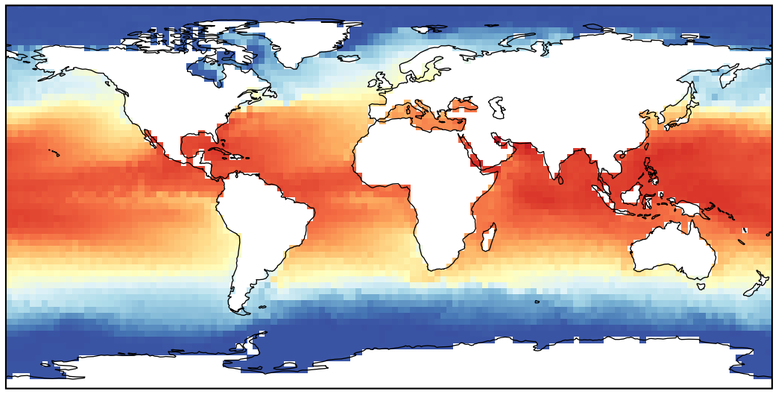



# Chain init: 2020-07-09 — Test (°C)

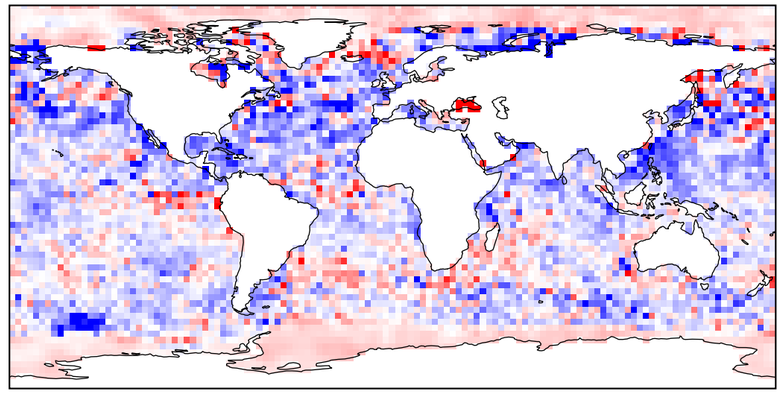
Actual — Step 1 (°C)



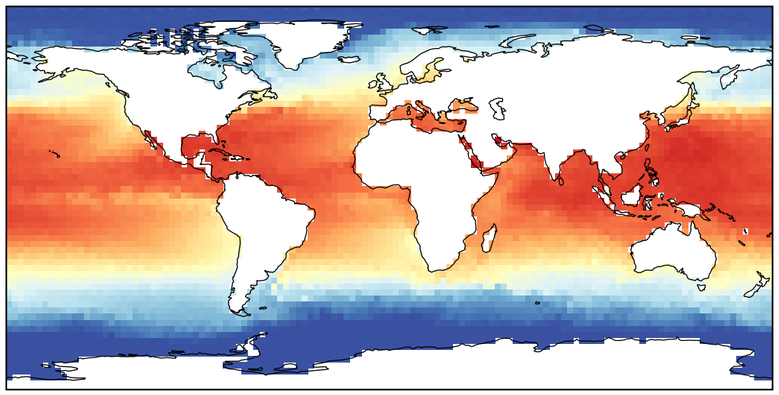
SFNO — Step 1 (°C)



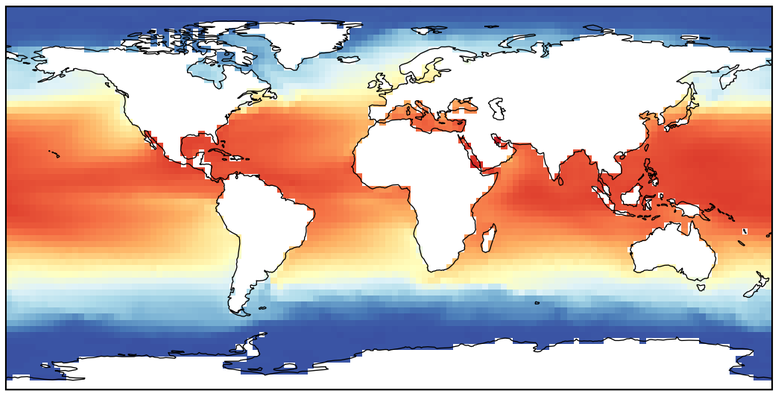
Error — Step 1 (°C)



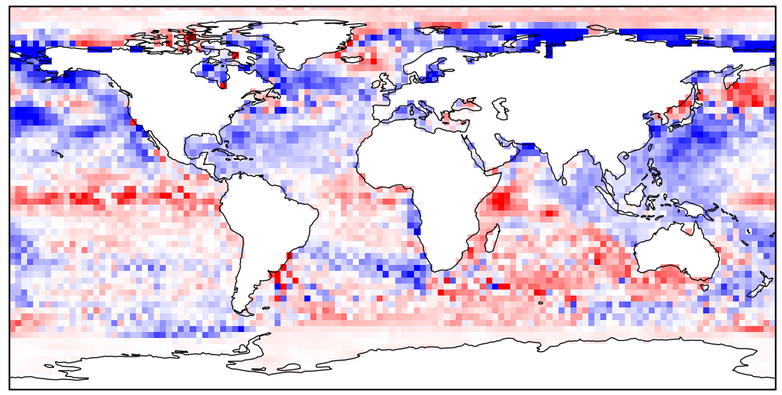
Actual — Step 46 (°C)



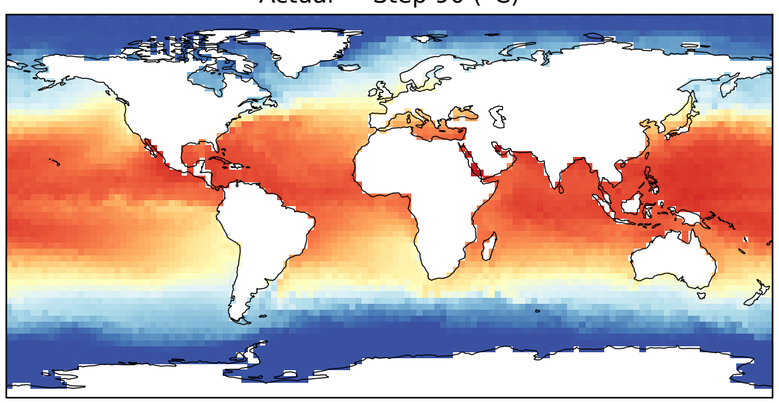
SFNO — Step 46 (°C)



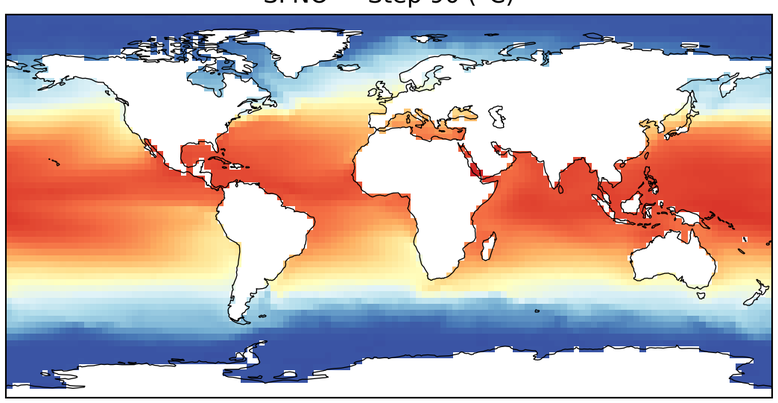
Error — Step 46 (°C)



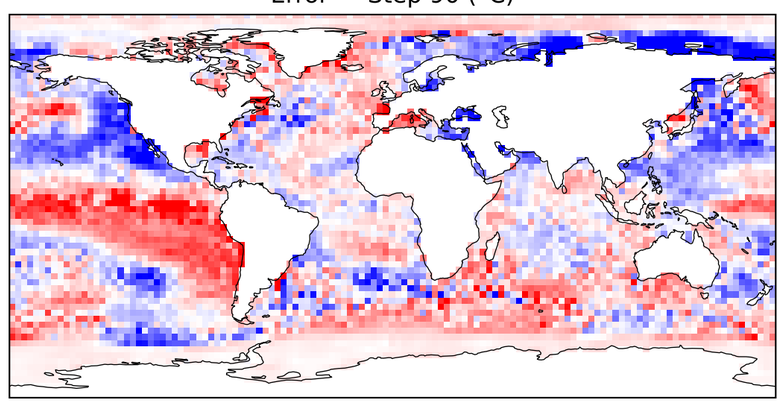
Actual — Step 90 (°C)

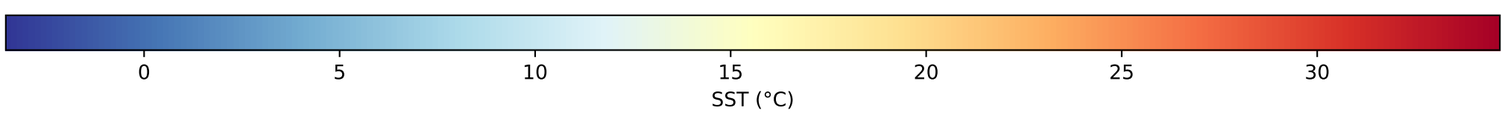


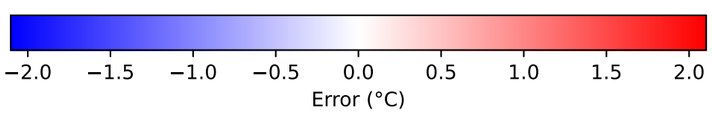
SFNO — Step 90 (°C)



Error — Step 90 (°C)

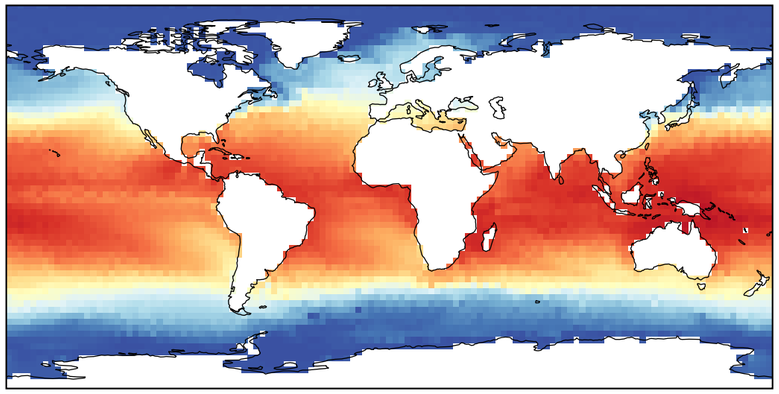




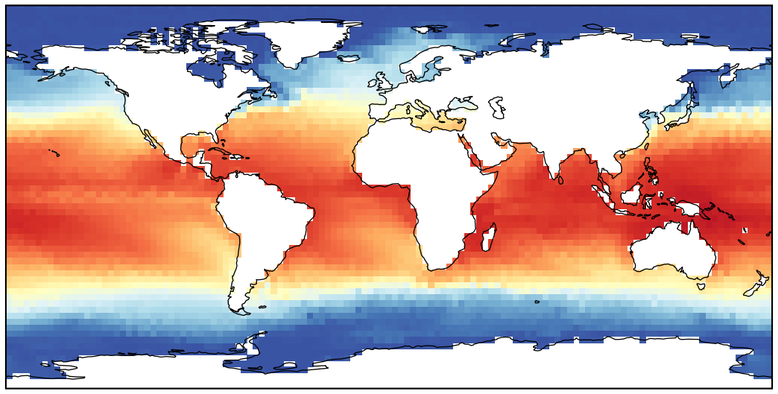



# Chain init: 2021-01-06 — Test (°C)

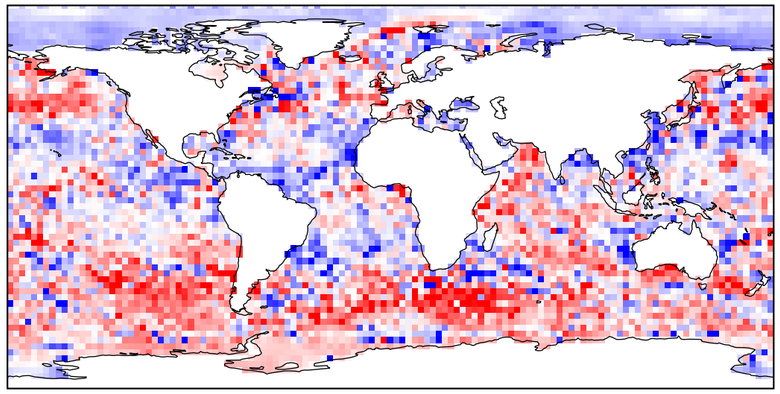
Actual — Step 1 (°C)



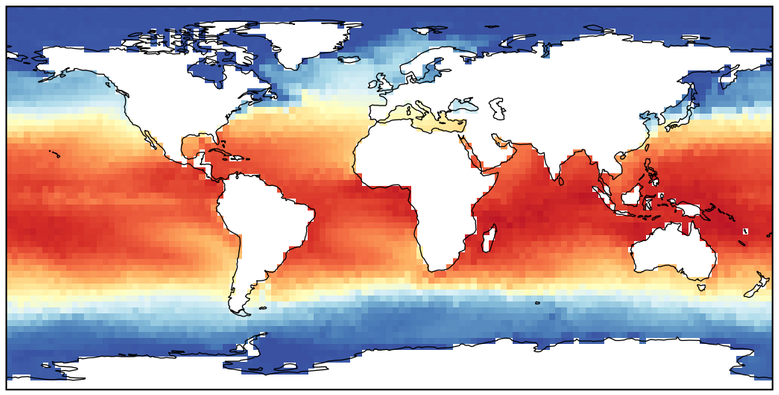
SFNO — Step 1 (°C)



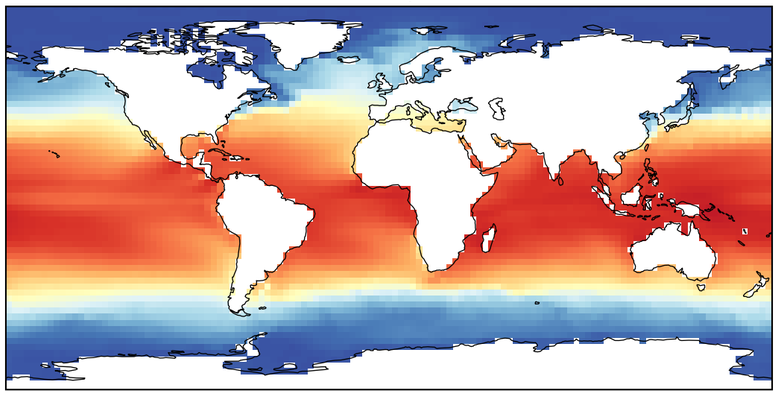
Error — Step 1 (°C)



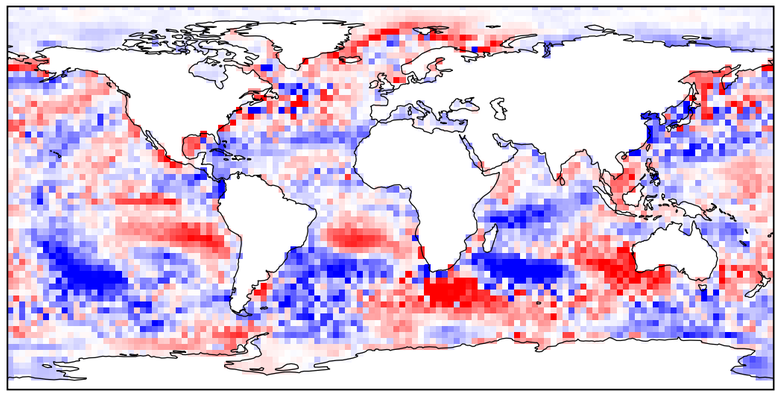
Actual — Step 46 (°C)



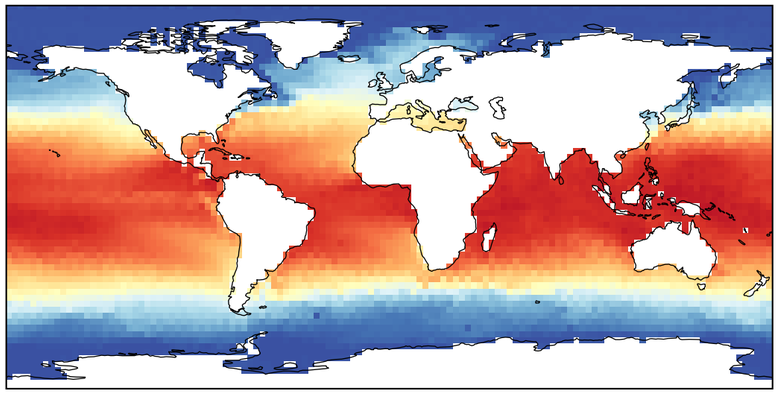
SFNO — Step 46 (°C)



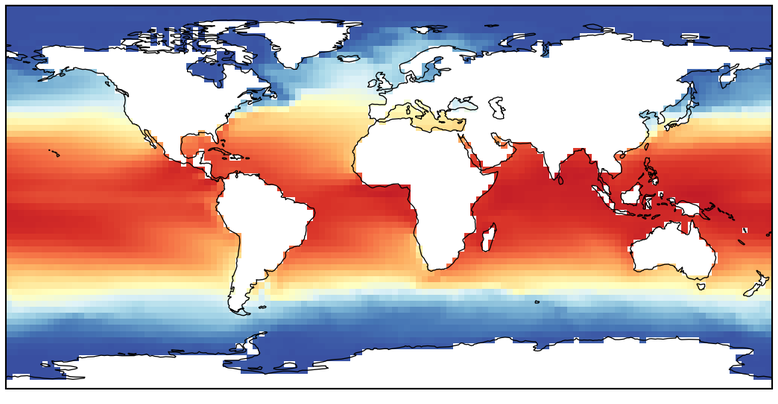
Error — Step 46 (°C)



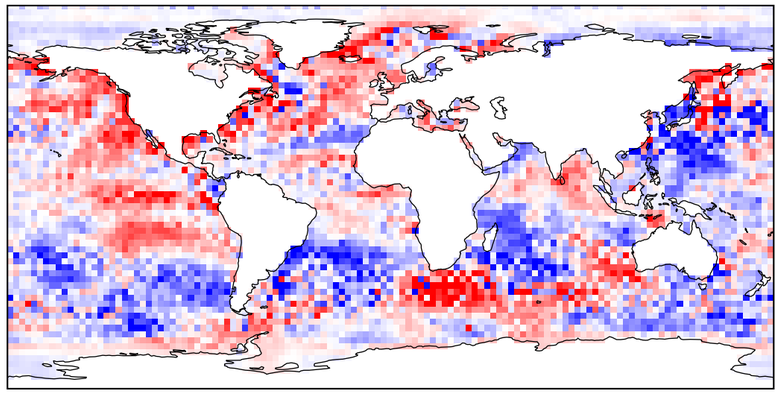
Actual — Step 90 (°C)

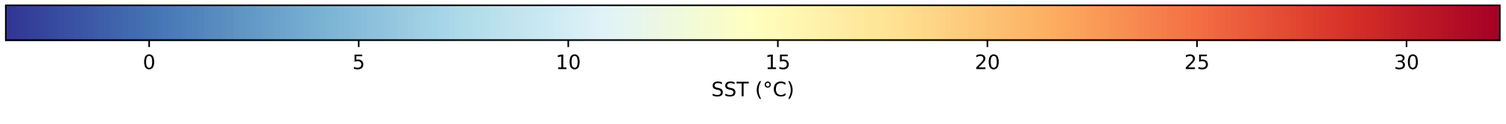


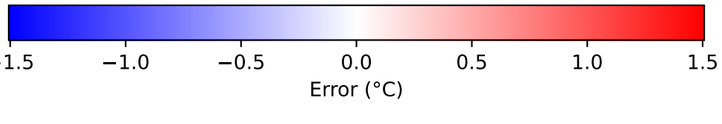
SFNO — Step 90 (°C)



Error — Step 90 (°C)

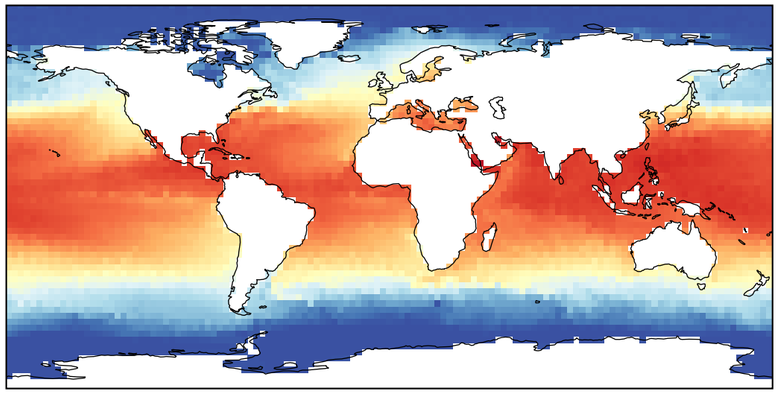




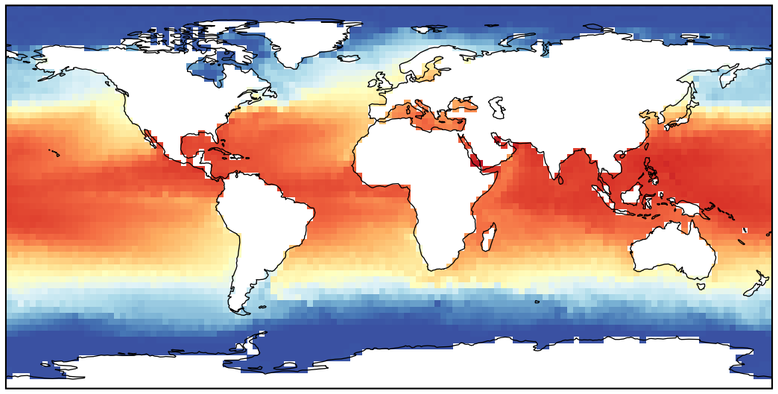



# Chain init: 2021-07-05 — Test (°C)

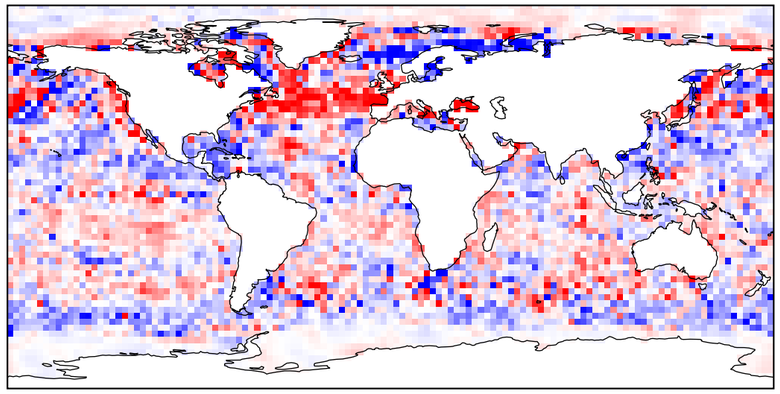
Actual — Step 1 (°C)



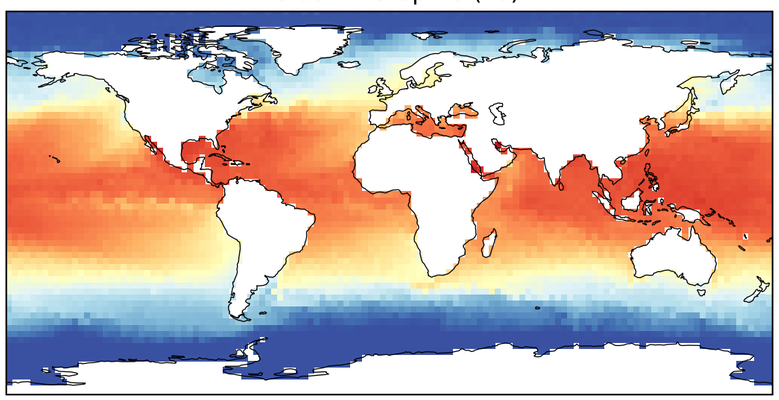
SFNO — Step 1 (°C)



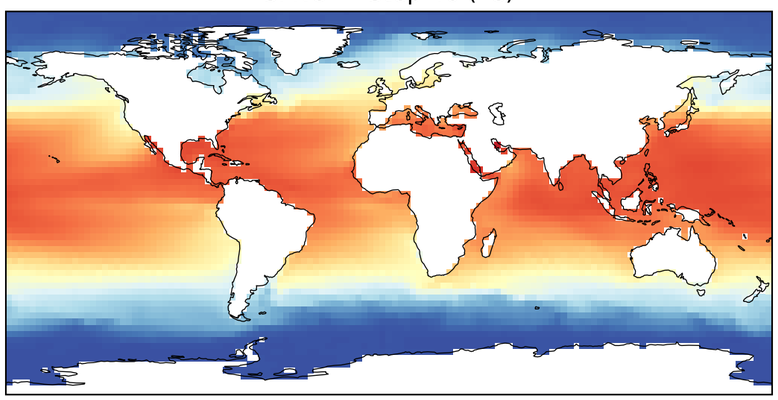
Error — Step 1 (°C)



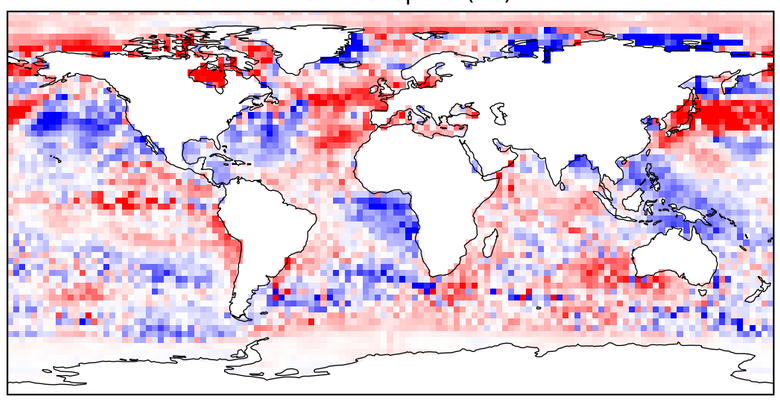
Actual — Step 46 (°C)



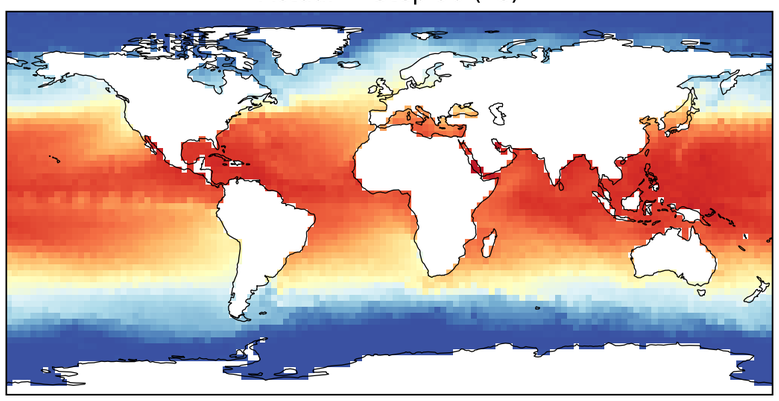
SFNO — Step 46 (°C)



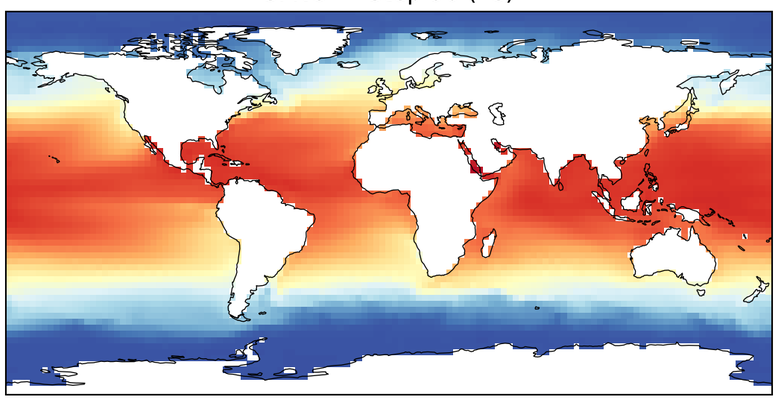
Error — Step 46 (°C)



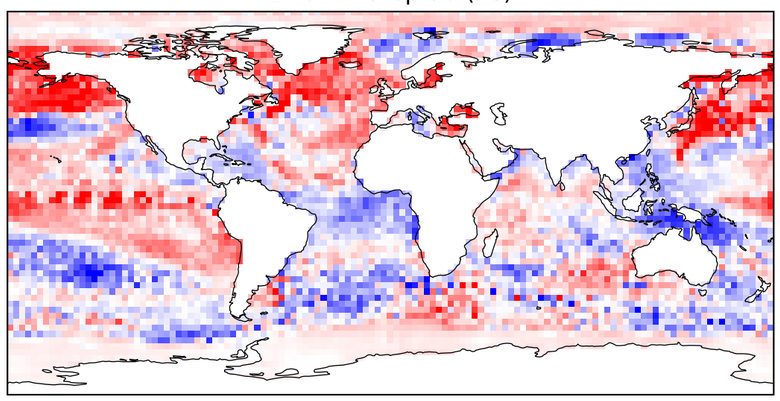
Actual — Step 90 (°C)

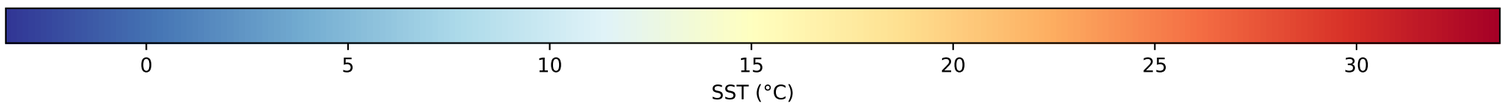


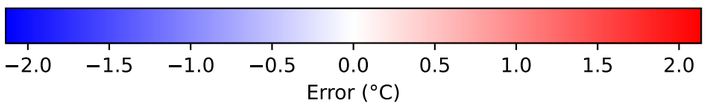
SFNO — Step 90 (°C)



Error — Step 90 (°C)

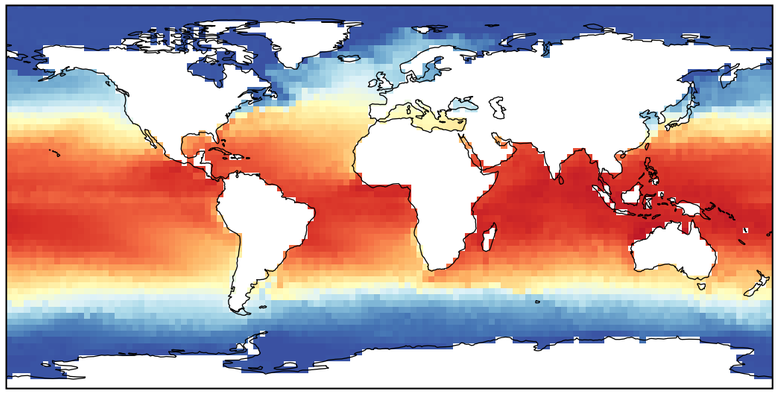




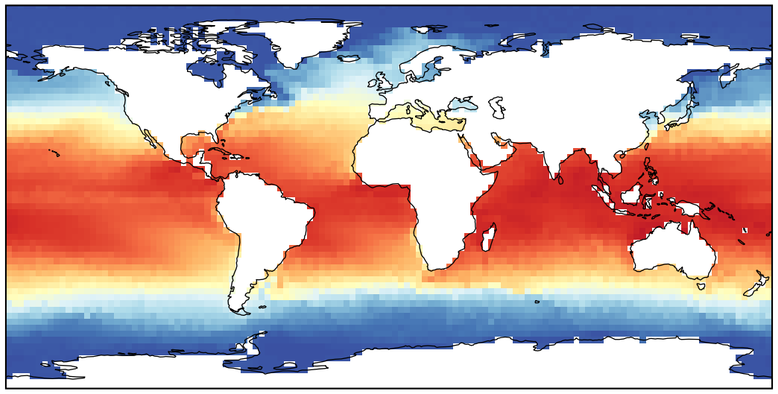



# Chain init: 2022-04-01 — Test (°C)

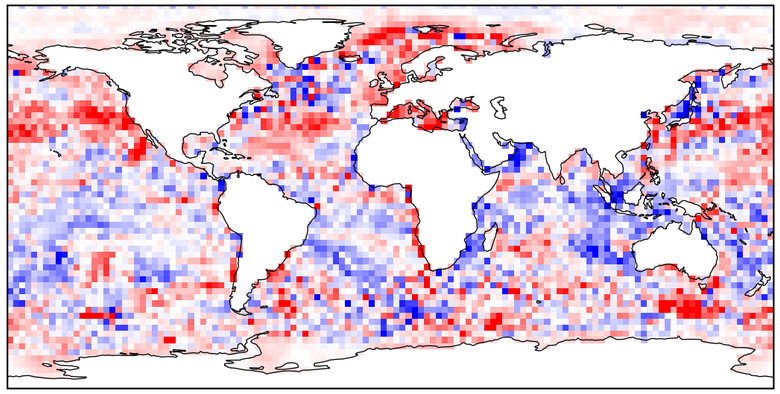
Actual — Step 1 (°C)



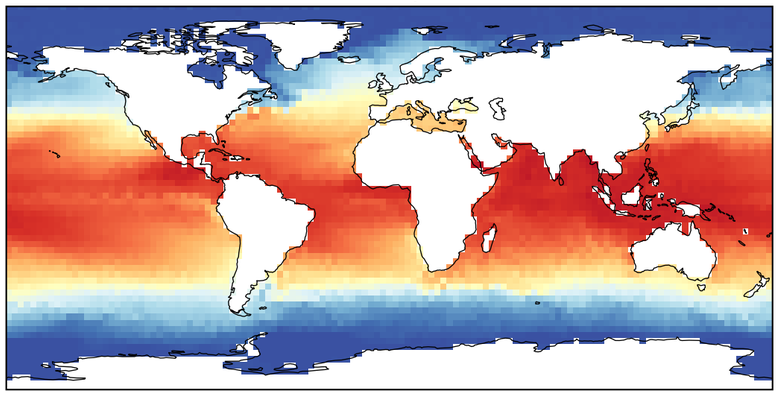
SFNO — Step 1 (°C)



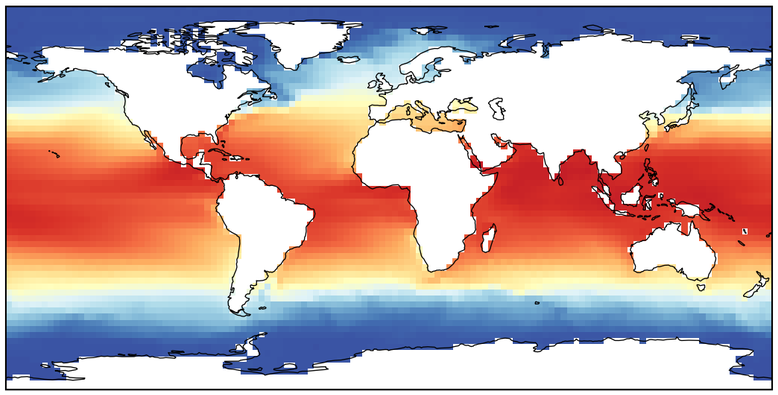
Error — Step 1 (°C)



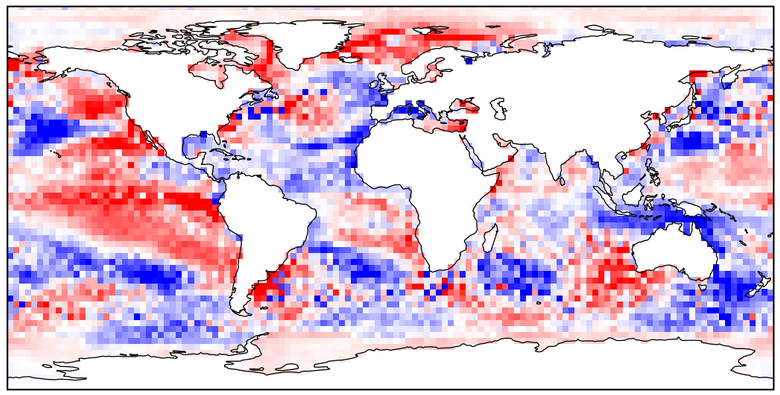
Actual — Step 46 (°C)



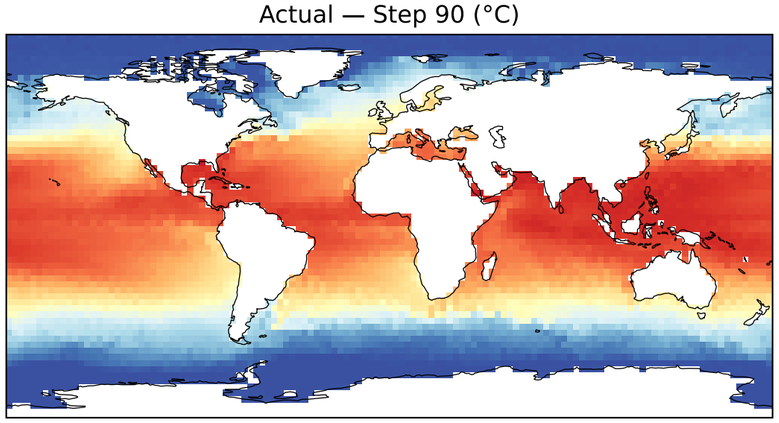
SFNO — Step 46 (°C)



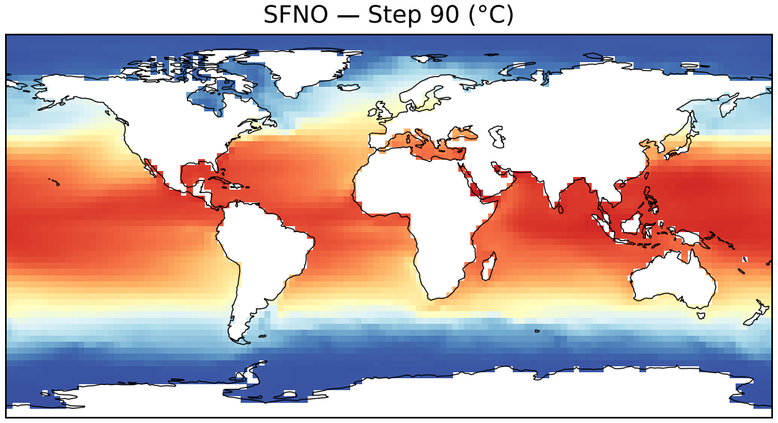
Error — Step 46 (°C)



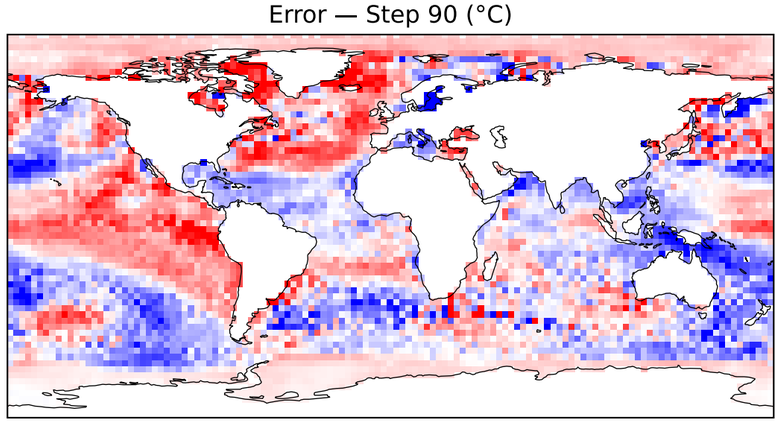
Actual — Step 90 (°C)

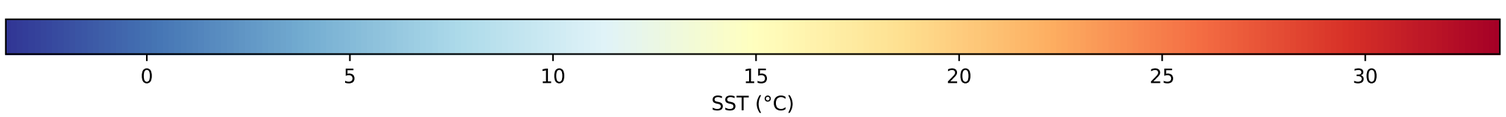


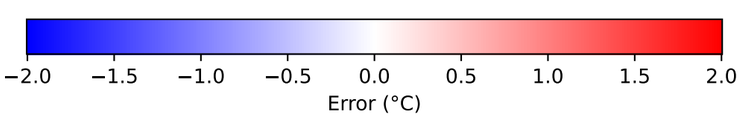
SFNO — Step 90 (°C)



Error — Step 90 (°C)

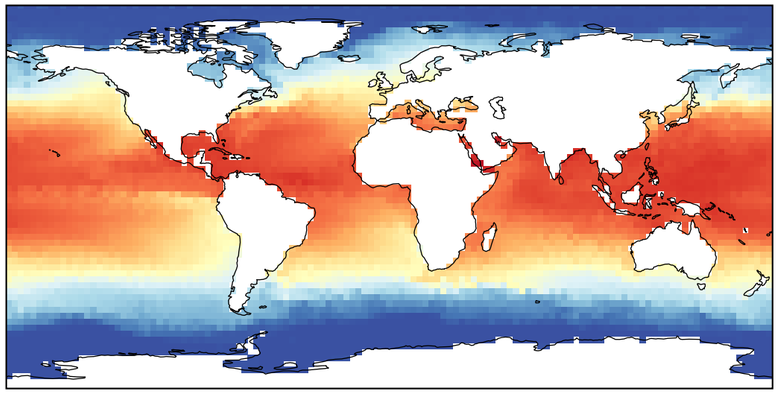




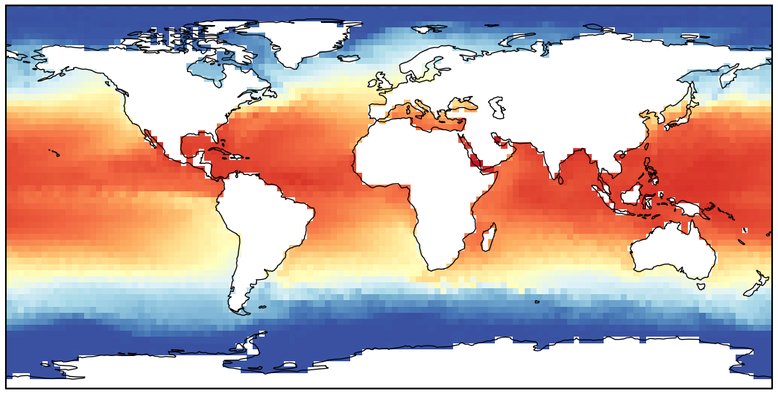



# Chain init: 2022-09-28 — Test (°C)

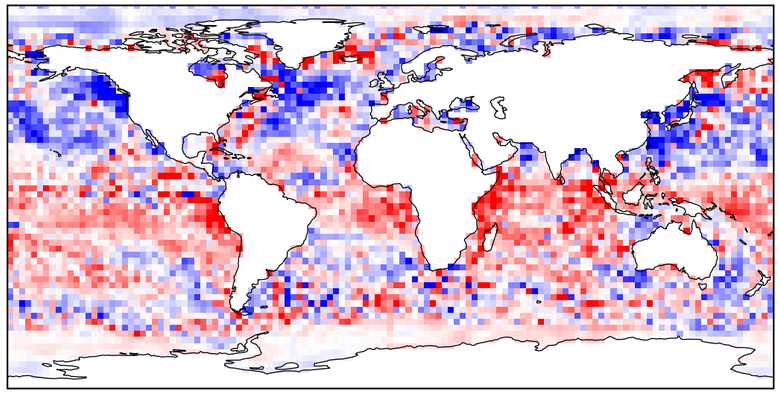
Actual — Step 1 (°C)



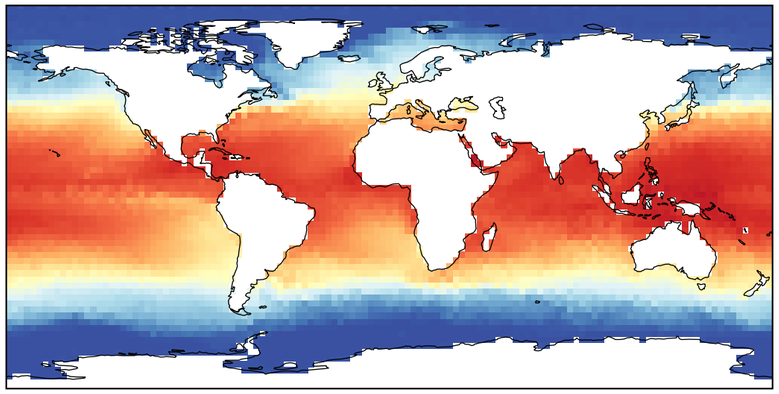
SFNO — Step 1 (°C)



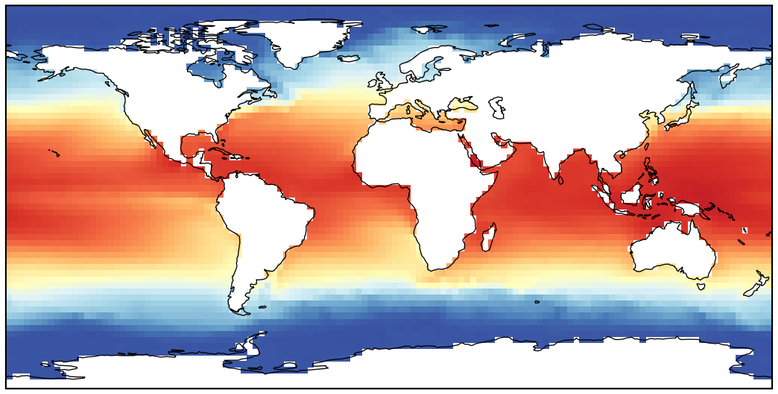
Error — Step 1 (°C)



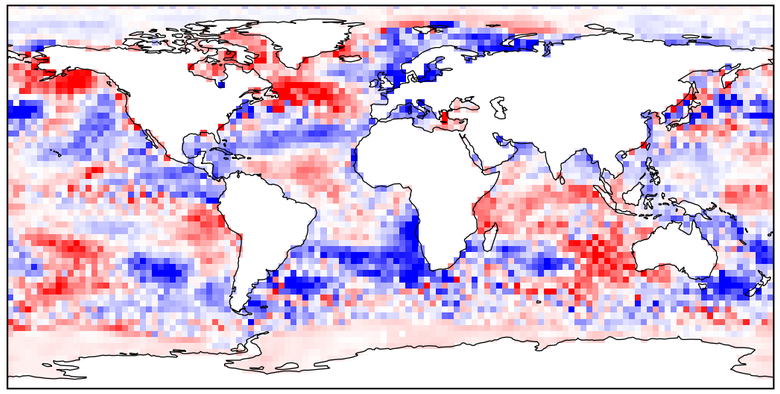
Actual — Step 46 (°C)



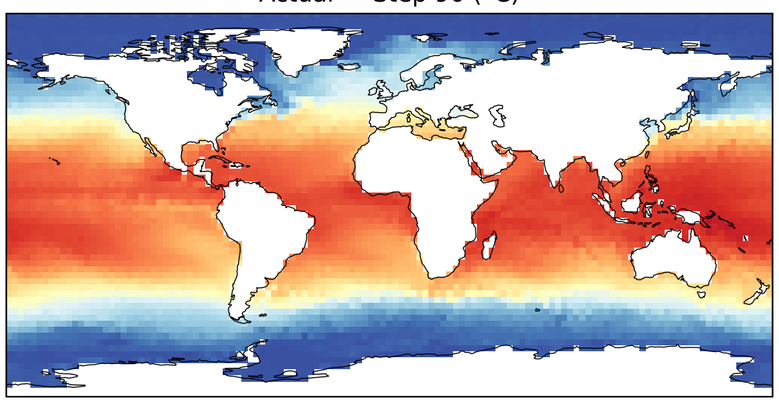
SFNO — Step 46 (°C)



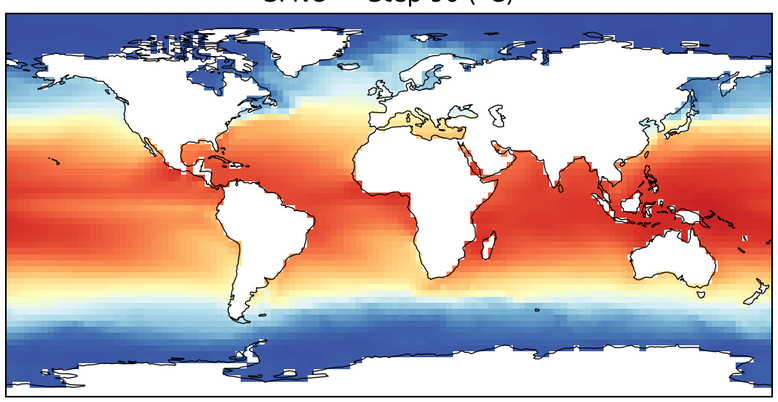
Error — Step 46 (°C)



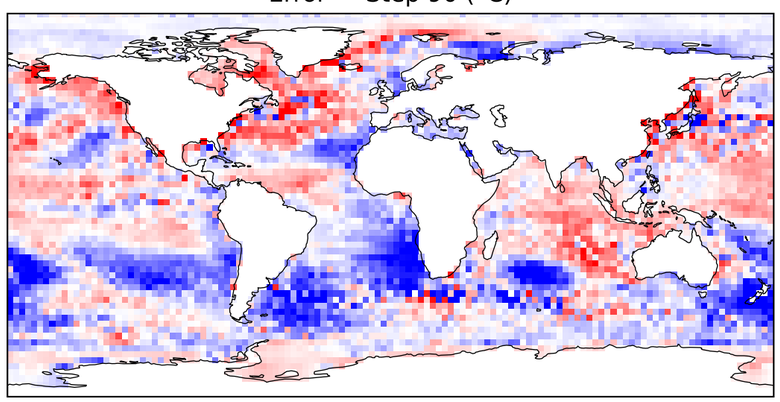
Actual — Step 90 (°C)

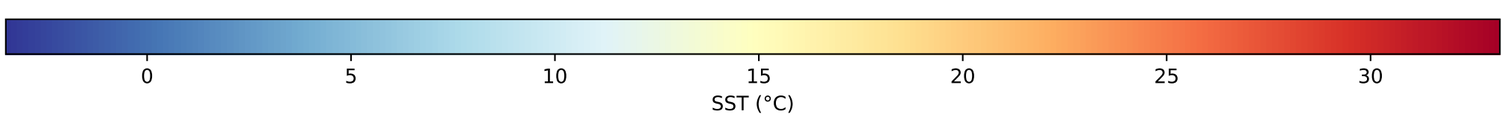


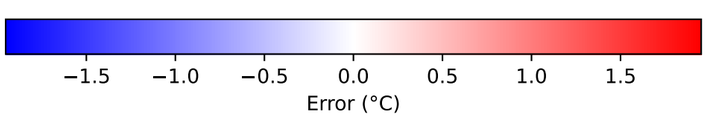
SFNO — Step 90 (°C)



Error — Step 90 (°C)

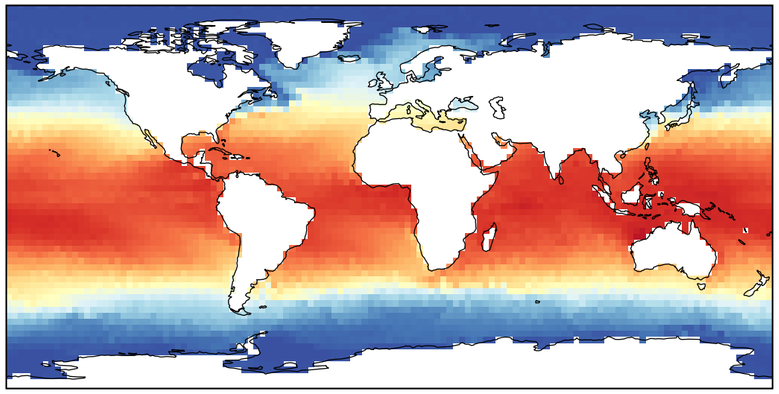




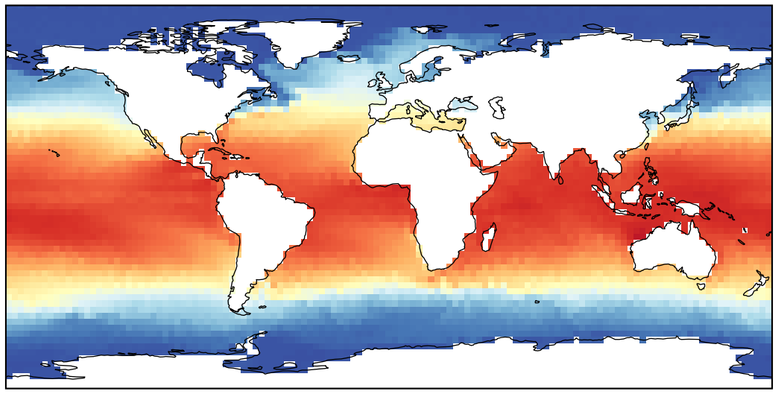



# Chain init: 2023-03-27 — Test (°C)

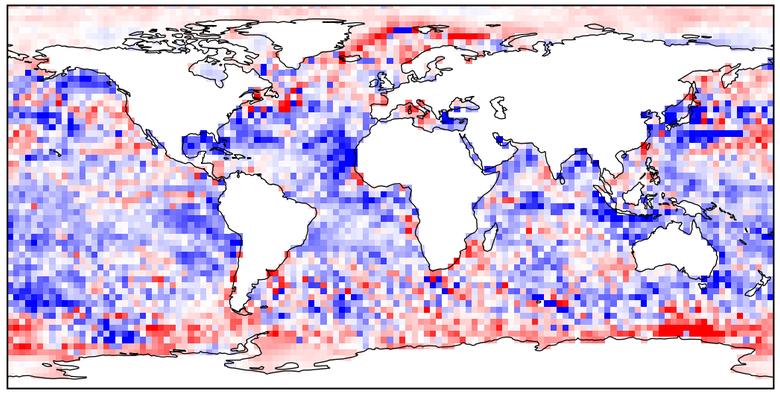
Actual — Step 1 (°C)



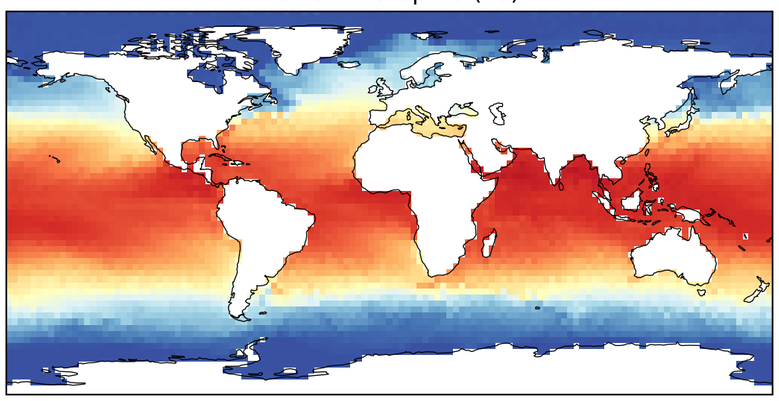
SFNO — Step 1 (°C)



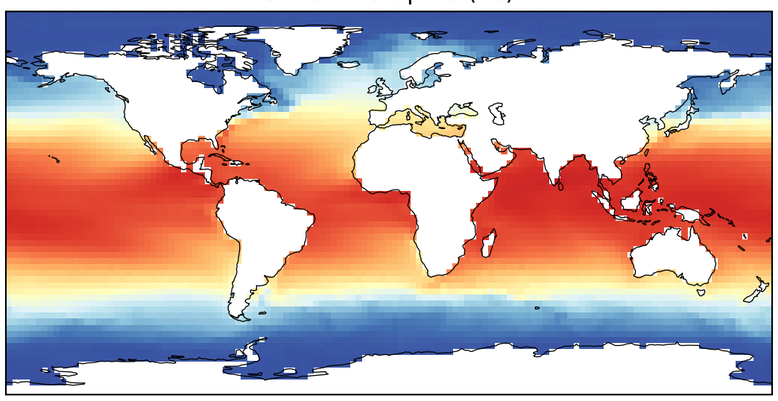
Error — Step 1 (°C)



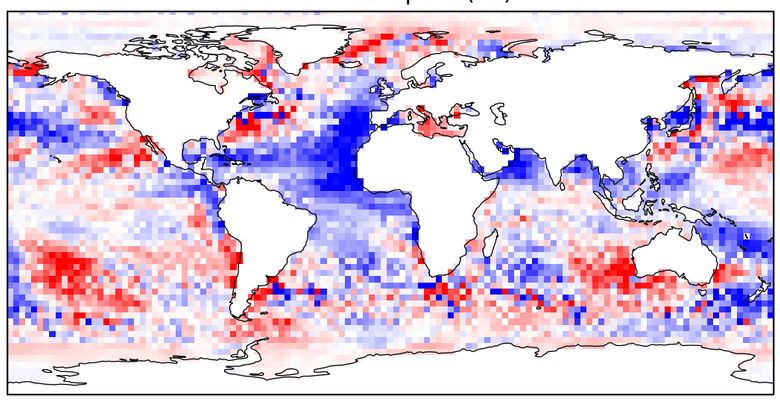
Actual — Step 46 (°C)



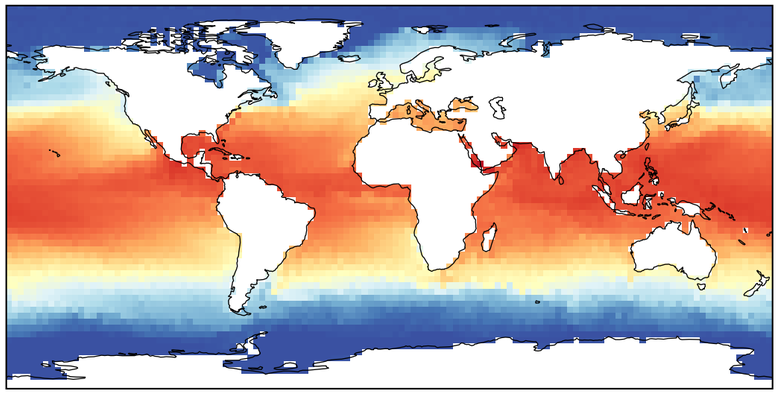
SFNO — Step 46 (°C)



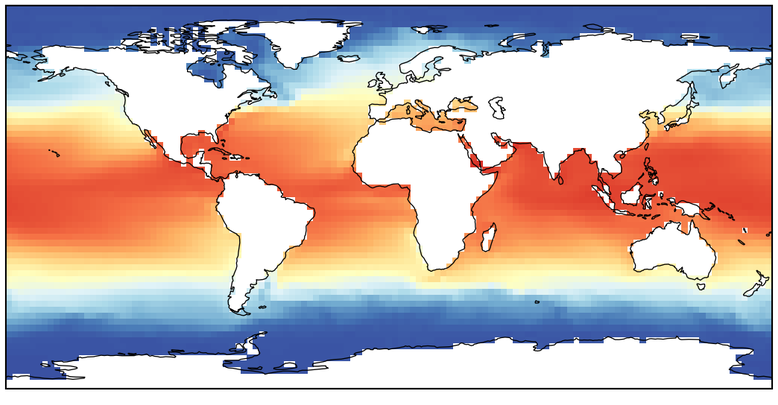
Error — Step 46 (°C)



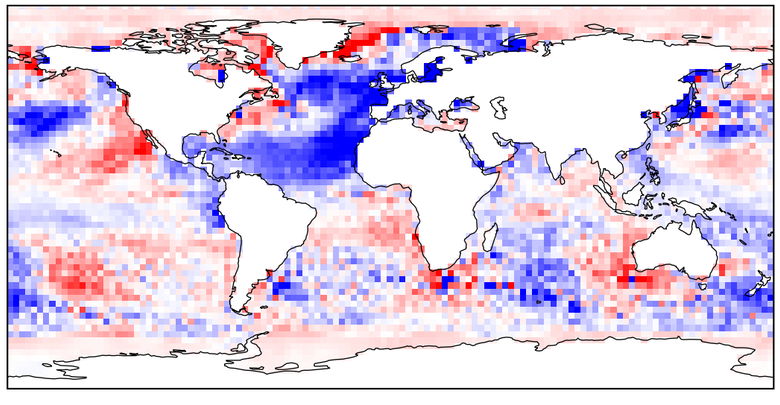
Actual — Step 90 (°C)

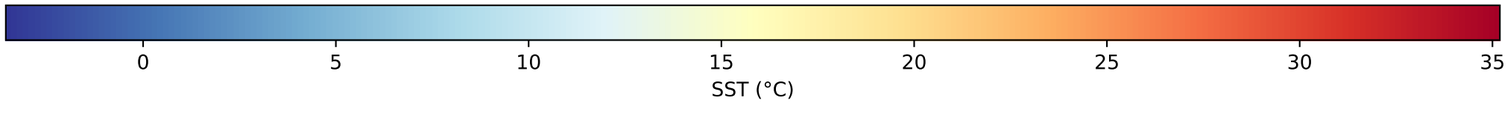


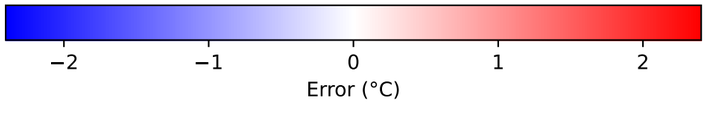
SFNO — Step 90 (°C)



Error — Step 90 (°C)

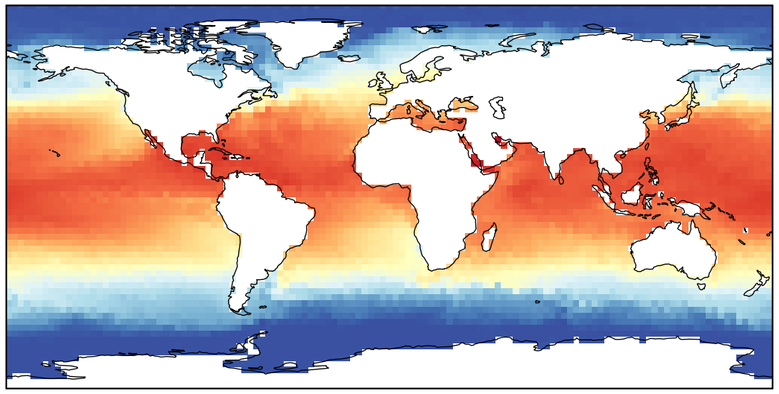




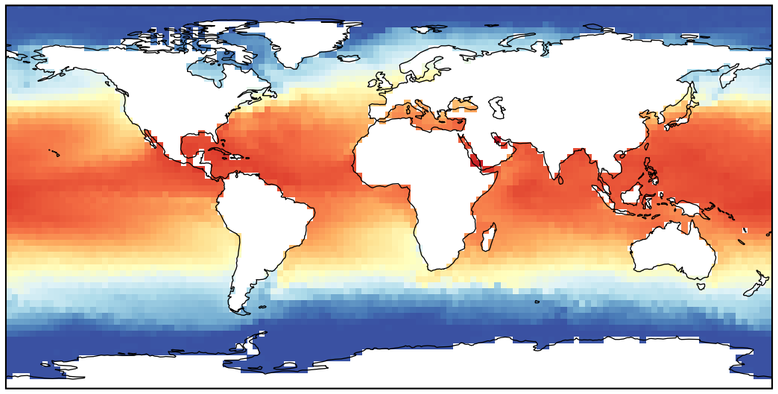



# Chain init: 2023-09-23 — Test (°C)

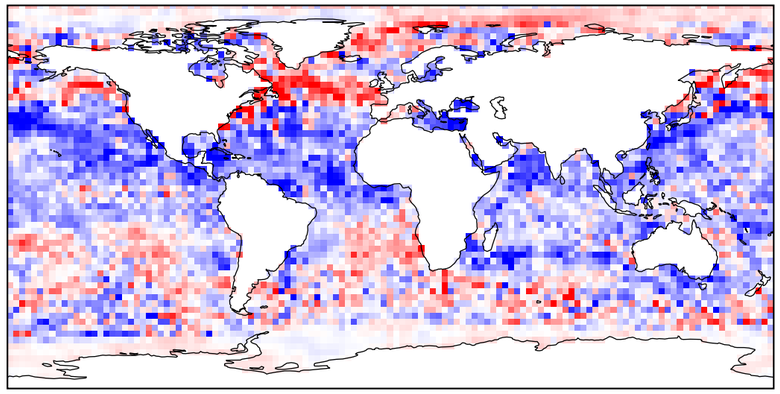
Actual — Step 1 (°C)



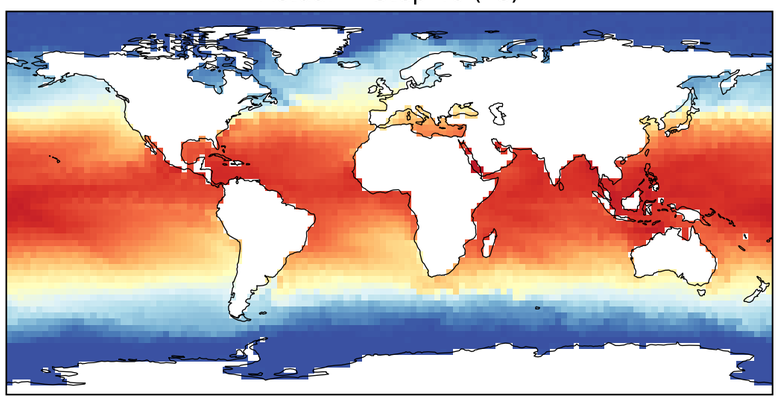
SFNO — Step 1 (°C)



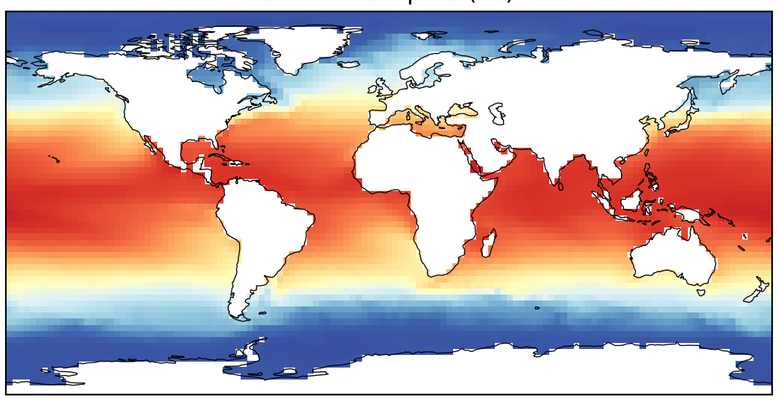
Error — Step 1 (°C)



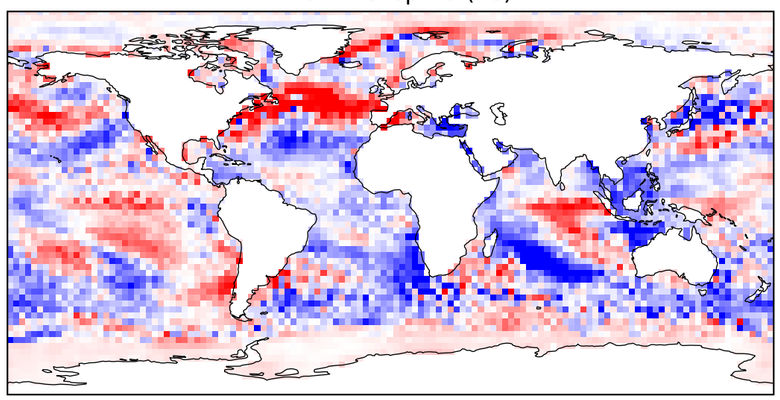
Actual — Step 46 (°C)



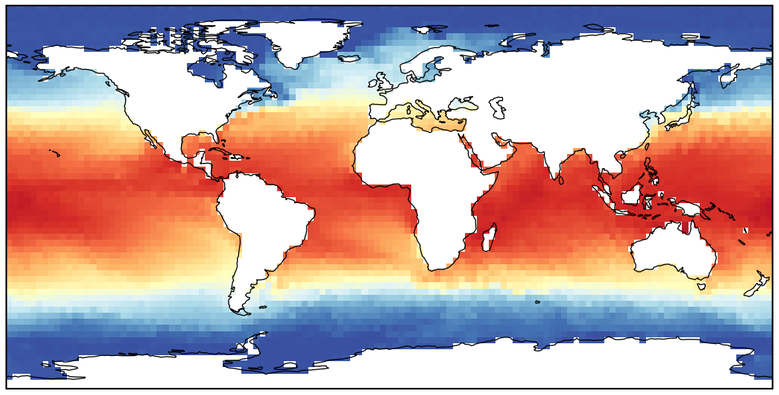
SFNO — Step 46 (°C)



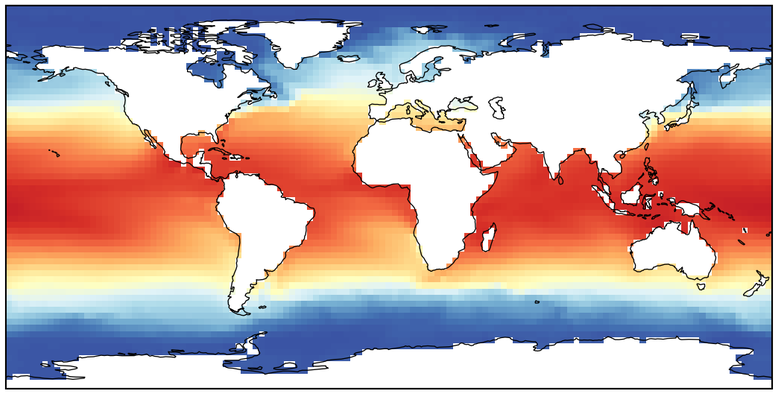
Error — Step 46 (°C)



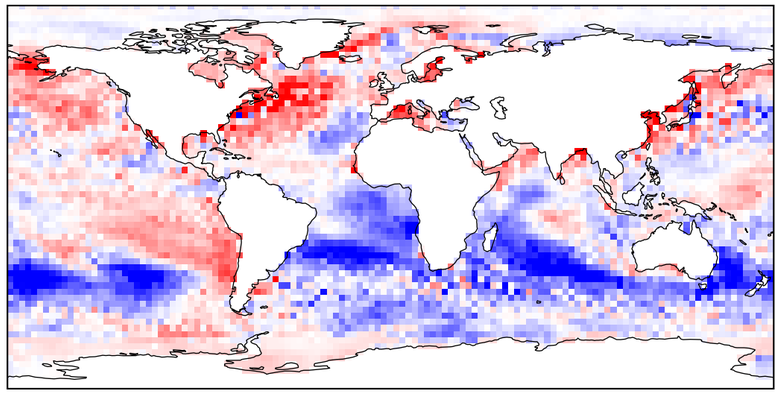
Actual — Step 90 (°C)

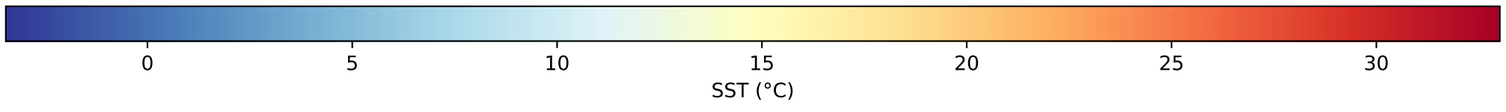


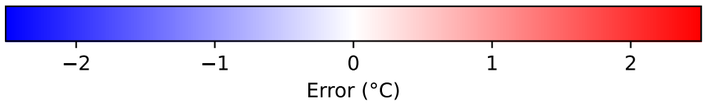
SFNO — Step 90 (°C)



Error — Step 90 (°C)

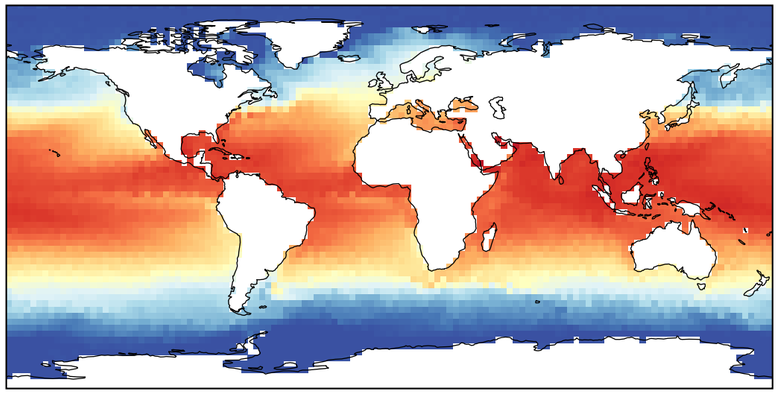




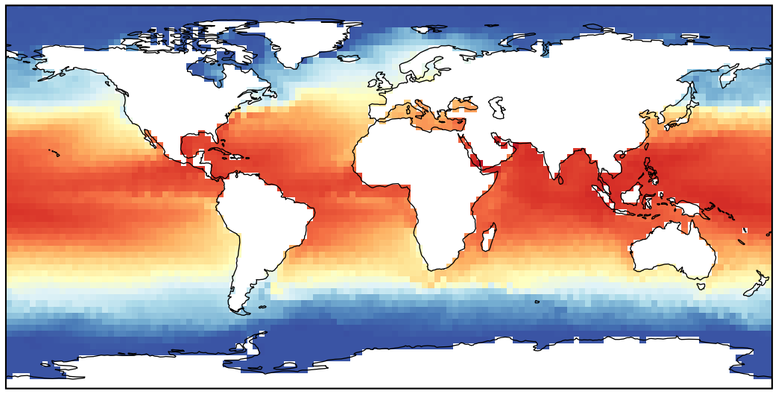



# Chain init: 2024-06-19 — Test (°C)

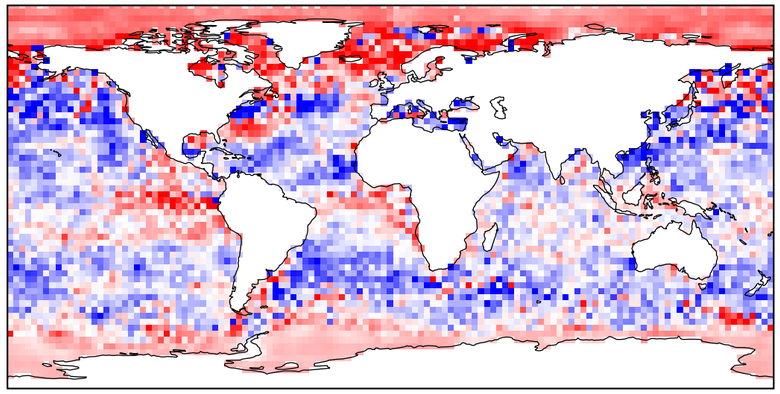
Actual — Step 1 (°C)



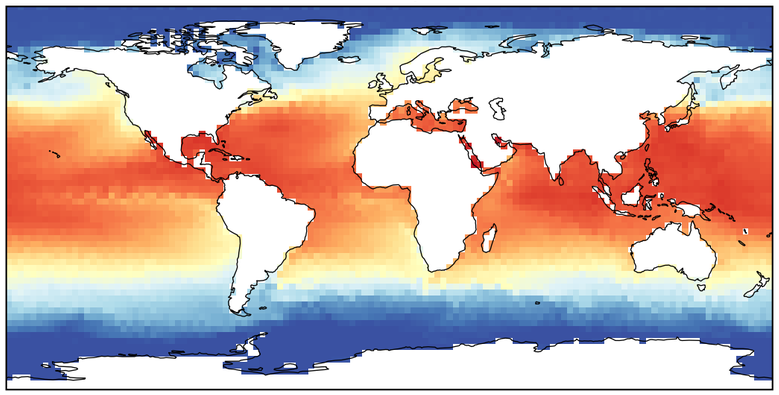
SFNO — Step 1 (°C)



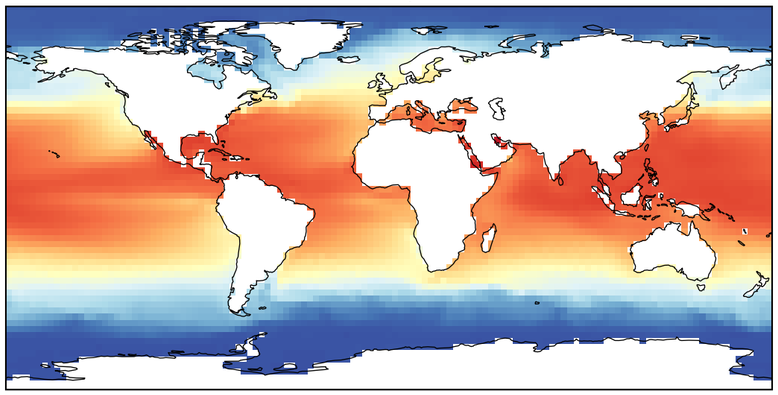
Error — Step 1 (°C)



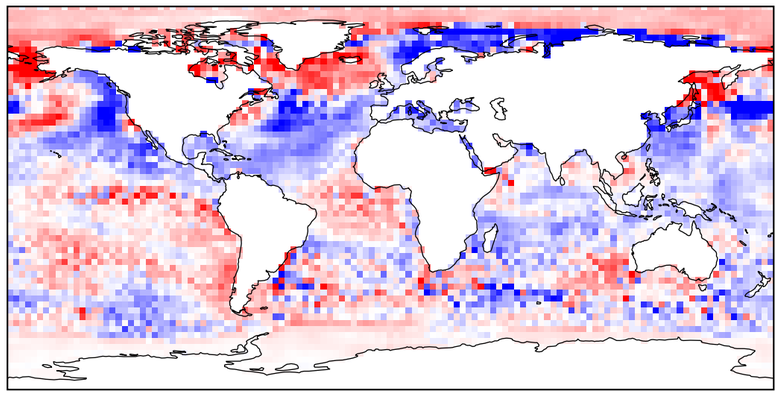
Actual — Step 46 (°C)



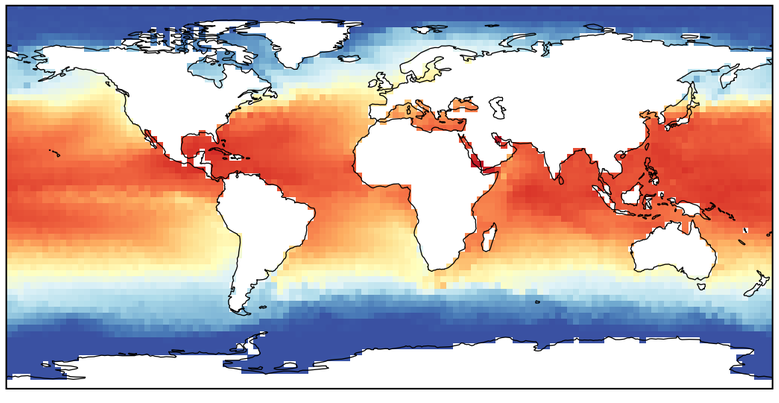
SFNO — Step 46 (°C)



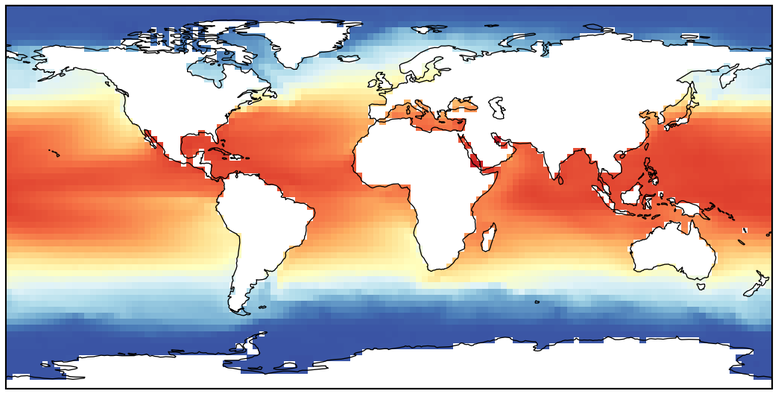
Error — Step 46 (°C)



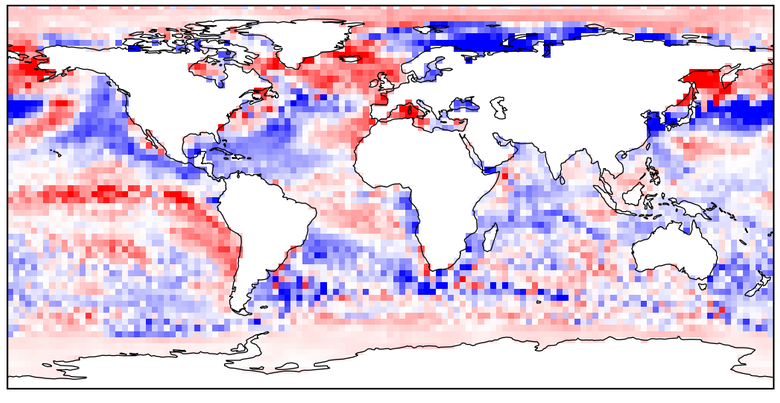
Actual — Step 90 (°C)

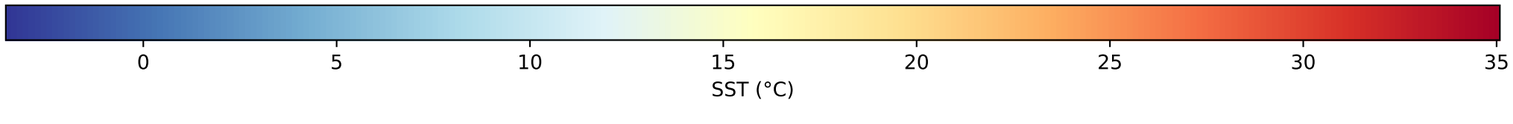


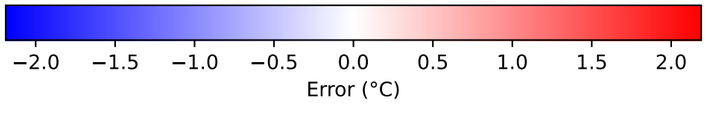
SFNO — Step 90 (°C)



Error — Step 90 (°C)



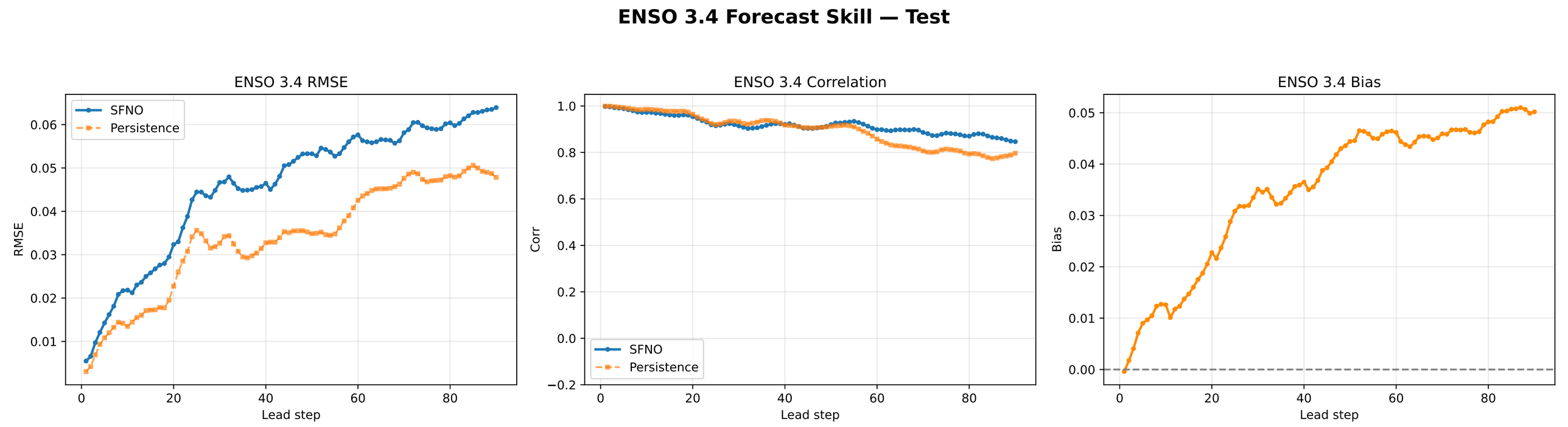


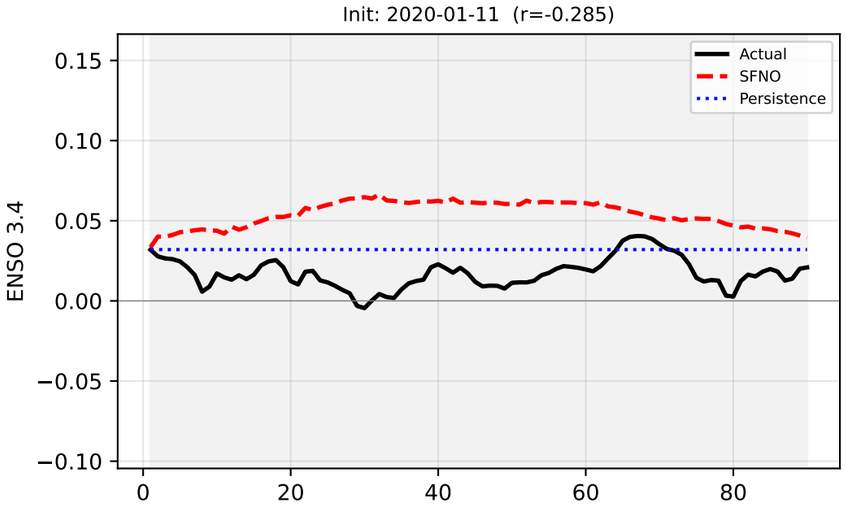
# Global-Mean SST — Test (°C)

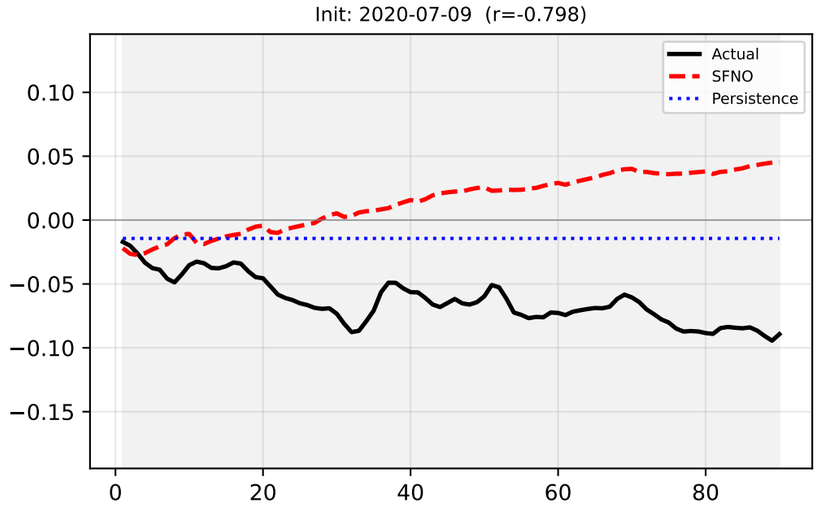
# ENSO 3.4 Metrics — Test

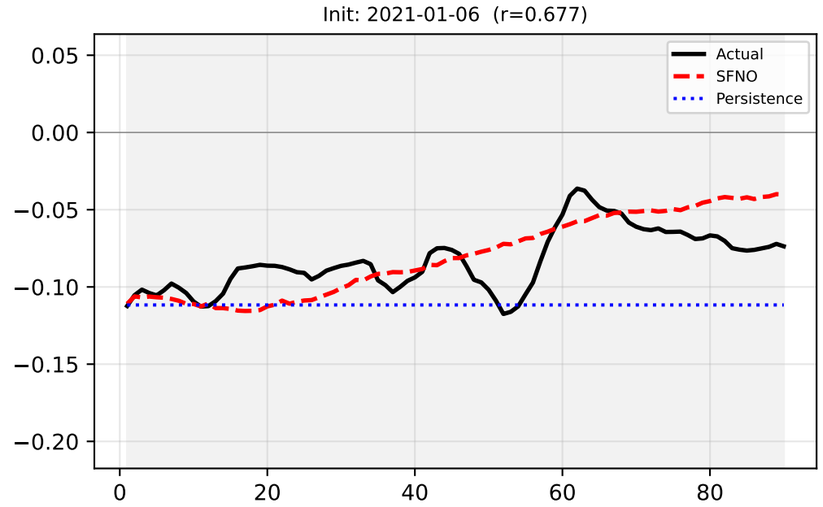
| Lead Step | SFNO RMSE | Persist RMSE | SFNO Corr | Persist Corr | SFNO Bias |
|---|---|---|---|---|---|
| 1 | 0.0055 | 0.0031 | 0.9976 | 0.9992 | -0.0004 |
| 2 | 0.0066 | 0.0042 | 0.9963 | 0.9989 | 0.0018 |
| 3 | 0.0097 | 0.0069 | 0.9925 | 0.9972 | 0.0041 |
| 4 | 0.0121 | 0.0093 | 0.9908 | 0.9950 | 0.0071 |
| 5 | 0.0143 | 0.0108 | 0.9883 | 0.9927 | 0.0090 |
| 6 | 0.0162 | 0.0120 | 0.9841 | 0.9896 | 0.0097 |
| 7 | 0.0181 | 0.0133 | 0.9795 | 0.9867 | 0.0105 |
| 8 | 0.0209 | 0.0144 | 0.9739 | 0.9842 | 0.0123 |
| 9 | 0.0217 | 0.0142 | 0.9721 | 0.9843 | 0.0127 |
| 10 | 0.0218 | 0.0135 | 0.9723 | 0.9857 | 0.0126 |
| 11 | 0.0212 | 0.0145 | 0.9722 | 0.9850 | 0.0101 |
| 12 | 0.0230 | 0.0155 | 0.9692 | 0.9833 | 0.0118 |
| 13 | 0.0236 | 0.0160 | 0.9675 | 0.9817 | 0.0123 |
| 14 | 0.0250 | 0.0171 | 0.9640 | 0.9784 | 0.0137 |
| 15 | 0.0258 | 0.0172 | 0.9614 | 0.9772 | 0.0147 |
| 16 | 0.0267 | 0.0173 | 0.9592 | 0.9774 | 0.0160 |
| 17 | 0.0276 | 0.0178 | 0.9594 | 0.9770 | 0.0175 |
| 18 | 0.0280 | 0.0177 | 0.9616 | 0.9777 | 0.0188 |
| 19 | 0.0295 | 0.0195 | 0.9603 | 0.9736 | 0.0205 |
| 20 | 0.0324 | 0.0227 | 0.9539 | 0.9644 | 0.0228 |
| 21 | 0.0330 | 0.0260 | 0.9466 | 0.9526 | 0.0216 |
| 22 | 0.0362 | 0.0285 | 0.9372 | 0.9440 | 0.0237 |
| 23 | 0.0388 | 0.0308 | 0.9307 | 0.9369 | 0.0259 |
| 24 | 0.0427 | 0.0341 | 0.9191 | 0.9251 | 0.0288 |
| 25 | 0.0445 | 0.0356 | 0.9149 | 0.9197 | 0.0308 |
| 26 | 0.0444 | 0.0349 | 0.9172 | 0.9226 | 0.0318 |
| 27 | 0.0436 | 0.0332 | 0.9215 | 0.9287 | 0.0318 |
| 28 | 0.0433 | 0.0315 | 0.9239 | 0.9351 | 0.0320 |
| 29 | 0.0448 | 0.0319 | 0.9199 | 0.9350 | 0.0335 |
| 30 | 0.0467 | 0.0326 | 0.9141 | 0.9322 | 0.0351 |
| 31 | 0.0468 | 0.0342 | 0.9086 | 0.9251 | 0.0345 |
| 32 | 0.0480 | 0.0344 | 0.9029 | 0.9217 | 0.0351 |
| 33 | 0.0465 | 0.0325 | 0.9045 | 0.9261 | 0.0335 |
| 34 | 0.0452 | 0.0308 | 0.9063 | 0.9305 | 0.0322 |
| 35 | 0.0448 | 0.0295 | 0.9113 | 0.9367 | 0.0324 |
| 36 | 0.0449 | 0.0293 | 0.9168 | 0.9386 | 0.0333 |
| 37 | 0.0450 | 0.0297 | 0.9218 | 0.9370 | 0.0344 |
| 38 | 0.0455 | 0.0304 | 0.9240 | 0.9336 | 0.0356 |
| 39 | 0.0457 | 0.0314 | 0.9233 | 0.9265 | 0.0359 |
| 40 | 0.0465 | 0.0327 | 0.9207 | 0.9182 | 0.0365 |
| 41 | 0.0450 | 0.0329 | 0.9237 | 0.9161 | 0.0350 |
| 42 | 0.0463 | 0.0329 | 0.9178 | 0.9151 | 0.0356 |
| 43 | 0.0481 | 0.0339 | 0.9119 | 0.9113 | 0.0368 |
| 44 | 0.0505 | 0.0353 | 0.9044 | 0.9066 | 0.0388 |
| 45 | 0.0508 | 0.0351 | 0.9036 | 0.9069 | 0.0393 |
| 46 | 0.0515 | 0.0355 | 0.9031 | 0.9049 | 0.0405 |
| 47 | 0.0524 | 0.0355 | 0.9056 | 0.9067 | 0.0418 |
| 48 | 0.0532 | 0.0355 | 0.9080 | 0.9080 | 0.0430 |
| 49 | 0.0533 | 0.0352 | 0.9122 | 0.9099 | 0.0436 |
| 50 | 0.0533 | 0.0349 | 0.9196 | 0.9127 | 0.0444 |
| 51 | 0.0528 | 0.0349 | 0.9270 | 0.9148 | 0.0446 |
| 52 | 0.0546 | 0.0352 | 0.9266 | 0.9157 | 0.0465 |
| 53 | 0.0543 | 0.0346 | 0.9295 | 0.9176 | 0.0464 |
| 54 | 0.0537 | 0.0345 | 0.9309 | 0.9155 | 0.0459 |
| 55 | 0.0527 | 0.0348 | 0.9340 | 0.9104 | 0.0450 |
| 56 | 0.0533 | 0.0362 | 0.9288 | 0.9010 | 0.0449 |
| 57 | 0.0547 | 0.0378 | 0.9219 | 0.8912 | 0.0458 |
| 58 | 0.0560 | 0.0390 | 0.9131 | 0.8831 | 0.0463 |
| 59 | 0.0571 | 0.0408 | 0.9040 | 0.8706 | 0.0464 |
| 60 | 0.0576 | 0.0425 | 0.8982 | 0.8575 | 0.0462 |
| 61 | 0.0563 | 0.0435 | 0.8981 | 0.8473 | 0.0444 |
| 62 | 0.0560 | 0.0441 | 0.8948 | 0.8398 | 0.0438 |
| 63 | 0.0558 | 0.0448 | 0.8931 | 0.8324 | 0.0434 |
| 64 | 0.0560 | 0.0452 | 0.8967 | 0.8290 | 0.0443 |
| 65 | 0.0566 | 0.0452 | 0.8977 | 0.8279 | 0.0453 |
| 66 | 0.0565 | 0.0452 | 0.8971 | 0.8256 | 0.0454 |
| 67 | 0.0564 | 0.0453 | 0.8964 | 0.8230 | 0.0454 |
| 68 | 0.0557 | 0.0457 | 0.8990 | 0.8187 | 0.0448 |
| 69 | 0.0563 | 0.0462 | 0.8962 | 0.8146 | 0.0451 |
| 70 | 0.0581 | 0.0476 | 0.8859 | 0.8065 | 0.0459 |
| 71 | 0.0589 | 0.0486 | 0.8810 | 0.8014 | 0.0458 |
| 72 | 0.0605 | 0.0490 | 0.8729 | 0.8005 | 0.0467 |
| 73 | 0.0605 | 0.0487 | 0.8724 | 0.8030 | 0.0467 |
| 74 | 0.0597 | 0.0474 | 0.8782 | 0.8113 | 0.0466 |
| 75 | 0.0593 | 0.0468 | 0.8832 | 0.8146 | 0.0467 |
| 76 | 0.0590 | 0.0470 | 0.8813 | 0.8124 | 0.0462 |
| 77 | 0.0589 | 0.0471 | 0.8796 | 0.8097 | 0.0461 |
| 78 | 0.0591 | 0.0472 | 0.8762 | 0.8062 | 0.0463 |
| 79 | 0.0602 | 0.0480 | 0.8713 | 0.7976 | 0.0476 |
| 80 | 0.0604 | 0.0482 | 0.8702 | 0.7935 | 0.0482 |
| 81 | 0.0598 | 0.0479 | 0.8775 | 0.7956 | 0.0483 |
| 82 | 0.0603 | 0.0482 | 0.8805 | 0.7931 | 0.0492 |
| 83 | 0.0613 | 0.0492 | 0.8789 | 0.7855 | 0.0503 |
| 84 | 0.0620 | 0.0500 | 0.8710 | 0.7786 | 0.0503 |
| 85 | 0.0628 | 0.0506 | 0.8651 | 0.7738 | 0.0507 |
| 86 | 0.0628 | 0.0500 | 0.8627 | 0.7761 | 0.0508 |
| 87 | 0.0631 | 0.0492 | 0.8601 | 0.7818 | 0.0510 |
| 88 | 0.0634 | 0.0490 | 0.8543 | 0.7847 | 0.0507 |
| 89 | 0.0635 | 0.0487 | 0.8489 | 0.7888 | 0.0499 |
| 90 | 0.0639 | 0.0478 | 0.8464 | 0.7967 | 0.0502 |

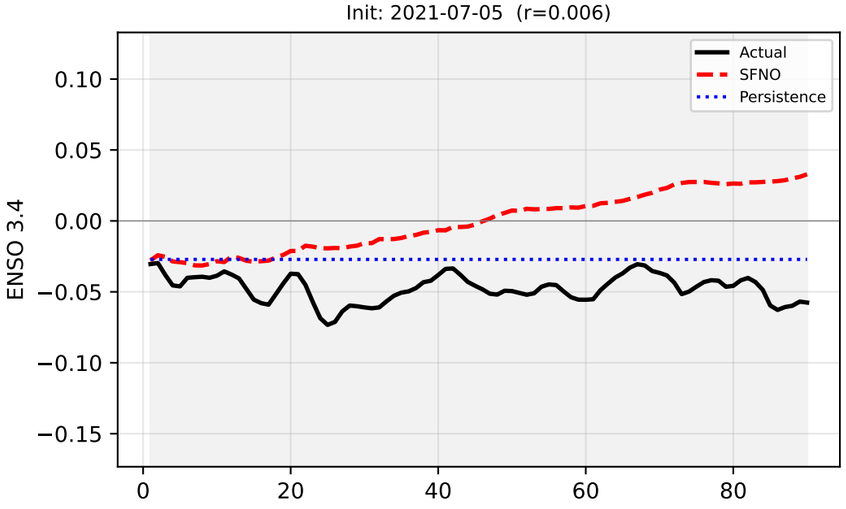

ENSO 3.4 Forecast Skill — Test
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

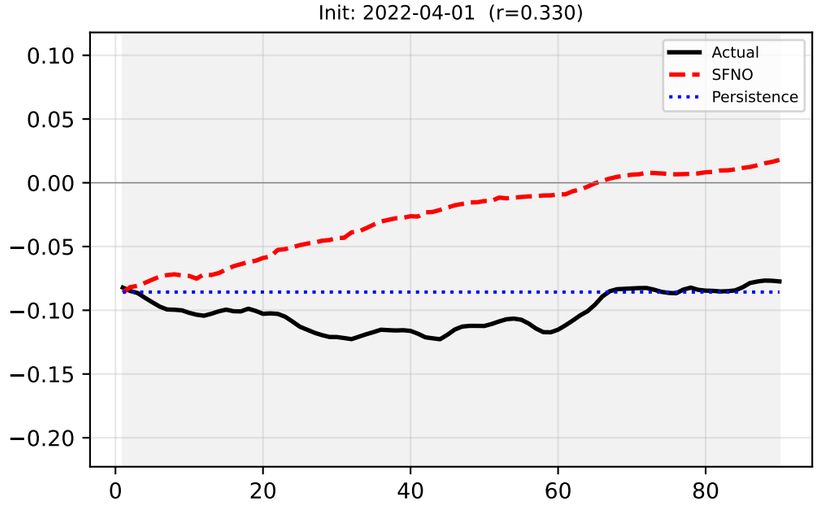
# ENSO 3.4 Index — Test

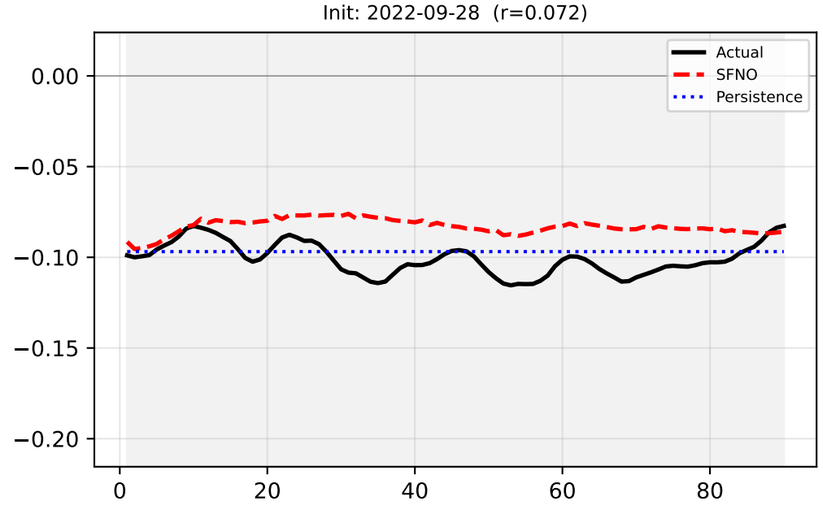


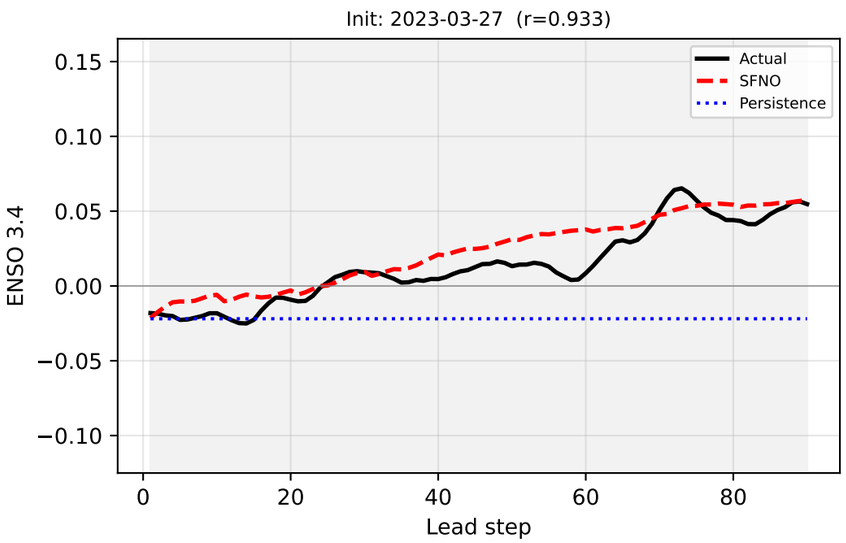



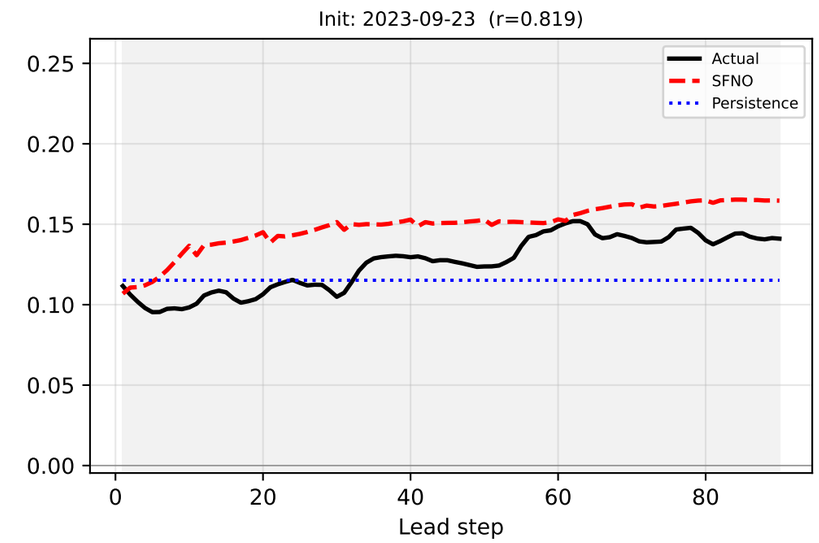



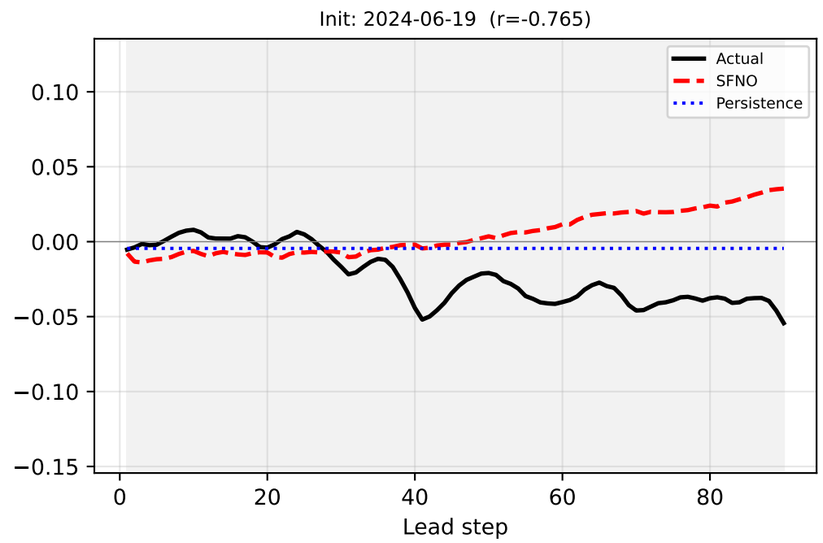



# ENSO 3.4 Scatter — Test